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RAMAN SPECTROSCOPY SYSTEM WITH STANDOFF DETECTION

Abstract

In one embodiment, a system includes a pump light source configured to produce a pump beam of light at a pump frequency and a Stokes light source configured to produce a Stokes beam of light at a Stokes frequency. The pump and Stokes frequencies are offset by a frequency offset Ω , and the pump and Stokes beams are directed to an object. The system further includes one or more optical elements configured to collect a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the object. The system also includes an optical receiver configured to detect the Raman signal. The optical receiver includes a probe light source configured to produce a probe beam of light at a probe frequency and an optical detector configured to coherently mix the Raman signal with the probe beam of light to produce a corresponding photocurrent signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. patent application Ser. No. 18/762,412, filed 2 Jul. 2024, which claims the benefit, under 35 U.S.C. § 119 (e), of U.S. Provisional Patent Application No. 63/560,521, filed 1 Mar. 2024, the entireties of which are incorporated by reference herein.

TECHNICAL FIELD

[0002] This disclosure generally relates to Raman spectroscopy systems and methods that use Raman scattering.

BACKGROUND

[0003] Raman spectroscopy is an optical measurement technique that can be applied to the study of molecular dynamics (e.g., to investigate vibrational and rotational states of molecules). Molecules typically exhibit molecular vibrations with frequencies ranging from less than 10 terahertz (THz) to approximately 100 THz, which corresponds to wavenumbers of approximately 300 cm.sup.-1 to 3000 cm.sup.-1 and wavelengths of approximately 30 to 3 micrometers (μm). Raman spectroscopy is based on the inelastic scattering of photons (referred to as Raman scattering) that occurs when light interacts with molecular vibrations or phonons in a sample. Raman scattering causes the energy (or equivalently, the frequency) of scattered light to be shifted, and this shift in energy can provide information about the vibrational modes of molecules in the sample.

[0004] Raman spectroscopy can be used in various chemical sensing applications to identify molecular components in a sample. Since many molecules exhibit a unique Raman scattering spectrum, the spectrum of Raman-scattered light produced when light interacts with a sample can serve as a fingerprint to sense or identify various molecular species within the sample. A sample illuminated with light may produce Raman scattered light at different wavelengths from the illumination light, and measurement of the spectrum of the Raman scattered light is typically performed in the optical domain. For example, the spectrum of Raman scattered light can be measured using an optical spectrometer which separates the Raman scattered light into its optical frequency components using a diffractive element, such as a diffraction grating.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIGS. 1-2 each illustrates an example Raman spectroscopy system.

[0006] FIGS. 3-4 each illustrates an example Raman signal produced by coherent Raman scattering.

[0007] FIG. 5 illustrates the example Raman signal of FIG. 4 along with a probe beam.

[0008] FIG. 6 illustrates an expanded view of a portion of the Raman signal of FIG. 5.

[0009] FIG. 7 illustrates an expanded view of a portion of the Raman signal of FIG. 6.

[0010] FIGS. 8-10 illustrate time-domain and frequency-domain plots of an example electronic signal resulting from coherent mixing of the Raman signal and probe beam of FIG. 7.

[0011] FIG. 11 illustrates an example Raman signal that is measured at multiple probe frequencies.

[0012] FIG. 12 illustrates an example Raman spectrum corresponding to the Raman signal of FIG. 11.

[0013] FIG. 13 illustrates another example Raman signal that is measured at multiple probe

frequencies.

[0014] FIGS. **14-15** each illustrates a second example Raman signal obtained by changing the frequency offset between a pump beam and a Stokes beam.

[0015] FIG. **16** illustrates an example Raman signal along with two probe beams of light.

[0016] FIG. **17** illustrates an example optical receiver for measuring the Raman signal and pump beam of light from FIG. **16**.

[0017] FIG. **18** illustrates an example Raman spectroscopy system for measuring a Raman signal produced by spontaneous Raman scattering.

[0018] FIG. **19** illustrates an example Raman signal produced by the Raman spectroscopy system of FIG. **18**.

[0019] FIG. **20** illustrates an example laser diode that produces a free-space beam of light.

[0020] FIG. **21** illustrates an example laser diode that produces seed light that is amplified by a semiconductor optical amplifier (SOA).

[0021] FIG. **22** illustrates an example laser diode that produces seed light that is amplified by a fiber-optic amplifier.

[0022] FIG. **23** illustrates an example sampled-grating distributed Bragg reflector (SG-DBR) laser.

[0023] FIG. **24** illustrates an example light source with multiple laser diodes and an optical multiplexer that combines light produced by the laser diodes into a single output beam of light.

[0024] FIG. **25** illustrates an example pump laser and Stokes laser with a fiber-optic combiner that produces a combined pump-Stokes beam coupled into an optical fiber.

[0025] FIG. **26** illustrates an example laser diode coupled to a waveguide of a photonic integrated circuit (PIC).

[0026] FIG. **27** illustrates an example pump laser and Stokes laser with a photonic integrated circuit (PIC) that produces a combined pump-Stokes beam coupled into an optical waveguide of the PIC.

[0027] FIG. **28** illustrates an example fiber-optic combiner that combines a Raman signal with a probe beam.

[0028] FIG. **29** illustrates an example photonic integrated circuit (PIC) with a waveguide combiner that combines a Raman signal with a probe beam.

[0029] FIGS. **30-35** each illustrates example frequency ranges of a pump beam and a Stokes beam.

[0030] FIG. **36** illustrates an example optical receiver with two detectors.

[0031] FIG. **37** illustrates an example optical receiver configured for polarization-sensitive detection of a Raman signal.

[0032] FIG. **38** illustrates an example optical receiver configured to detect in-phase and quadrature components of a Raman signal.

[0033] FIG. **39** illustrates an example optical receiver configured to detect polarization as well as in-phase and quadrature components of a Raman signal.

[0034] FIG. **40** illustrates an example Raman spectroscopy system with standoff detection.

[0035] FIG. **41** illustrates an example Raman spectroscopy system with a balanced-detection optical receiver.

[0036] FIG. **42** illustrates an example light detection and ranging (lidar) system.

[0037] FIG. **43** illustrates example pulses of light produced by the lidar system of FIG. **42**.

[0038] FIGS. **44-45** each illustrates an example Raman spectroscopy system and lidar system.

[0039] FIGS. **46-49** each illustrates an example combined Raman spectroscopy and lidar system.

[0040] FIG. **50** illustrates an example combined Raman spectroscopy and rangefinder system configured to perform a distance measurement with a Stokes pulse of light.

[0041] FIG. **51** illustrates an example combined Raman spectroscopy and rangefinder system configured to perform a distance measurement with a pump pulse of light.

[0042] FIGS. **52-54** each illustrates example time-domain plots of pump and Stokes beams of light.

[0043] FIGS. **55-56** each illustrates example time-domain plots of pump and Stokes beams of light

along with a Raman signal and a probe beam of light.

[0044] FIG. 57 illustrates an example Raman spectroscopy system that includes a visible light source.

[0045] FIG. 58 illustrates an example method for measuring a Raman signal.

[0046] FIG. 59 illustrates another example method for measuring a Raman signal.

[0047] FIG. 60 illustrates an example computer system.

DETAILED DESCRIPTION

[0048] FIGS. 1-2 each illustrates an example Raman spectroscopy system **100**. The Raman spectroscopy system **100** in each of FIGS. 1-2 may detect a Raman signal **160** by coherently mixing the Raman signal **160** with a probe beam of light **120_{pr}** produced by a probe light source **110_{pr}**. The Raman spectroscopy system **100** in each of FIGS. 1-2 may be referred to as a coherent Raman spectroscopy system, a coherent Raman spectroscopy system with heterodyne detection, or a high-resolution coherent Raman spectroscopy system. One or more of the systems or methods described herein may be applied to any suitable form of coherent Raman spectroscopy or coherent Raman scattering (CRS), such as for example, coherent anti-Stokes Raman scattering (CARS), stimulated Raman scattering (SRS), or Raman-induced Kerr effect (RIKE).

[0049] The Raman spectroscopy system **100** in FIG. 1 includes a pump light source **110_{pu}** that produces a pump beam of light **120_{pu}** and a Stokes light source **110_S** that produces a Stokes beam of light **120_S**. The pump light source **110_{pu}** produces the pump beam of light **120_{pu}** at a pump frequency, which may be referred to as a first frequency and may be represented by $\nu_{\text{sub.1}}$, $\nu_{\text{sub.pu}}$, $\omega_{\text{sub.1}}$, or $\omega_{\text{sub.pu}}$. The pump light source **110_{pu}** may be referred to as a first light source, and the pump beam of light **120_{pu}** may be referred to as a first beam of light. The Stokes light source **110_S** produces the Stokes beam of light **120_S** at a Stokes frequency, which may be referred to as a second frequency and may be represented by $\nu_{\text{sub.2}}$, $\nu_{\text{sub.S}}$, $\omega_{\text{sub.2}}$, or $\omega_{\text{sub.S}}$. The Stokes light source **110_S** may be referred to as a second light source, and the Stokes beam of light **120_S** may be referred to as a second beam of light. The pump and Stokes frequencies may be offset by a frequency offset Ω , where Ω equals $\nu_{\text{sub.pu}} - \nu_{\text{sub.S}}$ (or equivalently, $\Omega = \nu_{\text{sub.1}} - \nu_{\text{sub.2}}$). Generally, the pump frequency $\nu_{\text{sub.pu}}$ is greater than the Stokes frequency $\nu_{\text{sub.S}}$, and the frequency offset Ω is a positive value. The pump and Stokes light sources in FIG. 1 may each include a laser. For example, the Raman spectroscopy system **100** in FIG. 2 includes a pump laser **110_{pu}** that produces a pump beam of light **120_{pu}** and a Stokes laser **110_S** that produces a Stokes beam of light **120_S**.

[0050] In FIGS. 1-2, the pump beam **120_{pu}** and the Stokes beam **120_S** are directed to a sample **150**, and the sample **150** produces a Raman signal **160** in response to the pump and Stokes beams. For example, the Raman signal **160** may be produced by coherent Raman scattering of the pump and Stokes beams within the sample **150**. A Raman spectroscopy system may include one or more optical elements that direct the pump beam **120_{pu}** and the Stokes beam **120_S** to a sample **150**. Additionally, the optical elements may collect the Raman signal **160** produced by the sample **150** in response to the pump beam **120_{pu}** and Stokes beam **120_S** and may direct the Raman signal **160** to an optical receiver **200**. The optical elements may include free-space optics, fiber-optic components, waveguide-based optics, metamaterials, or any combination thereof. For example, the optical elements may include a mirror, lens, optical combiner (e.g., beamsplitter), optical fiber, photonic integrated circuit (PIC), optical waveguide, or metamaterial-based optic. As another example, the optical combiner **130_a** in each of FIGS. 1-2 may be a free-space dichroic beamsplitter that transmits light at the pump-beam wavelength and reflects light at the Stokes-beam wavelength. The combiner **130_a** may combine the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140** that is directed to the sample **150**. The pump and Stokes beams may be combined so that they are substantially overlapped with one another and propagate in the same direction along approximately the same optical axis. Alternatively, a Raman spectroscopy system **100** may not include a pump-Stokes optical combiner **130_a**, and the pump beam **120_{pu}** and

the Stokes beam **120S** may be directed to a sample **150** as separate beams (e.g., the pump and Stokes beams may be overlapped or combined at the sample rather than being combined earlier). In this embodiment, the pump and Stokes beams may enter the sample **150** from opposite sides (e.g., the pump and Stokes beams may propagate to the sample in opposite directions along approximately the same optical axis). As another example, the optical elements may include a lens (not illustrated in FIGS. **1-2**) that focuses the pump-Stokes beam **140** onto the sample **150**.

Additionally, the optical elements may include a lens (not illustrated in FIGS. **1-2**) that collects the Raman signal **160** to produce a Raman-signal beam that is directed to the optical receiver **200**.

[0051] In FIGS. **1-2** the combined pump-Stokes beam **140** is directed to one side of the sample **150**, and the Raman signal **160** is emitted from the opposite side of the sample. A Raman signal **160** that is collected and directed to an optical receiver **200** may be emitted from a sample **150** in any suitable direction. For example, a Raman signal **160** that is sent to an optical receiver **200** may be emitted from a sample **150** in a forward-scattered direction (e.g., as illustrated in FIGS. **1-2**), in a backward-scattered direction (e.g., back towards the pump or Stokes beam), or in a sideways-scattered direction (e.g., in a direction approximately orthogonal to the combined pump-Stokes beam **140** in FIGS. **1-2**).

[0052] The Raman spectroscopy system **100** in each of FIGS. **1-2** includes an optical receiver **200** that detects the Raman signal **160** produced by the sample **150**. The optical receiver **200** (which may be referred to as a heterodyne optical receiver or a high-resolution optical receiver) may detect the Raman signal **160** using an optical heterodyne technique in which the Raman signal **160** is coherently mixed with a probe beam of light **120pr**. The optical receiver **200** in FIG. **1** includes a probe light source **110pr** that produces a probe beam of light **120pr** at a probe frequency. The probe light source **110pr** may be referred to as a third light source, and the probe beam of light **120pr** may be referred to as a third beam of light. The probe frequency may be referred to as a third frequency and may be represented by $\nu_{\text{sub.3}}$, $\nu_{\text{sub.pr}}$, $\omega_{\text{sub.3}}$, or $\omega_{\text{sub.pr}}$. The probe light source **110pr** in FIG. **1** may include a laser. For example, the optical receiver **200** in FIG. **2** includes a probe laser **110pr** that produces a probe beam of light **120pr**.

[0053] The optical combiner **130b** in each of FIGS. **1-2** may be a dichroic or a non-dichroic beamsplitter that combines the Raman signal **160** and the probe beam **120pr** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. An optical receiver **200** may include one or more optical detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam **120pr** to produce an electronic signal. Each of the optical receivers **200** in FIGS. **1-2** includes one detector **220** that receives the combined probe-Raman signal **210**. The Raman signal **160** and the probe beam **120pr** are coherently mixed at the detector **220**, and this heterodyne mixing process produces an electronic signal, which in FIGS. **1-2** is indicated as analog photocurrent signal **i**. In FIGS. **1-2**, the detection electronics **230** receives the photocurrent signal **i** and produces a digital output signal **240** that corresponds to the photocurrent signal. The digital output signal **240** may be sent to a processor, and the processor may determine a characteristic of the analog photocurrent signal **i** based on the digital output signal **240**. For example, the characteristic of the analog photocurrent signal (or the characteristic of an analog voltage signal that corresponds to the photocurrent signal) determined by the processor may include one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, an area, a frequency, a phase, and a polarization. An analog photocurrent signal may be referred to as a photocurrent, a photocurrent signal, or a current signal, and an analog voltage signal may be referred to as a voltage signal. A processor of a Raman spectroscopy system **100** may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. A processor may be similar to the computer system **6000** illustrated in FIG. **60** and described herein.

[0054] In FIGS. **1-2**, the probe beam **120pr** does not travel through the sample **150**, and the probe

beam is combined with the Raman signal **160** after the Raman signal has exited the sample **150**. In other embodiments, a Raman spectroscopy system **100** may include a probe laser **110_{pr}** that produces a probe beam **120_{pr}** that is directed through the sample **150**. For example, the probe beam **120_{pr}** may be combined with the pump beam **120_{pu}** and the Stokes beam **120_S**, and all three beams may be directed to the sample **150**. The probe beam **120_{pr}** may travel through the sample **150** and may exit the sample along with the Raman signal **160**. The optical receiver **200** may not include a combiner **130_b**, since the probe beam **120_{pr}** and the Raman signal **160** may already be combined after they exit the sample **150**. The probe beam **120_{pr}** and the Raman signal **160** may be directed to a detector **220** without being transmitted or reflected by an optical combiner **130_b**.

[0055] The detection electronics **230** in FIG. 2 includes an electronic amplifier **232** and a digitizer **236**. The electronic amplifier **232** may include a transimpedance amplifier that amplifies the photocurrent signal *i* to produce an analog voltage signal **234** that corresponds to the photocurrent signal *i* (e.g., the photocurrent signal *i* and the analog voltage signal **234** may have similar temporal shapes or may include similar electronic frequency components). The electronic amplifier **232** may include an additional gain stage that further amplifies an intermediate voltage signal produced by the transimpedance amplifier to produce the voltage signal **234**. Additionally, the electronic amplifier **232** may include an electronic filter (e.g., a low-pass, high-pass, or band-pass filter) that filters the photocurrent signal or voltage signal. For example, the electronic amplifier **232** may include (i) a high-pass filter that removes a DC offset and low-frequency components (e.g., frequency components below 10 MHz) from the photocurrent signal or (ii) a band-pass filter that removes the DC and low-frequency components as well as high-frequency components (e.g., frequency components above 5 GHz). Herein, an electronic signal produced in response to coherent mixing may refer to a current signal (e.g., a photocurrent *i* produced by a detector **220**) or may refer to a corresponding voltage signal (e.g., a voltage signal **234** produced by an electronic amplifier that amplifies a photocurrent produced by a detector to produce the voltage signal). In FIG. 2, the digitizer **236** receives the voltage signal **234** and produces a digital output signal **240** that includes a digital representation of the voltage signal **234**. The digital output signal **240** may be a time-domain digital representation of the voltage signal **234**. The digital output signal **240** may be referred to as corresponding to or representing the voltage signal **234** or the photocurrent signal *i*. The digitizer **236** may include an analog-to-digital converter (ADC) that produces a digital version of the voltage signal **234**. Additionally or alternatively, the digitizer **236** may include a peak detector that determines a peak value of the voltage signal **234**.

[0056] A Raman spectroscopy system **100** may include one or more optical detectors **220**. An optical detector **220** (which may be referred to as a detector, photodetector, or photodiode) may include a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT). A PN photodiode refers to a photodiode structure formed by a p-doped semiconductor and an n-doped semiconductor, where the PN acronym refers to the structure having p-doped and n-doped regions. A PIN photodiode refers to a photodiode structure formed by an undoped intrinsic semiconductor region located between p-doped and n-doped regions, where the PIN acronym refers to the structure having p-doped, intrinsic, and n-doped regions.

[0057] A PN photodiode, PIN photodiode, APD, or SPAD may include any suitable semiconductor material, such as for example: silicon, germanium, gallium arsenide (GaAs), indium phosphide (InP), indium arsenide (InAs), aluminum arsenide (AlAs), indium antimonide (InSb), aluminum antimonide (AlSb), gallium antimonide (GaSb), aluminum gallium arsenide (AlGaAs), indium gallium arsenide (InGaAs), indium aluminum arsenide (InAlAs), indium arsenide antimonide (InAsSb), aluminum arsenide antimonide (AlAsSb), aluminum gallium antimonide (AlGaSb), gallium arsenide antimonide (GaAsSb), aluminum indium arsenide antimonide (AlInAsSb), indium gallium arsenide antimonide (InGaAsSb), indium gallium aluminum arsenide (InGaAlAs), aluminum gallium arsenide antimonide (AlGaAsSb), or silicon germanium (SiGe). For example,

the Raman signal **160** and the probe beam **120pr** in FIGS. **1-2** may each have a wavelength in the 400-1100 nanometer (nm) range, and the detector **220** may include a silicon PIN photodiode. As another example, the Raman and probe beams in FIGS. **1-2** may each have a wavelength in the 1000-1700 nm range, and the detector **220** may include an InGaAs PIN photodiode. As another example, a detector **220** may include an APD that includes a semiconductor material with antimonide (e.g., InSb, AlSb, GaSb, InAsSb, AlAsSb, AlGaSb, GaAsSb, AlInAsSb, InGaAsSb, or AlGaAsSb).

[0058] In order for a detector **220** to detect an optical signal, the wavelength of the optical signal must be within the detector's wavelength range of responsivity (e.g., approximately 400-1100 nm for a silicon detector, and approximately 1000-1700 nm for an InGaAs detector) and a frequency of an amplitude modulation of the optical signal must be within the electronic bandwidth of the detector. The electronic bandwidth (Δf) of a detector **220** refers to the range of electronic modulation frequencies over which a detector may detect an optical signal, where detection of the optical signal refers to (i) the detector producing a photocurrent signal i that corresponds to the optical signal and (ii) an electronic amplifier **232** producing a voltage signal **234** that corresponds to the photocurrent signal. If a silicon detector **220** has an electronic bandwidth that extends from 100 MHz to 5 GHz, then the detector may detect optical signals with (i) wavelengths between approximately 400 nm and 1100 nm and (ii) amplitude modulation between 100 MHz and 5 GHz. For example, a 900-nm optical signal with an amplitude modulation at a frequency between 100 MHz and 5 GHz may be detected by the silicon detector **220**. The silicon detector **220** may not detect a continuous-wave or substantially constant portion of a 900-nm optical signal (e.g., the substantially constant portion of the optical signal may produce a DC current in the detector that may be electronically filtered out), and the silicon detector **220** may not detect a portion of the optical signal with an amplitude modulation greater than approximately 5 GHz. Herein, reference to the electronic bandwidth (Δf) of a detector **220** may refer to (i) the electronic bandwidth of just the detector or (ii) the overall bandwidth of the detector in combination with an electronic amplifier **232**.

[0059] A detector **220** may have an electronic bandwidth Δf between approximately 100 MHz and approximately 50 GHz. For example, the detector **220** in each of FIGS. **1-2** may have an electronic bandwidth between 100 MHz and 10 GHz. As another example, the detector **220** in each of FIGS. **1-2** may have an electronic bandwidth that extends from a low-frequency cutoff to a high-frequency cutoff. The low-frequency cutoff may be approximately DC (i.e., zero hertz), 1 MHz, 10 MHz, 50 MHz, or 100 MHz, and the high-frequency cutoff may be approximately 500 MHz, 1 GHz, 2 GHz, 5 GHz, 10 GHz, 20 GHz, or 50 GHz. The electronic bandwidth of a detector **220** may refer to the bandwidth of only the detector **220**. For example, a detector **220** may have an electronic bandwidth that extends from DC to 10 GHz, and the detector may be referred to as having a 10-GHz bandwidth that extends from DC to 10 GHz. Alternatively, the electronic bandwidth of a detector **220** may refer to the overall bandwidth of the detector in combination with an electronic amplifier **232** that amplifies the photocurrent signal i produced by the detector. An electronic amplifier **232** may have a low-frequency cutoff (e.g., DC, 1 MHz, 10 MHz, 50 MHz, or 100 MHz) and a high-frequency cutoff (e.g., 500 MHz, 1 GHz, 2 GHz, 5 GHz, 10 GHz, 20 GHz, or 50 GHz), and a detector-amplifier combination may be referred to as having an electronic bandwidth that extends from the low-frequency cutoff to the high-frequency cutoff. For example, if the detector bandwidth extends from DC to 10 GHz, and the electronic amplifier bandwidth extends from DC to 5 GHz, then the detector (or, the detector-amplifier combination) may be referred to as having an electronic bandwidth of 5 GHz that extends from DC to 5 GHz. As another example, if the detector bandwidth extends from DC to 10 GHz, and the electronic amplifier bandwidth extends from 100 MHz to 5 GHz, then the detector may be referred to as having an electronic bandwidth of 4.9 GHz that extends from 100 MHz to 5 GHz.

[0060] A Raman spectroscopy system **100** may include one or more optical waveplates that change

or rotate the polarization of a beam of light. For example, a half-wave plate may be used to rotate a linearly polarized beam of light to a different polarization orientation (e.g., from vertically polarized to horizontally polarized), and a quarter-wave plate may be used to convert a linearly polarized beam of light to a circular or elliptical polarization. The pump laser **110pu** in FIG. 2 may produce linearly polarized light, and the waveplate **132a** may be a half-wave plate that rotates the polarization of the pump beam **120pu** prior to the pump beam being directed to the sample **150**. Alternatively, the waveplate **132a** may be a quarter-wave plate that converts the linearly polarized pump beam **120pu** to a circular or elliptical polarization prior to the pump beam being directed to the sample **150**. Similarly, the Stokes laser **110S** in FIG. 2 may produce linearly polarized light, and the waveplate **132b** may be (i) a half-wave plate that rotates the polarization of the Stokes beam **120S** or (ii) a quarter-wave plate that converts the Stokes beam **120S** to a circular or elliptical polarization.

[0061] The probe laser **110pr** in FIG. 2 may produce linearly polarized light, and the waveplate **132c** may be (i) a half-wave plate that rotates the polarization of the probe beam **120pr** or (ii) a quarter-wave plate that converts the probe beam **120pr** to a circular or elliptical polarization prior to the probe beam being combined with the Raman signal **160**. Changing the polarization of the probe beam **120pr** may allow the Raman signal **160** and the probe beam to be coherently mixed. The polarization of the probe beam **120pr** can be changed so that it has both horizontal and vertical polarization components, which ensures that at least a portion of the probe beam **120pr** and the Raman signal **160** have polarizations that are oriented in the same direction so that their electric fields may be added together.

[0062] Each of the optical waveplates **132** in FIG. 2 may be a free-space optical element, a fiber-optic component, a waveguide-based optical element, or a metamaterial-based optic. Additionally, each of the optical waveplates **132** in FIG. 2 may be a fixed waveplate or an adjustable waveplate. A fixed waveplate may have a fixed optical phase difference between the two axes of the waveplate (e.g., a quarter-wave plate may have a one-quarter wavelength phase difference, and a half-wave plate may have a one-half wavelength phase difference). An adjustable waveplate may allow for the phase difference between the two axes of the waveplate to be dynamically changed. For example, an electronically adjustable waveplate may include a Pockels cell, a liquid crystal device, or a photoelastic modulator that allows the phase difference to be adjusted electronically so that the waveplate can be dynamically configured to act as a waveplate having any suitable phase difference (e.g., a phase difference between zero wavelengths and one-half wavelength). An adjustable waveplate may switch between (i) applying no phase difference to incident light so that the transmitted light has the same polarization as the incident light and (ii) applying a one-quarter wavelength or one-half wavelength phase difference so that linearly polarized incident light is converted to circularly polarized light or is rotated to a different polarization. For example, the pump laser **110pu** in FIG. 2 may produce vertically polarized light, and the waveplate **132a** may be an adjustable waveplate that switches between (i) applying no polarization rotation to the pump beam **120pu** so that the pump beam remains vertically polarized and (ii) applying a 90-degree rotation to the pump-beam polarization so that the pump beam **120pu** after the waveplate **132a** is horizontally polarized. Dynamically changing the polarization of the pump beam **120pu** may allow the Raman spectroscopy system **100** in FIG. 2 to perform measurements at two different pump-beam polarizations, which may produce additional data for characterization of the sample **150**.

[0063] In some embodiments, an optical waveplate **132** may be a metamaterial-based waveplate. A metamaterial refers to an engineered material having features or repeating patterns at scales smaller than the wavelength of light interacting with the metamaterial. A metamaterial may be configured to act as a mirror, lens, waveplate, diffractive optical element, optical combiner, or optical waveguide. A metamaterial-based waveplate may affect the polarization of a beam of light based on wavelength. For example, the Raman spectroscopy system **100** in FIG. 2 may include a metamaterial-based waveplate (not illustrated in FIG. 2) located after the combiner **130a** and before

the sample **150**. The metamaterial waveplate may change the polarization of the pump beam **120_{pu}** from linear to circular while preserving the polarization of the Stokes beam **120_S** (e.g., the Stokes beam may remain linearly polarized and may not be significantly changed by the waveplate). [0064] A Raman spectroscopy system **100** may include an optical filter that transmits light at one or more wavelengths and blocks light at one or more other wavelengths. The optical filter **134** in FIG. 2 is located between the sample **150** and the optical receiver **200** and may be configured to substantially transmit one or more optical wavelengths associated with the Raman signal **160** and substantially block one or more wavelengths associated with the pump beam **120_{pu}** or the Stokes beam **120_S**. For example, the optical filter **134** may transmit greater than 90% of the Raman signal **160** and may block greater than 90% of both the pump and Stokes beams. As another example, the optical filter **134** may transmit greater than 90% of the Raman signal **160** and the Stokes beam **120_S**, and the optical filter **134** may block greater than 98% of the pump beam **120_{pu}**.

[0065] A Raman spectroscopy system **100** may include an optical polarizer that transmits light having a particular polarization (e.g., horizontal) and blocks light having an orthogonal polarization (e.g., vertical). The optical polarizer **136** in FIG. 2 is located between the sample **150** and the optical receiver **200** and may be oriented to transmit light with a polarization associated with the Raman signal. Additionally, the polarizer **136** may block a polarization associated with the pump or Stokes beams. For example, the Stokes beam **120_S** incident on the sample **150** may be vertically polarized, and the Raman signal **160** may be at least partially horizontally polarized. The polarizer **136** may be oriented to block vertically polarized light and transmit horizontally polarized light so that the Stokes beam **120_S** is blocked and the Raman signal **160** is at least partially transmitted by the polarizer. A Raman spectroscopy system **100** may include both an optical filter **134** and an optical polarizer **136**. For example, the optical filter **136** in FIG. 2 may be configured to transmit the Raman signal **160** and the Stokes beam **120_S** and block the pump beam **120_{pu}**. Additionally, the Raman signal **160** and the Stokes beam **120_S** may be orthogonally polarized, and the polarizer **136** may be oriented to transmit the Raman signal **160** and block the Stokes beam **120_S**. Using a filter or polarizer located after the sample **150** to block light from the pump beam **120_{pu}** or the Stokes beam **120_S** may reduce noise in the system by reducing the amount of unwanted background light that reaches the detector **220**.

[0066] The sample **150** in FIGS. 1-2 may be a solid, liquid, or gas, or any combination thereof. The sample **150** may include a biological material, an organic material, an inorganic material, a crystalline material, an amorphous solid material, or any other suitable material or combination of suitable materials. For example, the sample **150** may include a drug, mineral, food, contaminant, or explosive material that produces a Raman signal **160** in response to excitation by the pump and Stokes beams. As another example, the sample **150** may be a biological material (e.g., blood, urine, saliva, sweat, or cerebrospinal fluid) and a component or molecule (e.g., glucose or cortisol) that is part of the biological material may produce a Raman signal **160**. As another example, the sample **150** may be water or wastewater that may include a contaminant, virus, bacteria, or an indicator of an infectious disease. The Raman signal **160** may be produced by coherent Raman scattering of the pump and Stokes beams within the sample **150**, and the frequency offset Ω between the pump and Stokes beams may be approximately equal to a vibrational frequency or an electronic-transition frequency of a particular material that is part of the sample. The vibrational frequency of the particular material may correspond to a molecular vibration of a molecule, or in the case of a crystalline material, may correspond to a lattice vibration of a crystal. For example, the sample **150** may include glucose, and the frequency offset Ω may be approximately equal to a frequency of a molecular vibration of glucose.

[0067] The Raman spectroscopy systems **100** in FIGS. 1-2 may perform one or more measurements of a sample **150**, and each measurement may include determining a characteristic of an electronic signal that results from the coherent mixing of the Raman signal **160** and the probe beam **120_{pr}**. The electronic signal may include a photocurrent i or a corresponding voltage signal

234. The frequency offset Ω may be approximately equal to a vibrational frequency or electronic-transition frequency of a particular material, and based on the one or more measurements, a processor may determine whether the particular material is present in the sample **150**. Additionally or alternatively, the processor may determine an amount or a concentration of the particular material in the sample based on the measurements. For example, the frequency offset Ω may be approximately equal to a vibrational frequency of glucose, and, based on one or more optical heterodyne measurements of a Raman signal **160** produced by a sample **150**, the processor may determine (i) whether glucose is present in the sample or (ii) an amount or a concentration of glucose in the sample. The amount of glucose that is present in the sample **150** may be proportional to the amplitude of a photocurrent signal i produced by coherent mixing of the Raman signal **160** and the probe beam **120_{pr}**. Based on the amplitude of the photocurrent signal i , the processor may determine the amount or concentration of glucose in the sample **150**.

[0068] A technical advantage of a coherent Raman spectroscopy system **100** as described herein is a higher spectral resolution or a better chemical sensitivity than a conventional Raman spectroscopy system. As such, a Raman spectroscopy system **100** as described herein may be referred to as a high-resolution Raman spectroscopy system or as a high-resolution coherent Raman spectroscopy system. In a conventional Raman spectroscopy system, a Raman signal produced by a sample may be measured in the optical domain using an optical spectrometer. A spectrometer typically uses a dispersive optical element (e.g., a diffraction grating) to separate the Raman signal into its various spectral components. However, this type of measurement performed in the optical domain typically has a spectral resolution on the order of 1 cm.sup.-1 (or, about 30 GHz). In contrast, the spectral resolution of a coherent Raman spectroscopy system with heterodyne detection, as described herein, is determined primarily by the spectral linewidth $\Delta\nu_{\text{sub.pr}}$ of the probe beam **120_{pr}** that is coherently mixed with the Raman signal **160**. The probe laser **110_{pr}** may include a wavelength-tunable laser diode with a linewidth of 200 MHz or less, which corresponds to a spectral resolution of the Raman spectroscopy system **100** of less than 200 MHz (or, less than 0.007 cm.sup.-1). This 200-MHz spectral resolution is more than 100 times better than the 30-GHz spectral resolution of a conventional Raman spectroscopy system. In some embodiments, the probe laser **110_{pr}** may have a linewidth of 1 MHz or less, which corresponds to a spectral resolution of the Raman spectroscopy system **100** of less than 1 MHz (or, less than $33\times 10^{-6}\text{ cm.sup.-1}$). A related advantage of a coherent Raman spectroscopy system **100** is that the signal capture and analysis are performed in the electronic domain (e.g., at electronic frequencies between DC and 50 GHz) rather than in the optical domain (e.g., at optical frequencies between 60 THz and 1,000 THz). The coherent mixing of two optical signals (Raman signal **160** and probe beam **120_{pr}**) produces an electronic signal which can be analyzed with relatively high resolution compared to an optical signal. This electronic signal analysis, along with the relatively narrow spectral linewidth of the probe laser **110_{pr}**, provides a coherent Raman spectroscopy system **100** with a high spectral resolution. Additionally, the wavelength tunability of the probe laser **110_{pr}** allows a Raman spectrum of a material to be determined at multiple frequencies with high spectral resolution.

[0069] The higher spectral resolution of a coherent Raman spectroscopy system **100** may provide a corresponding improvement in the ability of the coherent Raman spectroscopy system to sense various chemical species. For example, a high-resolution coherent Raman spectroscopy system **100** may be able to distinguish between different chemical species that have Raman peaks located relatively close together, whereas a conventional Raman spectroscopy system may not be able to resolve spectral features below about 1 cm.sup.-1 . Additionally, the higher spectral resolution of a coherent Raman spectroscopy system **100** may allow for lower concentrations of materials to be detected, as compared to a conventional Raman spectroscopy system. For example, a coherent Raman spectroscopy system **100** may be able to detect small deviations in the chemical signature of a biological sample, which may indicate the presence of damage or a mutation, which in turn may be correlated with a disease or pathogen.

[0070] Another technical advantage of a coherent Raman spectroscopy system **100** as described herein is its relatively compact size. A coherent Raman spectroscopy system may be packaged in a relatively compact enclosure as compared to a conventional Raman spectroscopy system. Since the spectral resolution of an optical spectrometer scales inversely with the optical path length of the spectrometer (e.g., a longer path length provides better spectral resolution), an optical spectrometer with a spectral resolution around 1 cm^{-1} can be quite large or bulky. In contrast, since the spectral resolution of a coherent Raman spectroscopy system is determined primarily by the spectral linewidth of the probe laser **110pr**, a coherent Raman spectroscopy system does not require a long optical path length to provide high spectral resolution. Thus, an enclosure for a coherent Raman spectroscopy system may be significantly smaller than that for a conventional Raman spectroscopy system. In some embodiments, a coherent Raman spectroscopy system may be packaged as a compact device that may be referred to as a lab-on-a-chip or a spectrometer on a chip. For example, a coherent Raman spectroscopy system may be packaged as a wearable device that provides ongoing, continual monitoring for a person or an animal.

[0071] FIGS. **3-4** each illustrates an example Raman signal **160** produced by coherent Raman scattering. In a coherent Raman spectroscopy system **100**, the pump beam **120pu** and the Stokes beam **120S** are directed to a sample **150**, which produces a Raman signal **160** by coherent Raman scattering of the pump and Stokes beams within the sample. The frequency of the pump beam **120pu** is $\nu_{\text{sub.1}}$, and the frequency of the Stokes beam **120S** is $\nu_{\text{sub.2}}$. The frequency offset Ω between the pump and Stokes beams equals $\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$. The frequencies of the pump and Stokes beams may be set so that the frequency offset Ω is approximately equal to a vibrational frequency or an electronic-transition frequency of a particular material. A coherent Raman spectroscopy system **100** may measure a Raman signal **160** produced by a sample **150** to determine (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material within the sample.

[0072] The frequency offset Ω between the pump and Stokes beams may be any suitable fixed or adjustable value between approximately 5 terahertz (THz) and approximately 100 THz. Expressed in wavenumbers, this corresponds to the frequency offset Ω being between approximately 167 cm^{-1} and approximately 3336 cm^{-1} . For example, in a coherent Raman spectroscopy system **100**, the pump beam **120pu** may have a wavelength between approximately 1220 nanometers (nm) and approximately 1450 nm (which corresponds to a pump-beam frequency $\nu_{\text{sub.1}}$ between approximately 246 THz and approximately 207 THz), and the Stokes beam **120S** may have a wavelength between approximately 1490 nm and approximately 1570 nm (which corresponds to a Stokes-beam frequency $\nu_{\text{sub.2}}$ between approximately 201 THz and approximately 191 THz). This system may produce pump and Stokes beams having a frequency offset Ω between approximately 5.6 THz and approximately 54.8 THz (or, in wavenumbers, between approximately 185 cm^{-1} and 1827 cm^{-1}). For example, if the pump beam **120pu** has a wavelength of 1330 nm (or equivalently, a frequency $\nu_{\text{sub.1}}$ of 225.4 THz) and the Stokes beam **120S** has a wavelength of 1550 nm (or equivalently, a frequency $\nu_{\text{sub.2}}$ of 193.4 THz), then the frequency offset Ω between the pump and Stokes beams is approximately 32 THz (or, in wavenumbers, 1067 cm^{-1}). The pump and Stokes beams in a Raman spectroscopy system **100** may each have any suitable wavelength between approximately 300 nm and approximately 5,000 nm. For example, if the pump beam **120pu** has a wavelength of approximately 785 nm (or equivalently, a frequency $\nu_{\text{sub.1}}$ of 381.9 THz) and the Stokes beam **120S** has a wavelength of approximately 840 nm (or equivalently, a frequency $\nu_{\text{sub.2}}$ of 356.9 THz), then the frequency offset Ω between the pump and Stokes beams is approximately 25 THz (or, in wavenumbers, 834 cm^{-1}). As another example, the pump beam **120pu** may have a wavelength between approximately 700 nm and approximately 850 nm, between approximately 890 nm and approximately 920 nm, or between approximately 1000 nm and approximately 1100 nm.

[0073] The Raman signal **160** in each of FIGS. **3-4** is an optical signal with a spectral linewidth of

$\Delta v_{\text{sub.1}} - v_{\text{sub.2}}$. In FIG. 3, the Raman signal **160** has a center frequency approximately equal to $2v_{\text{sub.1}} - v_{\text{sub.2}}$ (which is equal to $v_{\text{sub.1}} + \Omega$, since $\Omega = v_{\text{sub.1}} - v_{\text{sub.2}}$). The center frequency of a Raman signal **160** may refer to the frequency of a central peak or the frequency of an approximate center of the Raman signal. Additionally or alternatively, the center frequency of a Raman signal **160** may correspond to a Raman peak of an associated Raman spectrum. The Raman signal **160** in FIG. 3 may be produced by coherent anti-Stokes Raman scattering (CARS) in which the pump and Stokes beams interact with a sample **150** to produce a Raman signal **160** at or around the frequency $2v_{\text{sub.1}} - v_{\text{sub.2}}$. For example, if the pump and Stokes beams have respective wavelengths of 1064 nm and 1550 nm (which corresponds to frequencies of approximately 281.8 THz and 193.4 THz), then the frequency offset Ω is 88.3 THz (or, 2947 cm^{-1}), and the anti-Stokes Raman signal **160** has a center wavelength of approximately 810 nm (which corresponds to a frequency of approximately 370 THz). The Raman signal **160** in FIG. 4, which overlaps the frequency of the Stokes beam **120S**, may be produced by stimulated Raman scattering. The Raman signal **160** in FIG. 4 may be centered at or near the frequency $v_{\text{sub.2}}$ of the Stokes beam **120S**. For example, the Raman signal **160** in FIG. 4 may have a center frequency that is within approximately 200 GHz (or, 6.7 cm^{-1}) of the Stokes-beam frequency $v_{\text{sub.2}}$.

[0074] FIG. 5 illustrates the example Raman signal **160** of FIG. 4 along with a probe beam **120pr**. The Raman signal **160** may be produced by stimulated Raman scattering of the pump beam **120pu** and Stokes beam **120S**, and the probe beam **120pr** may be used to measure the Raman signal. The Raman signal **160** is centered at or near the frequency $v_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $v_{\text{sub.3}}$ of the probe beam of light **120pr** overlaps the Raman signal **160** and is relatively close to the frequency $v_{\text{sub.2}}$ of the Stokes beam of light **120S** (e.g., the probe-beam frequency $v_{\text{sub.3}}$ may be within 200 GHz of the Stokes-beam frequency $v_{\text{sub.2}}$). In a coherent Raman spectroscopy system **100**, the pump frequency $v_{\text{sub.1}}$, Stokes frequency $v_{\text{sub.2}}$, and probe frequency $v_{\text{sub.3}}$ may each be between approximately 60 THz and approximately 1,000 THz (which corresponds to a wavelength between approximately 5,000 nm and approximately 300 nm). For example, in FIG. 5 the pump frequency $v_{\text{sub.1}}$ may be 291 THz (which corresponds to a wavelength of approximately 1030 nm), the Stokes frequency $v_{\text{sub.2}}$ may be 240.03 THz (which corresponds to a wavelength of approximately 1249.00 nm), and the probe frequency $v_{\text{sub.3}}$ may be 240.00 THz (which corresponds to a wavelength of approximately 1249.14 nm).

[0075] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include a wavelength-tunable light source. A wavelength-tunable light source refers to a light source **110** that can produce light at multiple different wavelengths within a range of wavelengths (or equivalently, at multiple different frequencies within a range of frequencies). For example, the probe light source **110pr** in FIG. 1 may be a wavelength-tunable light source where the wavelength of the probe beam **120pr** is adjustable over an 80-nm wavelength range from 1490 nm to 1570 nm (which corresponds to a 10.3 THz frequency range from approximately 201.2 THz to approximately 191.0 THz). At any given time, a wavelength-tunable light source **110** may operate at any one of the different wavelengths within its wavelength-tuning range. For example, during a first period of time, the probe light source **110pr** with a 1490-1570 nm wavelength-tuning range may produce a probe beam **120pr** at 1500 nm, and during a subsequent second period of time, the probe light source may be tuned to operate at 1560 nm. A wavelength-tunable light source may be adjustable over a frequency range that corresponds to a wavelength range having a width or span between approximately 10 nm and approximately 100 nm. For example, the width of the wavelength-tuning range of a wavelength-tunable light source may be approximately 10 nm, 20 nm, 30 nm, 40 nm, 50 nm, 60 nm, 80 nm, or 100 nm. A wavelength-tuning range from 1490 nm to 1570 nm may be referred to as having an 80-nm width or an 80-nm span. Around 1550 nm, a 10-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 1.25 THz (or equivalently, 41.6 cm^{-1}), and a 100-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 12.5 THz (or equivalently, 416 cm^{-1}). Around 1050 nm, a 10-nm

wavelength-tuning range corresponds to a frequency-tuning range of approximately 2.72 THz (or equivalently, 90.7 cm.sup.-1), and a 100-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 27.3 THz (or equivalently, 909 cm.sup.-1). A wavelength-tunable light source may be referred to as a frequency-tunable light source or a tunable light source.

[0076] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include one or more laser diodes, and each of the laser diodes may be a fixed-wavelength laser diode or a wavelength-tunable laser diode. A fixed-wavelength laser diode may operate at a single wavelength or within a relatively narrow wavelength range (e.g., within 0.1 nm of a particular wavelength). A fixed-wavelength laser diode may include a distributed feedback (DFB) laser diode, a distributed Bragg reflector (DBR) laser diode, a fiber-Bragg-grating (FBG) stabilized laser diode, a temperature-stabilized laser diode, or any other suitable fixed-wavelength laser diode. A wavelength-tunable laser diode may produce light at multiple different wavelengths within a range of wavelengths. For example, a wavelength-tunable laser diode may be configured to produce light at any wavelength within a wavelength range having a width between approximately 10 nm and approximately 100 nm. At any given time, a wavelength-tunable laser diode may operate at any one wavelength of multiple different wavelengths within a range of wavelengths. A wavelength-tunable laser diode may include an external-cavity laser diode, a thermally tuned laser diode, or a sampled-grating distributed Bragg reflector (SG-DBR) laser. For example, a wavelength-tunable SG-DBR laser may have a 40-nm wavelength-tuning range that extends from 1530 nm to 1570 nm, and the SG-DBR laser may be adjustable to operate at any single wavelength within the 40-nm wavelength range. A wavelength-tunable laser diode may be referred to as a frequency-tunable laser diode or a tunable laser diode. For example, a tunable laser with a 40-nm wavelength-tuning range that extends from 1530 nm to 1570 nm may also be referred to as a frequency-tunable laser with a 5.0-THz frequency-tuning range that extends from approximately 191 THz to approximately 196 THz.

[0077] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include one or more of the following: light-emitting diode (LED), super-luminescent light source, short-pulse laser, broadband light source, fiber laser, solid-state laser, quantum-cascade laser. For example, a light source that produces light over a relatively broad range of wavelengths (e.g., a super-luminescent light source, short-pulse laser, or broadband light source) may be used to investigate a sample over the broad range of wavelengths without having to use a wavelength-tunable light source.

[0078] FIG. **6** illustrates an expanded view of a portion of the Raman signal **160** of FIG. **5**. The portion of FIG. **5** enclosed by the dashed-line box is expanded in FIG. **6**. The peak of the Raman signal **160** in FIG. **6** is approximately coincident with the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** overlaps the Raman signal.

[0079] The Raman signal **160** in FIG. **6** is an optical signal with a spectral linewidth of $\Delta\nu_{\text{sub.R}}$. The spectral linewidth of a Raman signal **160** may have a value between approximately 30 GHz and 300 GHz (or, in wavenumbers, between approximately 1 cm.sup.-1 and approximately 10 cm.sup.-1). The spectral linewidth $\Delta\nu_{\text{sub.R}}$ of a Raman signal **160** may represent the spectral width of a peak of the Raman signal (e.g., a full-width-at-half-maximum of the peak) or may represent an approximate extent or width of the full Raman signal. For example, the spectral linewidth $\Delta\nu_{\text{sub.R}}$ of the Raman signal **160** in FIG. **13** corresponds to a full-width-at-half-maximum of a peak of the Raman signal. In FIG. **6**, the spectral linewidth $\Delta\nu_{\text{sub.R}}$ of the Raman signal **160** corresponds to an extent or width of the full Raman signal. For example, the spectral linewidth of a Raman signal **160** may equal a spectral width at which an envelope of the Raman signal has decreased to a particular level (e.g., to 50%, 20%, or 10% of a peak value). The envelope may be a curve that decreases monotonically away from a peak of the Raman signal and approximately follows an overall shape of the Raman signal. In FIG. **6**, the dashed-line curve that traces the peaks of the Raman signal **160** represents an envelope of the Raman signal, and the spectral linewidth $\Delta\nu_{\text{sub.R}}$ corresponds to a full-width-at-10% of the Raman-signal envelope (i.e.,

the points at which the envelope has decreased to 10% of its maximum value).

[0080] In a coherent Raman spectroscopy system **100**, the difference between the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be greater than a low-frequency limit $F_{\text{sub.1}}$ and less than a high-frequency limit $F_{\text{sub.2}}$ (i.e., $F_{\text{sub.1}} < |\nu_{\text{sub.2}} - \nu_{\text{sub.3}}| < F_{\text{sub.2}}$). For example, the low-frequency limit $F_{\text{sub.1}}$ may be approximately 1 MHz, 10 MHz, 50 MHz, 100 MHz, 200 MHz, 500 MHz, or 1 GHz, and the high-frequency limit $F_{\text{sub.2}}$ may be approximately 10 GHz, 20 GHz, 50 GHz, 100 GHz, 200 GHz, 500 GHz, or 1 THz. As another example, the low-frequency limit $F_{\text{sub.1}}$ may be related to the spectral linewidth $\Delta\nu_{\text{sub.pr}}$ of the probe beam **120pr** and the spectral linewidth $\Delta\nu_{\text{sub.S}}$ of the Stokes beam **120S** (e.g., $F_{\text{sub.1}}$ may be greater than $\Delta\nu_{\text{sub.pr}} + \Delta\nu_{\text{sub.S}}$). As another example, the high-frequency limit $F_{\text{sub.2}}$ may be related to the spectral linewidth $\Delta\nu_{\text{sub.R}}$ of a Raman signal **160** (e.g., $F_{\text{sub.2}}$ may be approximately equal $(0.5) \Delta\nu_{\text{sub.R}}$, $\Delta\nu_{\text{sub.R}}$, or $2\Delta\nu_{\text{sub.R}}$). As another example, $F_{\text{sub.1}}$ may be 100 MHz and $F_{\text{sub.2}}$ may be 200 GHz, which indicates that the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may differ by greater than 100 MHz and less than 200 GHz (i.e., $100 \text{ MHz} < |\nu_{\text{sub.2}} - \nu_{\text{sub.3}}| < 200 \text{ GHz}$). The frequencies $F_{\text{sub.1}}$ and $F_{\text{sub.2}}$ represent the frequency range with respect to the Stokes-beam frequency $\nu_{\text{sub.2}}$ over which the probe-beam frequency $\nu_{\text{sub.3}}$ may be scanned. The two hatched rectangles along the frequency axis in FIG. 6 represent the allowed frequency ranges for the probe beam **120pr**. The frequency of the probe beam **120pr** may be (i) between $\nu_{\text{sub.2}} - F_{\text{sub.2}}$ and $\nu_{\text{sub.2}} - F_{\text{sub.1}}$ or (ii) between $\nu_{\text{sub.2}} + F_{\text{sub.1}}$ and $\nu_{\text{sub.2}} + F_{\text{sub.2}}$. In an optical receiver **200**, the probe beam **120pr** and the Raman signal **160** may be coherently mixed together at an optical detector **220**. The frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be kept away from the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** by at least the low-frequency limit $F_{\text{sub.1}}$ (e.g., $|\nu_{\text{sub.2}} - \nu_{\text{sub.3}}| > F_{\text{sub.1}}$) to avoid mixing between the probe and Stokes beams, which could cause the detector to produce an unwanted electronic signal not related to the mixing between the probe beam and Raman signal. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be kept within the high-frequency limit $F_{\text{sub.2}}$ of the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** (e.g., $|\nu_{\text{sub.2}} - \nu_{\text{sub.3}}| < F_{\text{sub.2}}$), since measurements outside the high-frequency limit may not produce a significant electronic signal.

[0081] FIG. 7 illustrates an expanded view of a portion of the Raman signal **160** of FIG. 6. The portion of FIG. 6 enclosed by the dashed-line box is expanded in FIG. 7. The peak of the Raman signal **160** in FIG. 7 is located at frequency $\nu_{\text{sub.RP}}$, and the Stokes beam **120S** is located at the Stokes frequency $\nu_{\text{sub.2}}$. The Raman-signal peak frequency $\nu_{\text{sub.RP}}$ may correspond to a Raman peak of a material that is part of a sample **150** being measured by a Raman spectroscopy system **100**. For example, the difference between the pump-beam frequency $\nu_{\text{sub.1}}$ and the Raman-signal peak frequency $\nu_{\text{sub.RP}}$ may equal a vibrational frequency Ω of the material (e.g., $\nu_{\text{sub.1}} - \nu_{\text{sub.RP}} = \Omega$). In some embodiments, a Stokes light source **110S** may be operated so that the Stokes-beam frequency $\nu_{\text{sub.2}}$ is approximately equal to the Raman-signal peak frequency $\nu_{\text{sub.RP}}$, and in other embodiments (e.g., as illustrated in FIG. 7), the Stokes-beam frequency $\nu_{\text{sub.2}}$ may be slightly off-resonance or detuned with respect to the Raman-signal peak frequency $\nu_{\text{sub.RP}}$. For example, in FIG. 7, the Stokes-beam frequency $\nu_{\text{sub.2}}$ may differ from the Raman-signal peak frequency $\nu_{\text{sub.RP}}$ by less than or equal to 30 GHz, 10 GHz, 5 GHz, or 1 GHz.

[0082] The beams of light **120** produced by the pump, Stokes, and probe light sources **110** in a Raman spectroscopy system **100** may each have a spectral linewidth of less than 200 MHz. Additionally, one or more of the beams of light **120** may have a spectral linewidth of less than 1 MHz. For example, the spectral linewidth of a beam of light **120** may be less than 200 MHz, 100 MHz, 50 MHz, 10 MHz, 1 MHz, or 100 kHz. In FIG. 7, the spectral linewidth $\Delta\nu_{\text{sub.pr}}$ of the probe beam **120pr** and the spectral linewidth $\Delta\nu_{\text{sub.S}}$ of the Stokes beam **120S** may each be less than 200 MHz. In some embodiments, the probe light source **110pr** may be configured to produce a probe beam **120pr** having a relatively narrow spectral linewidth, which may allow a Raman

spectroscopy system **100** to measure a Raman spectrum with a high degree of spectral resolution. For example, the spectral linewidth $\Delta\nu_{\text{sub.pr}}$ of the probe beam **120pr** in FIG. 7 may be less than 1 MHz.

[0083] An optical receiver **200** of a coherent Raman spectroscopy system **100** may include one or more detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam of light **120pr** to produce an electronic signal. For an optical receiver **200** with a single detector **220**, all or most of the probe beam of light **120pr** may be mixed with the Raman signal **160**. For an optical receiver **200** with multiple detectors **220**, the probe beam of light **120pr** may be split so that a portion of the probe beam of light is sent to each of the detectors. For example, in an optical receiver **200** with four detectors **220**, the probe beam **120pr** may be split into four portions, and each detector may receive one of the four portions of the probe beam. Similarly, for an optical receiver **200** with a single detector **220**, all or most of the Raman signal **160** may be sent to the single detector, and for an optical receiver with multiple detectors **220**, the Raman signal **160** may be split so that a portion of the Raman signal is sent to each of the detectors.

[0084] Coherent mixing of a probe beam **120pr** and a Raman signal **160**, which may be referred to as heterodyne detection, may occur when the two optical signals are optically combined and then detected by a detector **220**. Optically combining the probe beam **120pr** and the Raman signal **160** may refer to combining the two optical signals so that their electric fields are summed together. For example, the probe beam and Raman signal may be combined (e.g., with an optical combiner **130**) so that the two signals are substantially coaxial and travel together in the same direction and along approximately the same optical path. Additionally, the probe beam and Raman signal may be combined so that at least a portion of their polarizations have the same orientation to allow at least a portion of their electric fields to be summed together. Once the probe beam and Raman signal are optically combined to produce a combined probe-Raman signal **210**, the probe beam and Raman signal may be coherently mixed at a detector **220**. The detector **220** may produce a photocurrent signal i corresponding to the coherent mixing of the probe beam **120pr** and a portion of the Raman signal **160**.

[0085] The portion of a Raman signal **160** that is coherently mixed with a probe beam of light **120pr** at a detector **220** to produce an electronic signal may refer to a spectral portion of the Raman signal. The spectral portion of a Raman signal **160** that is coherently mixed with a probe beam **120pr** may include optical frequency components of the Raman signal that are within a particular frequency range of the frequency $\nu_{\text{sub.3}}$ of the probe beam of light, where the particular frequency range is based on or depends on the electronic bandwidth Δf of the optical detector **220**. In FIG. 7, the hatched region around the probe frequency $\nu_{\text{sub.3}}$ represents the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr**. The particular frequency range illustrated by the hatched region extends from $\nu_{\text{sub.3}} - \Delta f$ to $\nu_{\text{sub.3}} + \Delta f$, where Δf is the electronic bandwidth of the detector. The particular frequency range includes the optical frequency components of the Raman signal **160** that are coherently mixed with the probe beam **120pr** to produce an electronic signal.

[0086] The electronic signal produced by the detector **220** in response to the coherent mixing of the portion of the Raman signal **160** and probe beam **120pr** may include one or more electronic frequency components, where each electronic frequency component has a frequency less than or equal to approximately Δf . For example, the electronic bandwidth Δf of the optical detector may be 10 GHz, and the electronic signal produced by the detector may include one or more electronic frequency components having frequencies less than or equal to approximately 10 GHz. Other optical frequency components of the Raman signal **160** that are outside the hatched region (e.g., optical frequencies less than $\nu_{\text{sub.3}} - \Delta f$ and greater than $\nu_{\text{sub.3}} + \Delta f$) may produce a coherent-mixing response in the detector **220**. However, since these Raman-signal optical frequency components would produce an electronic response at frequencies greater than Δf (which is outside

of the electronic bandwidth of the detector), these optical frequency components will not result in any significant contribution to the electronic signal. The electronic bandwidth of the detector **220** effectively limits or filters the optical frequency components of the Raman signal **160** that are measured by the optical receiver **200** to optical frequency components that are within a particular frequency range of the probe frequency. Accordingly, the electronic bandwidth of the detector **220**, in combination with the relatively narrow spectral linewidth of the probe beam **120pr**, may allow a Raman spectroscopy system **100** to measure a Raman signal **160** with a high degree of spectral resolution. Herein, an optical frequency or an optical frequency component refers to a signal in the optical domain between approximately 60 THz and approximately 1,000 THz, and an electronic frequency or an electronic frequency component refers to a signal in the electronic domain between 0 Hz and approximately 50 GHz.

[0087] The electronic signal that results from coherent mixing of a Raman signal **160** and a probe beam **120pr** may include a coherent-mixing term that is proportional to a product of (i) $E_{\text{sub.R}}$, the amplitude of the electric field of the Raman signal and (ii) $E_{\text{sub.pr}}$, the amplitude of the electric field of the probe beam. The photocurrent signal i produced by a detector **220** in response to the coherent mixing of a Raman signal **160** and a probe beam **120pr** may be proportional to the square of the summed electric fields of the probe beam **120pr** and a spectral portion of the Raman signal **160**. This type of detector **220** that produces a photocurrent signal i that is proportional to the square of a received electric field may be referred to as a square-law detector. The photocurrent signal i may be expressed as $i(t) = k|\epsilon_{\text{sub.R}}(t) + \epsilon_{\text{sub.pr}}(t)|^2$, where k is a constant (e.g., k may account for the responsivity of the detector **220** as well as other constant parameters or conversion factors). For clarity, the constant k or other constants (e.g., conversion constants or factors of 2 or 4) may be excluded from expressions herein related to the photocurrent i or the voltage signal **234**. In the above expression for $i(t)$, $\epsilon_{\text{sub.R}}(t)$ is the electric field of the Raman signal **160**, and $\epsilon_{\text{sub.pr}}(t)$ is the electric field of the probe beam **120pr**. The electric field of the Raman signal **160** may be expressed as $E_{\text{sub.R}} \cos[2\pi\nu_{\text{sub.R}}t + \phi_{\text{sub.R}}]$, where $E_{\text{sub.R}}$ is the amplitude of the electric field of the Raman signal. The electric field of the probe beam **120pr** may be expressed as $E_{\text{sub.pr}} \cos[2\pi\nu_{\text{sub.3}}t + \phi_{\text{sub.pr}}]$, where $E_{\text{sub.pr}}$ is the amplitude of the electric field of probe beam. The frequency $\nu_{\text{sub.R}}$ is the optical frequency of the electric field of the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr**. The frequency $\nu_{\text{sub.R}}$ may include the optical frequency components of the Raman signal **160** from $\nu_{\text{sub.3}} - \Delta f$ to $\nu_{\text{sub.3}} + \Delta f$, where Δf is the electronic bandwidth of the detector. The frequency $\nu_{\text{sub.3}}$ is the optical frequency of the electric field of the probe beam **120pr**. The term $\phi_{\text{sub.R}}$ is the phase of the electric field of the Raman signal **160**, and the term $\phi_{\text{sub.pr}}$ is the phase of the electric field of the probe beam **120pr**.

[0088] The above expression for the photocurrent signal i may be expanded and written as $i(t) = E_{\text{sub.R}}^2 + E_{\text{sub.pr}}^2 + 2E_{\text{sub.R}}E_{\text{sub.pr}} \cos[2\pi(\nu_{\text{sub.R}} - \nu_{\text{sub.3}})t + \Delta\phi]$, where, for clarity, the constant k is not included. In this expanded expression for the photocurrent signal $i(t)$, the first term $E_{\text{sub.R}}^2$ corresponds to the optical power ($P_{\text{sub.R}}$) of the Raman signal **160**, and the second term $E_{\text{sub.pr}}^2$ corresponds to the optical power ($P_{\text{sub.pr}}$) of the probe beam **120pr**. The third term in the above expression is $2E_{\text{sub.R}}E_{\text{sub.pr}} \cos[2\pi(\nu_{\text{sub.R}} - \nu_{\text{sub.3}})t + \Delta\phi]$ and may be referred to as a coherent-mixing term that represents coherent mixing between the electric fields of the Raman signal **160** and probe beam **120pr**. The phase difference $\Delta\phi$ is the phase difference between the electric fields of the Raman signal and the probe beam (e.g., $\Delta\phi = \phi_{\text{sub.R}} - \phi_{\text{sub.pr}}$). The coherent-mixing term is proportional to $E_{\text{sub.R}} \times E_{\text{sub.pr}}$, which is the product of the electric-field amplitudes of the Raman signal **160** and the probe beam **120pr**. Additionally, the coherent-mixing term includes a cosine function that varies in time based on the frequency difference ($\nu_{\text{sub.R}} - \nu_{\text{sub.3}}$) between the Raman signal **160** and the probe beam **120pr**. Since the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr** includes the optical frequency components of the Raman signal from $\nu_{\text{sub.3}} - \Delta f$ to

$v_{.3} + \Delta f$, the frequency-difference term ($v_{.R} - v_{.3}$) may include frequency components from zero to Δf . Accordingly, the coherent-mixing term may be referred to as including multiple electronic frequency components, where each electronic frequency component is proportional to $E_{RE.pr} \cos[2\pi ft + \Delta\phi]$. The frequency f , which may be referred to as an electronic frequency, is equal to the frequency difference ($v_{.R} - v_{.3}$) and has a value between zero and Δf . The coherent-mixing term may also be expressed as $2\sqrt{(P_{.R})}\sqrt{(P_{.pr})} \cos[2\pi ft + \Delta\phi]$, where $P_{.R}$ is the optical power of the Raman signal **160** and $P_{.pr}$ is the optical power of the probe beam **120pr**.

[0089] FIGS. **8-10** illustrate time-domain and frequency-domain plots of an example electronic signal resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. 7. The electronic signal produced by a detector **220** may be a photocurrent signal i , and an electronic amplifier **232** may produce a voltage signal **234** that corresponds to the photocurrent signal i . The voltage signal **234** may be approximately proportional to the photocurrent signal i , and both the voltage and photocurrent signals may be proportional to $P_{.R} + P_{.pr} + 2\sqrt{(P_{.R})}\sqrt{(P_{.pr})} \cos[2\pi ft + \Delta\phi]$, where the electronic frequency f is equal to the frequency difference ($v_{.R} - v_{.3}$). The electronic frequency f may take on multiple values (or a continuous range of values) between zero and Δf , and so, the voltage signal **234** in the time domain may be viewed as a summation over these multiple frequency components. In the frequency domain, the voltage signal **234** may have frequency components that extend from DC to Δf (or, for an amplifier **232** with a low-frequency cutoff filter, from the low-frequency cutoff to Δf). The two terms $P_{.R}$ and $P_{.pr}$ represent the optical powers of the Raman signal and probe beam. Over the time duration of a measurement, the powers of the Raman signal and probe beam may be approximately constant, and so the contribution of $P_{.R} + P_{.pr}$ in the above expression may correspond to a DC offset with little or no time variation. An electronic amplifier **232** may include a high-pass or band-pass filter that removes or attenuates the DC or low-frequency components, resulting in an electronic signal with little or no DC offset (e.g., as illustrated by the time-domain voltage signal **234** in FIG. **8**).

[0090] The signal characteristic **162** in each of FIGS. **7-10** represents a characteristic of an electronic signal (e.g., a characteristic of a photocurrent signal i or a characteristic of a corresponding voltage signal **234**) associated with a Raman signal **160**. A signal characteristic **162** may be determined from a digital signal **240**, where the digital signal is a digital representation of a photocurrent signal i or a voltage signal **234**. A characteristic **162** of an electronic signal may be determined by a processor of a Raman spectroscopy system **100** and may include one or more data points **163**, where each data point represents: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, an area, a frequency (e.g., an electronic frequency or an optical frequency), a phase, or a polarization. A Raman spectroscopy system **100** may measure one or more signal characteristics **162** associated with one or more Raman signals **160**, and based on the measured characteristics, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0091] FIG. **8** illustrates an example time-domain plot of a voltage signal **234** resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. 7. A digitizer **236** (e.g., an ADC) may produce a digital signal **240** that represents the voltage signal **234** in the time domain, and a processor may determine a characteristic **162** of the electronic signal based on the digital signal. The signal characteristic **162** in FIG. **8** includes a single data point **163t** which represents a peak amplitude Δt of the time-domain voltage signal **234**. The signal characteristic **162** may also include the optical frequency $v_{.3}$ that the probe beam **120pr** was set to when the electronic signal was obtained. Other time-domain-type data points **163** associated with the signal characteristic **162** in FIG. **8** may include (i) an amplitude of the time-domain voltage signal **234** at a particular time (e.g., at a temporal center) or (ii) an area associated with the voltage signal. For an

area associated with the voltage signal **234**, a processor may first convert the voltage signal into a non-negative signal (e.g., by taking the absolute value or by squaring the values of the voltage signal) and then integrate the non-negative signal to determine an area under the curve.

[0092] A processor of a Raman spectroscopy system **100** may determine a Fourier transform of a digital signal **240**. A voltage signal **234** may be a time-domain signal, and the digital signal **240** may be a time-domain digital representation of the voltage signal **234**. The Fourier transform of the digital signal **240** may produce a frequency-domain representation of the voltage signal **234**. From the Fourier transform, the processor may determine one or more electronic frequency components of the voltage signal **234**. Determining a frequency component of a voltage signal **234** may include determining an amplitude of the frequency-domain voltage signal at a particular frequency (e.g., at 4 GHz). FIGS. **9-10** each illustrates an example frequency-domain plot of a voltage signal **234** resulting from coherent mixing of the Raman signal **160** and probe beam **120_{pr}** of FIG. 7. The frequency-domain plots may be determined by taking a Fourier transform (e.g., a discrete Fourier transform or a fast Fourier transform) of a time-domain digital signal **240** that represents a time-domain voltage signal **234**. The x-axis of each of the frequency-domain plots in FIGS. **9-10** is labeled as “electronic frequency” to clarify and distinguish it from the x-axis of the frequency-domain plots in FIGS. **3-7**. The x-axis of each of the frequency-domain plots in FIGS. **3-7** may be referred to as an “optical frequency,” where the pump beam **120_{pu}**, Stokes beam **120_S**, Raman signal **160**, and probe beam **120_{pr}** each have frequencies in the optical domain (e.g., in the range of 60-1,000 THz). In contrast, the x-axis of each of the frequency-domain plots in FIGS. **9-10** includes frequencies in the electronic domain (e.g., in the range of DC to 50 GHz). Coherent mixing of a Raman signal **160** and probe beam **120_{pr}** (which are signals in the optical domain) produces an electronic signal that can be detected and analyzed using electronic techniques. The y-axis of each of the plots in FIGS. **8-10** is labeled “electronic signal amplitude” and may have units of voltage, current, or electrical power (e.g., watts).

[0093] In FIG. **9**, the frequency-domain voltage signal **234** has non-zero frequency components that extend from DC (i.e., zero hertz) to the detector-bandwidth frequency Δf . In FIG. **10**, the frequency-domain voltage signal drops off before reaching zero hertz, indicating that the corresponding voltage signal **234** may have been AC-coupled or high-pass filtered to remove the DC or low-frequency components. The signal characteristic **162** in FIG. **9** includes a single data point **163_f** which represents a peak amplitude $A_{\text{sub.p}}$ of the frequency-domain voltage signal **234**. The data point **163_f** may include the peak amplitude $A_{\text{sub.p}}$ or the frequency $f_{\text{sub.p}}$ at which the peak amplitude is located. The signal characteristic **162** in FIG. **10** includes five data points: data point **163_a** at frequency $f_{\text{sub.a}}$, data point **163_b** at frequency $f_{\text{sub.b}}$, data point **163_c** at frequency $f_{\text{sub.c}}$, data point **163_d** at frequency $f_{\text{sub.d}}$, and data point **163_e** at frequency $f_{\text{sub.e}}$. Each data point **163** in FIG. **10** may include an amplitude value (e.g., amplitude $A_{\text{sub.a}}$ may be associated with data point **163_a**) or a frequency (e.g., frequency f_a may be associated with data point **163_a**). The frequencies associated with the data points **163** in FIG. **10** may have particular values. For example, the detector bandwidth Δf may be 5 GHz, and the frequencies $f_{\text{sub.a}}$, $f_{\text{sub.b}}$, $f_{\text{sub.c}}$, $f_{\text{sub.d}}$, and $f_{\text{sub.e}}$ may have respective values of approximately 1 GHz, 2 GHz, 3 GHz, 4 GHz, and 5 GHz. Other frequency-domain-type data points **163** associated with the signal characteristic **162** in FIG. **9** or **10** may include (i) an amplitude of the frequency-domain voltage signal **234** at a particular frequency or at a center frequency or (ii) an area or an average amplitude associated with the frequency-domain voltage signal **234**. Additionally, the signal characteristic **162** in FIG. **9** or **10** may include the optical frequency $\nu_{\text{sub.3}}$ that the probe beam **120_{pr}** was set to when the voltage signal **234** was obtained.

[0094] A processor of a coherent Raman spectroscopy system **100** may associate a determined signal characteristic **162** with a Raman frequency shift. For example, the signal characteristic **162** in FIG. **7** may be associated with a Raman frequency shift having a frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3}}$, which is the difference between the pump-beam frequency $\nu_{\text{sub.1}}$ and the probe-beam frequency

v.sub.3. A sample **150** under investigation may include a material with a Raman spectrum having a peak at a frequency Ω , which is approximately equal to v.sub.1–v.sub.RP, the difference between the pump-beam frequency v.sub.1 and the Raman-signal peak frequency v.sub.RP. The peak frequency Ω may be approximately equal to a vibrational frequency of the material and may correspond to a Raman frequency shift of Ω (e.g., when excited with light at a frequency ν , the material may produce Raman-shifted light at the frequencies $\nu+\Omega$ and $\nu-\Omega$). In FIG. 7, the probe beam **120pr** is being used to measure the Raman signal **160** at the probe-frequency v.sub.3, and the resulting signal characteristic **162** that is determined may be associated with a Raman spectrum of the material at the frequency v.sub.1–v.sub.3, which may be referred to as a Raman frequency shift of v.sub.1–v.sub.3.

[0095] FIG. 11 illustrates an example Raman signal **160** that is measured at multiple probe frequencies v.sub.3. The probe beam **120pr** is tuned to multiple different probe frequencies (v.sub.3-1, v.sub.3-2, v.sub.3-3, . . . v.sub.3-n) to measure multiple respective signal characteristics (**162-1**, **162-2**, **162-3**, . . . **162-n**) of the Raman signal **160**. The Raman signal **160** is produced by coherent Raman scattering of pump and Stokes beams within a sample **150**. Throughout the measurements of the multiple signal characteristics **162** in FIG. 11, the pump-beam frequency v.sub.1 and the Stokes-beam frequency v.sub.2 may remain substantially constant so that the resulting Raman signal **160** also remains substantially constant (e.g., the Raman signal may exhibit substantially the same shape and amplitude throughout the measurements). The number n of signal characteristics **162** that are measured may be 1, 5, 10, 50, 100, 500, 1,000, or any other suitable number of signal characteristics. At each of the n probe frequencies, the probe beam **120pr** may be coherently mixed with a spectral portion of the Raman signal **160** that is within a particular frequency range of the probe frequency v.sub.3 (e.g., within Δf of the probe frequency, as illustrated in FIG. 7) to measure one signal characteristic **162**. Each signal characteristic **162** may provide information about the Raman signal **160** in the frequency region from v.sub.3– Δf to v.sub.3+ Δf . Based on one or more measured signal characteristics **162**, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0096] At each of the n probe frequencies, a single measurement may be performed, or multiple measurements may be performed. For example, with the probe beam **120pr** in FIG. 11 set to the probe-beam frequency v.sub.3-1, an optical receiver **200** may measure a single photocurrent signal i and produce a corresponding single digital output signal **240**. From that one output signal **240**, a processor may determine a signal characteristic **162-1**. Alternatively, with the probe beam **120pr** in FIG. 11 set to the probe-beam frequency v.sub.3-1, an optical receiver **200** may measure a series of multiple photocurrent signals i (e.g., 2, 4, 10, 20, 50, 100, or any other suitable number of signals) and produce multiple corresponding digital output signals **240**. A processor may determine one signal characteristic **162-1** from the multiple digital output signals. For example, the processor may average or otherwise combine the multiple digital output signals **240** to produce one signal characteristic **162-1**. Measurement of a series of multiple photocurrent signals i may improve the measurement accuracy by averaging out noise or removing outliers from the measurements.

[0097] By tuning the probe-beam frequency v.sub.3 to multiple frequencies across at least a portion of a Raman signal **160**, a coherent Raman spectroscopy system **100** may measure the Raman signal at multiple points **162**. For example, the probe-beam frequency v.sub.3 in FIG. 6 may be tuned across at least a portion of the frequency range between v.sub.2–F.sub.2 and v.sub.2–F.sub.1 or between v.sub.2+F.sub.1 and v.sub.2+F.sub.2. Each of the different probe-beam frequencies to which the probe beam **120pr** is changed may be offset from an adjacent probe-beam frequency by a frequency increment ΔF between approximately 10 MHz and approximately 10 GHz. For example, the probe-beam frequency in FIG. 11 may be changed in frequency increments ΔF of approximately 5 GHz as it is tuned across at least a portion of the Raman signal **160**. Each signal characteristic **162** in FIG. 11 is plotted along the x-axis at the probe frequency v.sub.3 at which the

signal characteristic was obtained. The y-axis in FIG. 11 may represent an amplitude or area associated with the signal characteristics 162. For example, each signal characteristic 162 in FIG. 11 may be plotted along the y-axis at a value corresponding to an amplitude or an area of an electronic signal from which the signal characteristic was obtained.

[0098] A probe light source 110_{pr} of a coherent Raman spectroscopy system 100 may include a wavelength-tunable laser, where the frequency $\nu_{\text{sub.3}}$ of the probe beam 120_{pr} is adjustable by changing the wavelength of light produced by the wavelength-tunable laser. For example, a wavelength-tunable laser may be adjustable over a wavelength range having a width between approximately 10 nm and approximately 100 nm. As another example, the wavelength-tuning range of a wavelength-tunable laser may be between approximately 1000 nm and approximately 1100 nm, between approximately 1490 nm and approximately 1570 nm, or between approximately 1600 nm and approximately 1690 nm. A wavelength-tunable laser may be continuously tunable over a wavelength-tuning range or may be tunable to multiple discrete wavelengths within a wavelength-tuning range. For example, a wavelength-tunable laser may be continuously tunable to any wavelength between 1530 nm and 1570 nm. Alternatively, a wavelength-tunable laser may be tunable to a set of approximately 10, 100, or 1,000 discrete wavelengths between 1530 nm and 1570 nm (e.g., the wavelengths may be separated from one another by approximately 4 nm, 0.4 nm, or 0.04 nm, respectively). A probe light source 110_{pr} may include a wavelength-tunable laser that sequentially changes the probe-beam frequency $\nu_{\text{sub.3}}$ to multiple different frequencies. For example, the probe-beam frequency in FIG. 6 may be tuned to approximately 100 different frequencies between the frequencies $\nu_{\text{sub.2}} - F_{\text{sub.2}}$ and $\nu_{\text{sub.2}} - F_{\text{sub.1}}$. At each of the different probe-beam frequencies, the probe beam 120_{pr} and a spectral portion of the Raman signal 160 may be coherently mixed at a detector 220 to produce a corresponding electronic signal, and a processor may determine a signal characteristic 162 based on the electronic signal.

[0099] In FIG. 11, the probe beam 120_{pr} may initially be set to the probe-beam frequency $\nu_{\text{sub.3-1}}$ and coherently mixed at a detector 220 with a spectral portion of the Raman signal 160 around the probe frequency $\nu_{\text{sub.3-1}}$. For example, the spectral portion of the Raman signal 160 may include optical frequency components of the Raman signal from $\nu_{\text{sub.3-1}} - \Delta f$ to $\nu_{\text{sub.3-1}} + \Delta f$, where Δf is the electronic bandwidth of the optical detector 220. A processor may determine a signal characteristic 162-1 based on the electronic signal resulting from the coherent mixing of the probe beam 120_{pr} at frequency $\nu_{\text{sub.3-1}}$ and the associated spectral portion of the Raman signal 160. After measuring the Raman signal 160 around the probe frequency $\nu_{\text{sub.3-1}}$, the probe light source 110_{pr} may change the probe-beam frequency by a frequency change ΔF to the frequency $\nu_{\text{sub.3-2}}$. The probe frequency $\nu_{\text{sub.3-2}}$ is equal to $\nu_{\text{sub.3-1}} + \Delta F$, and the frequency change ΔF between adjacent frequencies may be between approximately 10 MHz and approximately 10 GHz. The frequency change ΔF may be a fixed value or may be dynamically adjusted during a measurement of a Raman signal 160. After the probe-beam frequency is changed to $\nu_{\text{sub.3-2}}$, the probe beam 120_{pr} may be coherently mixed with the spectral portion of the Raman signal 160 around the probe frequency $\nu_{\text{sub.3-2}}$ (e.g., the spectral portion may include optical frequency components of the Raman signal from $\nu_{\text{sub.3-2}} - \Delta f$ to $\nu_{\text{sub.3-2}} + \Delta f$). A processor may determine a signal characteristic 162-2 based on the electronic signal resulting from the coherent mixing of the probe beam 120_{pr} at frequency $\nu_{\text{sub.3-2}}$ and the associated spectral portion of the Raman signal 160. After measuring the Raman signal 160 around the probe frequency $\nu_{\text{sub.3-2}}$, the probe light source 110_{pr} may change the probe-beam frequency by the frequency change ΔF to the frequency $\nu_{\text{sub.3-3}}$, and a measurement of the Raman signal 160 around the frequency $\nu_{\text{sub.3-3}}$ may be performed. The Raman spectroscopy system 100 may sequentially change the frequency $\nu_{\text{sub.3}}$ of the probe light source and measure a spectral portion of the Raman signal 160 at each frequency until reaching the final probe frequency $\nu_{\text{sub.3-n}}$. During the measurements, the probe-beam frequency $\nu_{\text{sub.3}}$ may be tuned so that it avoids overlapping with the frequency $\nu_{\text{sub.2}}$ of the Stokes beam to prevent mixing between the probe and Stokes beams.

[0100] FIG. 12 illustrates an example Raman spectrum corresponding to the Raman signal of FIG. 11. A Raman spectroscopy system 100 may measure a sample 150 and determine the signal characteristics 162 of the Raman signal 160 in FIG. 11, and the corresponding Raman spectrum in FIG. 12 may represent the Raman spectrum of one or more materials that are part of the sample 150. For example, the sample 150 may include glucose and the peak frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.RP}}$ of the Raman spectrum may be approximately equal to a frequency Ω of a molecular vibration of glucose. A processor of the Raman spectroscopy system 100 may determine the Raman spectrum in FIG. 12 based on the signal characteristics 162 (and the associated probe-beam frequencies $\nu_{\text{sub.3}}$) of the Raman signal 160 in FIG. 11. Each signal characteristic 162 of a Raman signal 160 measured at a probe frequency $\nu_{\text{sub.3}}$ may be associated with a Raman frequency shift having a frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3}}$, where $\nu_{\text{sub.1}}$ is the pump-beam frequency. To determine the Raman spectrum, each signal characteristic 162 may be transformed from its Raman-signal frequency $\nu_{\text{sub.3}}$ to a corresponding Raman-shift frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3}}$. For example, the processor may associate the signal characteristic 162-1 at frequency $\nu_{\text{sub.3-1}}$ in FIG. 11 with a Raman shift having the frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3-1}}$ in FIG. 12. Similarly, the signal characteristic 162-2 at frequency $\nu_{\text{sub.3-2}}$ in FIG. 11 may be associated with a Raman shift having the frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3-2}}$ in FIG. 12, and the signal characteristic 162-3 at frequency $\nu_{\text{sub.3-3}}$ in FIG. 11 may be associated with a Raman shift having the frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3-3}}$ in FIG. 12. Additionally, the peak frequency $\nu_{\text{sub.RP}}$ of the Raman signal 160 in FIG. 11 may correspond to the Raman-signal peak in FIG. 12 with a frequency of $\nu_{\text{sub.1}} - \nu_{\text{sub.RP}}$, which in turn may correspond to the vibrational frequency Ω of a particular material.

[0101] Based on the Raman spectrum in FIG. 12, a processor may determine (i) whether a particular material is present in a sample 150 or (ii) an amount or a concentration of the particular material in the sample 150. For example, the processor may compare one or more peaks, signal characteristics 162, or other features of the Raman spectrum in FIG. 12 to a previously determined Raman spectrum for glucose. If one or more peaks or characteristics 162 of the Raman spectrum in FIG. 12 match or line up with peaks or characteristics of the Raman spectrum for glucose, then the processor may determine that glucose is present in the sample 150. Alternatively, if the Raman spectrum in FIG. 12 is missing one or more peaks or characteristics of the Raman spectrum for glucose, then the processor may determine that little or no glucose is present in the sample 150. As another example, the processor may determine the amount or concentration of glucose in the sample 150 based on the Raman spectrum in FIG. 12. The concentration of glucose in the sample 150 may be related to the amplitude or height of one or more peaks of the Raman spectrum in FIG. 12 (e.g., the glucose concentration may be approximately proportional to the height of one or more Raman peaks). The concentration of glucose may then be determined based at least in part on the height of the Raman peak located at the Raman-shift frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.RP}}$.

[0102] FIG. 13 illustrates another example Raman signal 160 that is measured at multiple probe frequencies. The probe beam 120_{pr} is tuned to multiple different probe frequencies ($\nu_{\text{sub.3-1}}$, $\nu_{\text{sub.3-2}}$, $\nu_{\text{sub.3-3}}$, . . . $\nu_{\text{sub.3-n}}$) to measure multiple respective signal characteristics (162-1, 162-2, 162-3, . . . 162-n) of the Raman signal 160. The Raman signal 160 in FIG. 13 has a single peak centered at frequency $\nu_{\text{sub.RP}}$ with a spectral linewidth of $\Delta\nu_{\text{sub.R}}$. The Raman signal 160 in FIG. 11 has one main peak located at frequency $\nu_{\text{sub.RP}}$ along with multiple smaller peaks located on either side of the main peak. A Raman signal 160 may include a single peak (e.g., as illustrated in FIG. 13) or may include multiple peaks (e.g., as illustrated in FIG. 11).

[0103] FIGS. 14-15 each illustrates a second example Raman signal obtained by changing the frequency offset Ω between a pump beam 120_{pu} and a Stokes beam 120_S. A coherent Raman spectroscopy system 100 may include a pump light source 110_{pu} or a Stokes light source 110_S with a wavelength-tunable laser, and the frequency offset Ω may be adjustable by changing the wavelength of the wavelength-tunable laser. A wavelength-tunable pump or Stokes laser may be continuously tunable over a wavelength-tuning range or may be tunable to multiple discrete

wavelengths within a wavelength-tuning range. For example, Stokes beam **120** in FIG. **14** may be produced by a continuously tunable laser diode that can produce light at the frequencies $\nu_{\text{sub.2}}$ and $\nu'_{\text{sub.2}}$. As another example, a Stokes laser **110S** may include two fixed-wavelength laser diodes that operate at the respective frequencies $\nu_{\text{sub.2}}$ and $\nu'_{\text{sub.2}}$. Similarly, pump beam **120pu** in FIG. **15** may be produced by a continuously tunable laser diode that can produce light at the frequencies $\nu_{\text{sub.1}}$ and $\nu'_{\text{sub.1}}$, or a pump laser **110pu** may include two fixed-wavelength laser diodes that operate at the respective frequencies $\nu_{\text{sub.1}}$ and $\nu'_{\text{sub.1}}$.

[0104] In FIG. **14**, the pump frequency $\nu_{\text{sub.1}}$ is fixed, and the Stokes frequency is changed from $\nu_{\text{sub.2}}$ to $\nu'_{\text{sub.2}}$ to change the frequency offset from $\Omega_{\text{sub.1}}$ to $\Omega_{\text{sub.2}}$. Frequency offset $\Omega_{\text{sub.1}}$ equals $\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$, and frequency offset $\Omega_{\text{sub.2}}$ equals $\nu_{\text{sub.1}} - \nu'_{\text{sub.2}}$. In FIG. **15**, the Stokes frequency $\nu_{\text{sub.2}}$ is fixed, and the pump frequency is changed from $\nu_{\text{sub.1}}$ to $\nu'_{\text{sub.1}}$ to change the frequency offset from $\Omega_{\text{sub.1}}$ to $\Omega_{\text{sub.2}}$. Frequency offset $\Omega_{\text{sub.1}}$ equals $\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$, and frequency offset $\Omega_{\text{sub.2}}$ equals $\nu'_{\text{sub.1}} - \nu_{\text{sub.2}}$. In other embodiments, both the pump and Stokes frequencies may be changed to change the frequency offset Ω from one value to another.

[0105] FIG. **14** includes two Raman signals **160a** and **160b**. Initially, a Stokes light source **110S** may produce a Stokes beam **120S** at the frequency $\nu_{\text{sub.2}}$ to produce the frequency offset (21 between the Stokes beam and the pump beam **120pu**, where $221 = \nu_{\text{sub.1}} - \nu_{\text{sub.2}}$. For example, the Stokes beam **120S** may have a frequency $\nu_{\text{sub.2}}$ of 193 THz, and the pump beam **120pu** may have a frequency $\nu_{\text{sub.1}}$ of 207 THz, which corresponds to a frequency offset $\Omega_{\text{sub.1}}$ of 14 THz (or, 467 cm^{-1}). The Raman signal **160a** is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu** within a sample **150**. The Raman signal **160a** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** overlaps the Raman signal **160a** and is relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** (e.g., the probe-beam frequency $\nu_{\text{sub.3}}$ may be within 200 GHz of the Stokes-beam frequency $\nu_{\text{sub.2}}$). The probe beam **120pr** and a spectral portion of the Raman signal **160a** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160a** to measure multiple signal characteristics **162** associated with the Raman signal.

[0106] After measuring the first Raman signal **160a**, the Stokes light source **110S** may change the frequency of the Stokes beam to produce a Stokes beam **120S'** at the frequency $\nu'_{\text{sub.2}}$, resulting in a frequency offset of $\Omega_{\text{sub.2}}$ between the Stokes beam **120S'** and the pump beam **120pu**, where $\Omega_{\text{sub.2}} = \nu_{\text{sub.1}} - \nu'_{\text{sub.2}}$. For example, the Stokes beam **120S** may be changed to a frequency $\nu'_{\text{sub.2}}$ of 182 THz, and the pump-beam frequency $\nu_{\text{sub.1}}$ may remain at 207 THz, which corresponds to a frequency offset $\Omega_{\text{sub.1}}$ of 25 THz (or, 834 cm^{-1}). The second Raman signal **160b** is produced by coherent Raman scattering of the Stokes beam **120S'** and pump beam **120pu** within the sample **150**. The Raman signal **160b** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S'**. Additionally, a probe light source **110pr** may change the frequency of the probe beam to produce a probe beam **120pr'** at a frequency $\nu'_{\text{sub.3}}$ that overlaps the Raman signal **160b** and is relatively close to the frequency $\nu'_{\text{sub.2}}$ of the Stokes beam **120S'** (e.g., the probe-beam frequency $\nu'_{\text{sub.3}}$ may be within 200 GHz of the Stokes-beam frequency $\nu'_{\text{sub.2}}$).

[0107] The probe beam **120pr'** and a spectral portion of the Raman signal **160b** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu'_{\text{sub.3}}$ of the probe beam **120pr'** may be tuned across at least a portion of the second Raman signal **160b** to measure multiple signal characteristics **162** associated with the Raman signal.

[0108] The two Raman signals **160a** and **160b** in FIG. **14** may correspond to two respective peaks of a Raman spectrum of a material that is in the sample **150**. One Raman peak, associated with Raman signal **160a**, is located at a frequency of approximately $\Omega_{\text{sub.1}}$, and the frequency $\Omega_{\text{sub.1}}$

may correspond to a vibrational frequency of the material. The other Raman peak, associated with Raman signal **160b**, is located at a frequency of approximately $\Omega_{\text{sub.2}}$, and the frequency $\Omega_{\text{sub.2}}$ may correspond to another vibrational frequency of the material. A Raman spectroscopy system **100** may perform measurements of signal characteristics **162** at 1, 2, 4, 10, 20, or 50 different values of the frequency offset Ω . The frequency offset Ω between the Stokes beam **120S** and the pump beam **120pu** may be set to each of the different frequency-offset values to produce an associated Raman signal **160**. At each value of Ω , the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal. Then, the frequency of the Stokes beam or pump beam may be adjusted to the next value of Ω , where another measurement of an associated Raman signal **160** is performed. Based on the signal characteristics **162** associated with each of the values of the frequency offset Ω , a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0109] FIG. **15** includes the Raman signal **160c**. A first Raman signal (similar to Raman signal **160a** in FIG. **14**) may be approximately overlapped with the second Raman signal **160c** in FIG. **15**, and for clarity, the first Raman signal is not included in FIG. **15**. Initially, a Stokes light source **110S** may produce a Stokes beam **120S** at the frequency $\nu_{\text{sub.2}}$, and a pump light source **110pu** may produce a pump beam **120pu** at the frequency $\nu_{\text{sub.1}}$, which results in the frequency offset $\Omega_{\text{sub.1}}$ between the Stokes and pump beams in FIG. **15**. For example, the Stokes beam **120S** may have a frequency $\nu_{\text{sub.2}}$ of 193 THz, and the pump beam **120pu** may have a frequency $\nu_{\text{sub.1}}$ of 207 THz, which corresponds to a frequency offset $\Omega_{\text{sub.1}}$ of 14 THz (or, 467 $\text{cm}^{\text{sup.}}\text{-1}$). A first Raman signal (similar to Raman signal **160a** in FIG. **14**) is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu** within a sample **150**. The first Raman signal may be centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may overlap the first Raman signal and may be relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam of light **120S**. The probe beam **120pr** and a spectral portion of the first Raman signal may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be tuned across at least a portion of the first Raman signal to measure multiple signal characteristics **162** associated with the first Raman signal.

[0110] In FIG. **15**, after measuring the first Raman signal, the pump light source **110pu** may change the frequency of the pump beam **120pu** from $\nu_{\text{sub.1}}$ to $\nu_{\text{sub.1}}$. This results in a frequency offset of $\Omega_{\text{sub.2}}$ between the Stokes beam **120S** and the pump beam **120pu'**, where $\Omega_{\text{sub.2}} = \nu_{\text{sub.1}} - \nu_{\text{sub.2}}$. For example, the pump beam may be changed to a frequency $\nu_{\text{sub.1}}$ of 218 THz, and the Stokes-beam frequency $\nu_{\text{sub.2}}$ may remain at 193 THz, which corresponds to a frequency offset $\Omega_{\text{sub.1}}$ of 25 THz (or, 834 $\text{cm}^{\text{sup.}}\text{-1}$). The second Raman signal **160c** is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu'** within the sample **150**. As with the first Raman signal, the Raman signal **160c** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** overlaps the Raman signal **160c** and is relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam of light **120S** (e.g., the probe-beam frequency $\nu_{\text{sub.3}}$ may be within 200 GHz of the Stokes-beam frequency $\nu_{\text{sub.2}}$). The probe beam **120pr** and a spectral portion of the Raman signal **160c** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160c** to measure multiple signal characteristics **162** associated with the Raman signal.

[0111] In FIG. **15**, the first Raman signal (which may be similar to Raman signal **160a** in FIG. **14**) and the second Raman signal **160c** may correspond to two different peaks of a Raman spectrum of a material that is in the sample **150**. One Raman peak, associated with the first Raman signal, is located at a frequency of approximately $\Omega_{\text{sub.1}}$, and the frequency $\Omega_{\text{sub.1}}$ may correspond to a

vibrational frequency of the material. The other Raman peak, associated with Raman signal **160c**, is located at a frequency of approximately $\Omega_{\text{sub.2}}$, and the frequency $\Omega_{\text{sub.2}}$ may correspond to another vibrational frequency of the material. Based at least in part on the signal characteristics **162** associated with the two frequency offsets $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0112] FIG. **16** illustrates an example Raman signal **160** along with two probe beams of light **120pr-1** and **120pr-2**. The Raman signal **160** may be produced by coherent Raman scattering of the pump beam **120pu** and Stokes beam **120S** within a sample **150**. The frequency of the pump beam **120pu** is $\nu_{\text{sub.1}}$, and the frequency of the Stokes beam **120S** is $\nu_{\text{sub.2}}$, which corresponds to a frequency offset Ω equal to $\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$. The Raman signal **160** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** (e.g., the center frequency of the Raman signal **160** may be within 200 GHz of $\nu_{\text{sub.2}}$). The first probe beam **120pr-1** may be used to measure the Raman signal **160**, and the second probe beam **120pr-2** may be used to measure the pump beam **120pu**.

[0113] The frequency $\nu_{\text{sub.3}}$ of first probe beam **120pr-1** overlaps the Raman signal **160** and is relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** (e.g., $\nu_{\text{sub.3}}$ may be within 200 GHz of $\nu_{\text{sub.2}}$). The probe beam **120pr-1** and a spectral portion of the Raman signal **160** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr-1** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal.

[0114] The frequency $\nu_{\text{sub.4}}$ of the second probe beam **120pr-2** is relatively close to the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu** (e.g., $\nu_{\text{sub.4}}$ may be within 50 GHz of $\nu_{\text{sub.1}}$). For example, the frequency $\nu_{\text{sub.4}}$ of the second probe beam **120pr-2** may be offset from the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu** by approximately 10 GHz, 5 GHz, or 1 GHz. Alternatively, the frequency $\nu_{\text{sub.4}}$ of the second probe beam **120pr-2** may be approximately equal to the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu**. After the pump beam **120pu** has interacted with the sample, the probe beam **120pr-2** may be coherently mixed with the pump beam. For example, after the pump and Stokes beams have produced the Raman signal **160** and after the pump beam has exited the sample, the pump and probe beams may be coherently mixed together at a detector **220** to produce an electronic signal, from which a signal characteristic may be determined.

[0115] FIG. **17** illustrates an example optical receiver **200** for measuring the Raman signal **160** and pump beam of light **120pu** from FIG. **16**. The combined pump-Stokes beam **140** (which includes pump beam **120pu** and Stokes beam **120S**) is directed to a sample **150**, which produces a Raman signal **160** in response to the pump and Stokes beams. The Raman signal **160** may be collected by one or more optical elements and directed to the optical receiver **200**. Additionally, residual light from the pump beam **120pu** may be collected and directed to the optical receiver **200** as a residual pump beam **120pu-2**. The residual pump beam **120pu-2** may include light from the pump beam **120pu** after the pump beam has interacted with and exited the sample **150**. The optical receiver **200** in FIG. **17** may be referred to as a two-channel optical receiver that includes two parallel measurement channels for separately detecting and measuring the Raman signal **160** and the residual pump beam **120pu-2**. The first measurement channel detects the Raman signal **160** and includes probe laser **110pr-1**, detector **220-1**, and detection electronics **230-1**. The second measurement channel detects the residual pump beam **120pu-2** and includes probe laser **110pr-2**, detector **220-2**, and detection electronics **230-2**.

[0116] The Raman signal **160** and the residual pump beam **120pu-2** are directed to the combiner **130c**, which may be a dichroic beamsplitter, and the combiner **130c** reflects the Raman signal **160** and transmits the residual pump beam **120pu-2**. The combiner **130c** also transmits at least a portion of the probe beam **120pr-1** produced by the probe laser **110pr-1** and combines the probe beam

120pr-1 with the Raman signal **160** to produce a combined probe-Raman signal **210-1**. The probe-Raman signal **210-1** is sent to the detector **220-1**, where the probe beam **120pr-1** and a spectral portion of the Raman signal **160** are coherently mixed to produce a photocurrent signal i.sub.1. The detection electronics **230-1** receives the photocurrent signal h and produces a digital output signal **240-1** that corresponds to the photocurrent signal i.sub.1.

[0117] The combiner **130d** (which may be a dichroic or a non-dichroic beamsplitter) reflects at least a portion of the residual pump beam **120pu-2** and transmits at least a portion of the probe beam **120pr-2** produced by the probe laser **110pr-2**. The combiner **130d** combines the probe beam **120pr-2** with the residual pump beam **120pu-2** to produce a combined probe-pump signal **210-2**, which is sent to the detector **220-2**. The probe beam **120pr-2** and the residual pump beam **120pu-2** are coherently mixed at the detector **220-2** to produce a photocurrent signal i.sub.2, and the detection electronics **230-2** produces a digital output signal **240-2** that corresponds to the photocurrent signal i.sub.2.

[0118] The two digital output signals **240-1** and **240-2** may be sent to a processor which determines a signal characteristic **162** of each of the photocurrent signals i.sub.1 and i.sub.2 based on the digital output signals. Additionally, the frequency of the first probe beam **120pr-1** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal. If the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu** remains fixed, the frequency $\nu_{\text{sub.4}}$ of the second probe beam **120pr-2** may also remain fixed. Alternatively, if the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu** is changed (e.g., to switch to a different frequency offset Ω) the frequency of the probe beam **120pr-2** may also be switched to maintain a particular frequency offset between the pump and probe frequencies.

[0119] Measurement of the residual pump beam **120pu-2** may be performed two or more times to determine how the power of the pump beam changes when the Raman signal **160** is produced. For example, the residual pump beam **120pu-2** may be measured once when the Stokes beam **120S** is turned off (and no Raman signal **160** is produced) and another time when the Stokes beam is turned on (and the Raman signal **160** is produced). A processor may determine the change in the power of the residual pump beam **120pu-2** associated with the Stokes beam **120S** being turned off and on. Since at least part of the Raman signal **160** may be produced by Stokes-shifted photons from the pump beam **120pu**, a decrease in the power of the residual pump beam **120pu-2** may correspond to the power of the Raman signal.

[0120] In another embodiment of a two-channel optical receiver, the optical receiver may not include a second probe laser **110pr-2**. Instead, the residual pump beam **120pu-2** may be sent to a detector **220** for direct detection without mixing the residual pump beam with another signal.

[0121] FIG. **18** illustrates an example Raman spectroscopy system **100** for measuring a Raman signal **160** produced by spontaneous Raman scattering. Instead of producing a Raman signal by coherent Raman scattering of pump and Stokes beams within a sample (e.g., as illustrated in FIGS. **1-2**), the Raman signal **160** in FIG. **18** is produced by spontaneous Raman scattering of the pump beam of light **120pu** within the sample **150**. While the production of the Raman signal **160** in FIG. **18** is different from that of the Raman spectroscopy systems of FIGS. **1-2**, the optical receiver **200** and the Raman-signal detection technique in FIG. **18** is similar to that of FIGS. **1-2**. In FIG. **18**, the Raman signal **160** is detected by coherently mixing the Raman signal with a probe beam of light **120pr**.

[0122] The Raman spectroscopy system **100** in FIG. **18** includes a pump light source **110pu** that produces a pump beam of light **120pu** at a pump frequency $\nu_{\text{sub.1}}$. The pump beam **120pu** is directed to a sample **150** (e.g., by one or more optical elements), and the sample **150** produces a Raman signal **160** by spontaneous Raman scattering of light from the pump beam. The spontaneous Raman signal **160** is collected (e.g., by one or more optical elements) and directed to the optical receiver **200**. The optical receiver **200** includes a probe light source **110pr** that produces a probe beam of light **120pr** at a probe frequency $\nu_{\text{sub.3}}$, where the probe frequency overlaps the Raman

signal. The optical combiner **130b**, which may be a dichroic or a non-dichroic beamsplitter, combines the Raman signal **160** and the probe beam **120pr** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. The detector **220** coherently mixes a spectral portion of the Raman signal **160** with the probe beam **120pr** to produce a photocurrent signal *i*. The detection electronics **230** may produce (i) an analog voltage signal that corresponds to the photocurrent signal *i* and (ii) a digital output signal **240** that corresponds to the photocurrent signal or the voltage signal. The digital output signal **240** may be sent to a processor, and the processor may determine a signal characteristic **162** of the photocurrent signal or voltage signal based on the digital output signal **240**.

[0123] FIG. **19** illustrates an example Raman signal produced by the Raman spectroscopy system of FIG. **18**. The Raman signal **160** produced by spontaneous Raman scattering of the pump beam **120pu** has a peak frequency of $\nu_{\text{sub.RP}}$. The frequency offset Ω between the pump beam and the peak frequency of the Raman signal equals $\nu_{\text{sub.1}} - \nu_{\text{sub.RP}}$, and the frequency offset Ω may correspond to a vibrational frequency of a material that is part of the sample **150**. In FIG. **19**, the probe beam **120pr** may be coherently mixed with a spectral portion of the Raman signal **160** that is within a particular frequency range of the probe frequency $\nu_{\text{sub.3}}$ to measure one signal characteristic **162**. Additionally, the probe light source **110pr** in FIG. **18** may include a wavelength-tunable laser that tunes the probe beam **120pr** to multiple frequencies across at least a portion of the Raman signal **160**, and the optical receiver **200** may measure multiple respective signal characteristics **162** associated with the Raman signal.

[0124] FIG. **20** illustrates an example laser diode **110** that produces a free-space beam of light **120**. The laser diode **110** in FIG. **20** may be part of a pump light source **110pu**, a Stokes light source **110S**, or a probe light source **110pr**, and the free-space beam **120** may be a pump beam **120pu**, a Stokes beam **120S**, or a probe beam **120pr**. For example, the probe laser **110pr** in FIG. **2** may be similar to the laser diode **110** in FIG. **20**, and the probe laser **110pr** may produce a free-space probe beam **120pr** that is combined with the Raman signal **160** by a free-space beam combiner **130b**. In FIG. **20**, the electronic driver **112** supplies laser-diode drive current *I* to the laser diode **110**, and the laser diode produces output light that is collimated by a lens **114** to produce a collimated free-space beam **120**. In other embodiments, a lens **114** may produce a focused free-space beam **120** (e.g., the lens may focus the free-space beam onto a sample **150**). The laser current *I* supplied to the laser diode **110** may be a substantially constant DC current resulting in an output beam of light **120** having a substantially constant optical power. Additionally or alternatively, the laser current *I* may include pulses of current resulting in an output beam **120** that includes corresponding pulses of light.

[0125] The output beam of light **120** produced by the laser diode **110** may have a spectral linewidth of less than approximately 200 MHz, 100 MHz, 50 MHz, 10 MHz, 1 MHz, or 100 kHz. The laser diode **110** in FIG. **20** may be a wavelength-tunable laser diode where the wavelength of the output beam **120** is adjustable over a wavelength range having a width between approximately 10 nm and approximately 100 nm. For example, the operating wavelength of the laser diode **110** may be tunable over at least a portion of one of the following wavelength ranges: 1000 nm to 1100 nm; 1220 nm to 1450 nm; 1490 nm to 1570 nm; 1600 nm to 1690 nm. Alternatively, the laser diode **110** in FIG. **20** may be a fixed-wavelength laser diode. For example, the laser diode **110** in FIG. **20** may be a distributed feedback (DFB) laser diode with a spectral linewidth of less than 1 MHz, and the output beam **120** may have any suitable substantially fixed wavelength between approximately 600 nm and approximately 2000 nm.

[0126] FIG. **21** illustrates an example laser diode **110** that produces seed light **122** that is amplified by a semiconductor optical amplifier (SOA) **124**. Instead of directly emitting an output beam **120** (e.g., as illustrated in FIG. **20**), the light from a laser diode **110** may first be amplified by an optical amplifier. The laser diode **110** in FIG. **21** acts as a seed laser that produces seed light **122** that is coupled into the input end of the waveguide **125** of the SOA **124**. The SOA waveguide **125** in FIG.

21 is indicated by the cross-hatched region within the SOA **124**. The SOA **124** amplifies the seed light as it propagates within the waveguide from the input end to the output end, and the output beam of light **120** is emitted from the output end of the SOA. The optical gain provided by the SOA may come from electrical current that is supplied to the SOA by an electronic driver (not illustrated in FIG. **21**). For example, an electronic driver **112** may supply substantially constant DC electrical current to the laser diode **110** and to the SOA **124**, and the resulting output beam **120** may have substantially constant optical power. The output beam **120** may be a free-space beam, or the output beam **120** may be coupled into an optical fiber or into a waveguide of a photonic integrated circuit (PIC).

[0127] The laser diode **110** and the SOA **124** in FIG. **21** may be fabricated or integrated together on the same chip so that seed light **122** from the laser diode is directly coupled into the waveguide **125** of the SOA. The waveguide **125** of the SOA **124** may be a tapered optical waveguide (as illustrated in FIG. **21**) with a width that increases along a lateral direction from the input end that receives the seed light **122** to the output end that emits the output beam **120**. A light source that includes a seed laser diode **110** that supplies seed light **122** that is amplified by a SOA **124** (as illustrated in FIG. **21**) may be referred to as a master-oscillator power-amplifier laser (MOPA laser). The seed laser diode **110** may be referred to as a master oscillator, and the SOA **124** may be referred to as a power amplifier. The MOPA laser in FIG. **21** may be part of a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the output beam of light **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**.

[0128] FIG. **22** illustrates an example laser diode **110** that produces seed light **122** that is amplified by a fiber-optic amplifier **126**. The laser diode **110** in FIG. **22** acts as a seed laser that produces seed light **122** that is coupled into an optical fiber **116**. The optical fiber **116** directs the seed light **122** to the fiber-optic amplifier **126**, and the fiber-optic amplifier amplifies the seed light as it propagates through an optical gain fiber of the fiber-optic amplifier. The optical gain fiber may be doped with rare-earth ions (e.g., neodymium, erbium, or ytterbium) that provide the optical gain to the seed light. One or more pump lasers may optically pump the rare-earth ions, which in turn provide optical amplification to the seed light **122** propagating through the gain fiber. The amplified seed light produced by the fiber-optic amplifier **126** propagates in an optical fiber as a fiber-coupled output beam **120**. The laser diode **110** and fiber-optic amplifier **126** in FIG. **22** may be part of a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the fiber-coupled output beam **120** may be directed to an optical combiner **130**, a sample **150**, or a detector **220**.

[0129] A pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}** may include a seed laser diode **110** followed by an optical amplifier. The seed laser diode **110** may produce seed light **122** that is amplified by the optical amplifier to produce an output beam of light **120**. An optical amplifier may include a SOA **124** (e.g., as illustrated in FIG. **21**) or a fiber-optic amplifier **126** (e.g., as illustrated in FIG. **22**). In some embodiments, an optical amplifier may include a SOA **124** followed by a fiber-optic amplifier **126**. For example, a seed laser diode **110** may produce seed light **122** that is first amplified by a SOA **124** and then further amplified by a fiber-optic amplifier **126**.

[0130] In some embodiments, a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}** may include a laser diode **110** and an optical fiber **116** and may not include a fiber-optic amplifier. For example, light produced by a laser diode **110** may be coupled into an optical fiber **116** to produce a fiber-coupled beam **120**, and the optical fiber may direct the laser-diode light to an optical combiner **130**, a sample **150**, or a detector **220**. A laser diode **110** that produces a fiber-coupled beam **120** may be referred to as a fiber-coupled laser diode.

[0131] FIG. **23** illustrates an example sampled-grating distributed Bragg reflector (SG-DBR) laser **110**. An SG-DBR laser **110** is a wavelength-tunable laser diode that produces an output beam **120** that can be tuned over a wavelength range having a width of between 20 nm and 50 nm. For

example, an SG-DBR laser **110** may have a 40-nm wavelength-tuning range from approximately 1530 nm to approximately 1570 nm or from approximately 1630 nm to approximately 1670 nm. The SG-DBR laser **110** in FIG. **23** may be part of a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the output beam **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**. For example, an SG-DBR laser **110** may be part of a probe light source **110_{pr}** that produces the probe beam **120_{pr}** in FIG. **6**, and the SG-DBR laser may tune the frequency $\nu_{\text{sub.3}}$ of the probe beam across at least a portion of the Raman signal **160**. An SG-DBR laser **110** may produce a free-space beam **120**, a fiber-coupled beam **120**, or a beam that is coupled into a waveguide of a PIC. Alternatively, an SG-DBR laser **110** may be integrated with a SOA **124** that amplifies the light produced by the SG-DBR laser (e.g., as illustrated in FIG. **21**).

[0132] The SG-DBR laser **110** in FIG. **23** includes a back mirror **180**, a phase section **182**, a gain section **184**, and a front mirror **186**, where the phase and gain sections are located between the front and back mirrors. The laser diode current I supplied to the SG-DBR laser **110** includes the following: current $I_{\text{sub.b}}$ supplied to the back mirror **180**, current $I_{\text{sub.p}}$ supplied to the phase section **182**, current $I_{\text{sub.g}}$ supplied to the gain section **184**, and current $I_{\text{sub.f}}$ supplied to the front mirror **186**. The gain current $I_{\text{sub.g}}$ provides optical gain to the optical waveguide **188** of the SG-DBR laser **110**, and the other currents may be used to set the wavelength of the output beam **120** produced by the SG-DBR laser. The electronic driver **112** may supply particular combinations of electrical currents to the back mirror **180**, phase section **182**, gain section **184**, and front mirror **186**, where each particular combination of electrical currents causes the SG-DBR laser **110** to produce an output beam **120** at a particular wavelength. For example, an SG-DBR laser **110** may be part of a wavelength-tunable probe light source **110_{pr}** that produces the probe beam **120_{pr}** in FIG. **11**, and the electronic driver **112** may supply particular different combinations of the electrical currents $I_{\text{sub.b}}$, $I_{\text{sub.p}}$, $I_{\text{sub.g}}$, and $I_{\text{sub.f}}$ to produce the different probe frequencies $\nu_{\text{sub.3-1}}$, $\nu_{\text{sub.3-2}}$, $\nu_{\text{sub.3-3}}$, . . . and $\nu_{\text{sub.3-n}}$ to tune the probe beam across at least a portion of the Raman signal **160** in FIG. **11**.

[0133] FIG. **24** illustrates an example light source **110** with multiple laser diodes **110** and an optical multiplexer **118** that combines light produced by the laser diodes into a single output beam of light **120**. Each of the laser diodes **110-1**, **110-2**, . . . **110-N** produces a respective output beam **120-1**, **120-2**, . . . **120-N**, and the optical multiplexer **118** combines the output beams into the output beam of light **120**. The light source **110** may be configured to switch between operating the N laser diodes one at a time so that, at any given time, only one laser diode produces light, and the output beam **120** includes just the light produced by that one laser diode.

[0134] The optical multiplexer **118** may be a free-space device, a fiber-optic device, a waveguide-based device, or a metamaterial-based device, and the multiplexer may combine N different wavelengths of light from the N laser diodes into a single output beam **120**. The optical multiplexer **118** may include one or more of the following: a free-space diffraction grating; an arrayed waveguide grating (AWG); a metamaterial that acts as a diffractive optical element; one or more optical filters; one or more optical combiners; one or more optical switches (e.g., thermo-optic switches, liquid crystal switches, or microelectromechanical systems (MEMS) switches); a series of two or more fiber Bragg gratings with optical circulators.

[0135] The light source **110** in FIG. **24** may be a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the output beam **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**. The output beam **120** may be a free-space beam, or the output beam **120** may be coupled into an optical fiber or into a waveguide of a photonic integrated circuit (PIC). For example, each of the laser diodes **110-1**, **110-2**, . . . **110-N** may produce a fiber-coupled beam **120-1**, **120-2**, . . . **120-N**, and the optical multiplexer **118** may be a fiber-optic device that produces a fiber-coupled output beam **120**. The output beam **120** may be sent to an optical combiner **130**, a sample **150**, or a detector **220**. Alternatively, the light source **110** may include an optical amplifier (e.g., an SOA or a fiber-optic amplifier) located after the optical multiplexer **118**,

and the output beam **120** may be coupled from the multiplexer to an optical amplifier that provides optical amplification to the output beam.

[0136] The light source **110** in FIG. **24** includes N laser diodes **110-1**, **110-2**, . . . **110-N**, where N is an integer greater than or equal to 2. Each of the N laser diodes **110** in FIG. **24** may be a wavelength-tunable laser diode or a fixed-wavelength laser diode. For example, the light source in FIG. **24** may include (i) N wavelength-tunable laser diodes, (ii) N fixed-wavelength laser diodes, or (iii) one or more wavelength-tunable laser diodes and one or more fixed-wavelength laser diodes. The light source **110** in FIG. **24** may be referred to as a wavelength-tunable light source, a frequency-tunable light source, or a tunable light source. A wavelength-tunable light source may include one or more continuously tunable laser diodes (e.g., SG-DBR laser diodes); multiple fixed-wavelength laser diodes (e.g., multiple DFB laser diodes), each laser diode operating at a different wavelength; or any combination thereof.

[0137] The light source **110** in FIG. **24** may be a pump light source **110_{pu}** or a Stokes light source **110_S** that includes N fixed-wavelength laser diodes, each laser diode having a different operating wavelength. The wavelength of the output beam **120** produced by the wavelength-tunable light source **110** in FIG. **24** may be adjustable to any wavelength of N different wavelengths by selecting one of the N fixed-wavelength laser diodes for operation. The frequency offset Ω between the pump beam **120_{pu}** and Stokes beam **120_S** may be adjustable by selecting one of the fixed-wavelength laser diodes for operation. For example, the Stokes beams **120_S** and **120_{S'}** in FIG. **14** may be produced by the light source **110** in FIG. **24**. Laser diode **110-1** may be a fixed-wavelength laser diode operating at the frequency $\nu_{\text{sub.2}}$, and laser diode **110-2** may be a fixed-wavelength laser diode operating at the frequency $\nu_{\text{sub.2}}$. Selecting laser diode **110-1** for operation produces the frequency offset $\Omega_{\text{sub.1}}$ in FIG. **14**, and selecting laser diode **110-2** for operation produces the frequency offset $\Omega_{\text{sub.2}}$.

[0138] The light source **110** in FIG. **24** may operate only one of the N laser diodes **110** at any given time. Each of the laser diodes **110** may operate at a particular wavelength or over a particular range of wavelengths, and one of the laser diodes may be selected for operation based on the wavelength that is needed to perform a particular measurement. For example, the pump beams **120_{pu}** and **120_{pu'}** in FIG. **15** may be produced by the light source **110** in FIG. **24**. Laser diode **110-1** may be a fixed-wavelength laser diode operating at the frequency $\nu_{\text{sub.1}}$, and laser diode **110-2** may be a fixed-wavelength laser diode operating at the frequency $\nu'_{\text{sub.1}}$. During a first measurement period, laser diode **110-1** may be operated to produce output beam **120-1** at the frequency $\nu_{\text{sub.1}}$, and the other laser diodes **110-2** to **110-N** may be turned off or otherwise configured to not produce light. The multiplexer **118** receives the output beam **120-1** from the laser diode **110-1** and directs it to the output of the multiplexer to produce the output beam **120** having a frequency $\nu_{\text{sub.1}}$. During a second measurement period, the Raman signal **160_c** in FIG. **15** may be measured, and laser diode **110-2** may be operated to produce output beam **120-2** at the frequency $\nu'_{\text{sub.1}}$. The other laser diodes (i.e., laser diodes **110-1** to **110-N**, excluding laser diode **110-2**) may be turned off or otherwise configured to not produce light. The multiplexer **118** receives the output beam **120-2** from the laser diode **110-2** and directs it to the output of the multiplexer to produce the output beam **120** having a frequency $\nu'_{\text{sub.1}}$.

[0139] The light source **110** in FIG. **24** may be a probe light source that includes N wavelength-tunable laser diodes. The probe light source may be configured to tune over one or more wavelength ranges having a total width between $p \cdot N \cdot \Delta\lambda_{\text{sub.av}}$ and $N \cdot \Delta\lambda_{\text{sub.av}}$, where $\Delta\lambda_{\text{sub.av}}$ is an average wavelength-tuning range of the N laser diodes **110**, and p is a wavelength-overlap parameter between 0.5 and 1. For example, if the overlap parameter p has a value of 0.7, then the combined wavelength-tuning range of the N wavelength-tunable laser diodes may be between $(0.7)N \cdot \Delta\lambda_{\text{sub.av}}$ and $N \cdot \Delta\lambda_{\text{sub.av}}$. The wavelength-overlap parameter p represents the amount of wavelength overlap between adjacent wavelength-tuning ranges (e.g., an overlap value p of 1 indicates that there is no wavelength overlap between the wavelength-

tuning ranges). For example, the light source **110** in FIG. **24** may be a probe light source that includes three SG-DBR laser diodes having respective wavelength-tuning ranges of 1490-1530 nm, 1520-1560 nm, and 1550-1590 nm. Each of the SG-DBR laser diodes has a 40-nm tuning range with a 10-nm overlap between adjacent tuning ranges, which results in the light source **110** in FIG. **24** having a 100-nm wavelength-tuning range from 1490 nm to 1590 nm. In this case, the average wavelength-tuning range $\Delta\lambda_{\text{sub.av}}$ of the three laser diodes is 40 nm, and the total wavelength-tuning range has a width of 100 nm, which corresponds to the wavelength-overlap parameter p having a value of approximately 0.83. The probe light source **110** may produce an output beam **120** having any wavelength from 1490 nm to 1590 nm by selecting one of the three SG-DBR laser diodes for operation and tuning the laser diode to the desired wavelength. As another example, the light source **110** in FIG. **24** may be a probe light source **110** that includes three SG-DBR laser diodes having respective wavelength-tuning ranges of 1490-1530 nm, 1530-1560, and 1640-1680 nm. Each of the SG-DBR laser diodes has a 40-nm tuning range with no wavelength overlap between adjacent tuning ranges. The average wavelength-tuning range $\Delta\lambda_{\text{sub.av}}$ of the three laser diodes is 40 nm, and the total wavelength-tuning range has a width of 120 nm. This corresponds to the wavelength-overlap parameter p having a value of 1, indicating that there is no wavelength overlap between the tuning ranges of the three laser diodes.

[0140] FIG. **25** illustrates an example pump laser **110_{pu}** and Stokes laser **110_S** with a fiber-optic combiner **130** that produces a combined pump-Stokes beam **140** coupled into an optical fiber **116**. A Raman spectroscopy system **100** may include one or more optical elements that direct the pump and Stokes beams to a sample **150**. The optical elements may include a combiner **130** that combines the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140** that is directed to a sample **150**. In FIG. **1**, the combiner **130_a** may be a free-space optical combiner, and the pump beam **120_{pu}**, Stokes beam **120_S**, and combined beam **140** may each be free-space beams. The combiner **130** in FIG. **25** is a fiber-optic combiner that receives the pump and Stokes beams via two input optical fibers **116** and combines the two beams into a combined pump-Stokes beam **140** that propagates in an output optical fiber **116**. The pump laser **110_{pu}** may be a fiber-coupled laser diode that produces a pump beam **120_{pu}** that is directed to the fiber-optic combiner **130** via an input optical fiber **116**. Similarly, the Stokes laser **110_S** may be a fiber-coupled laser diode that produces a Stokes beam **120_S** that is directed to the fiber-optic combiner **130** via another input optical fiber **116**. The pump laser **110_{pu}** or the Stokes laser **110_S** may be followed by an optical amplifier (not illustrated in FIG. **25**) that amplifies the pump beam **120_{pu}** or Stokes beam **120_S** prior to directing the light to the combiner **130**. After the fiber-optic combiner **130** combines the pump and Stokes beams, the output optical fiber **116** may direct the combined pump-Stokes beam **140** to a sample **150**.

[0141] The fiber-optic combiner **130** in FIG. **25** may include a fiber-optic wavelength division multiplexer (WDM) with two input optical fibers (for the pump and Stokes beam) and one output optical fiber for the output beam **120**. The WDM may include a dichroic beamsplitter or a fused fiber coupler. Each of the input or output optical fiber **116** may be a single-mode optical fiber or a multi-mode optical fiber.

[0142] In some embodiments, instead of using a single pump laser or a single Stokes laser (as illustrated in FIG. **25**), a Raman spectroscopy system may use a pump or Stokes light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. **24**). For example, the light source **110** in FIG. **24** may be a pump light source that produces a fiber-coupled output beam **120** that is coupled to an input fiber of the fiber-optic combiner **130** in FIG. **25**. One of the laser diodes of the pump light source may be selected for operation, and the light from the selected laser diode may be directed by the multiplexer **118** to an optical fiber that is coupled to the combiner **130**. Additionally or alternatively, the light source **110** in FIG. **24** may be a Stokes light source that produces a fiber-coupled output beam **120** that is coupled to an input fiber of the fiber-optic combiner **130** in FIG. **25**.

[0143] FIG. 26 illustrates an example laser diode **110** coupled to a waveguide **172** of a photonic integrated circuit (PIC) **170**. A PIC **170** (which may be referred to as a planar lightwave circuit (PLC), a waveguide-based device, an integrated-optic device, an integrated optoelectronic device, or a silicon optical bench) may be fabricated from a substrate that includes silicon, indium phosphide, glass (e.g., silica), a polymer, or an electro-optic material (e.g., lithium niobate (LiNbO₃) or lithium tantalate (LiTaO₃)). A PIC **170** may include one or more optical waveguides **172** that confine and guide a beam of light. An optical waveguide **172** that is part of a PIC **170** may be a passive optical waveguide formed in the PIC, and the waveguide may convey light from one optical element to another with relatively low optical loss.

[0144] In FIG. 26, light from the laser diode **110** is coupled into the PIC waveguide **172** to produce a waveguide-coupled beam **120**. The waveguide **172** may convey the beam of light **120** from the laser diode **110** to another optical element (e.g., an optical combiner **130**, a sample **150**, or a detector **220**). Light from the laser diode **110** in FIG. 26 may be coupled into the waveguide **172** using one or more lenses, or the laser diode **110** may be butt-coupled to an input of the waveguide so that the light from the laser diode is directly coupled into the waveguide. The laser diode **110** may be mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached. For example, the laser diode **110** may be attached using epoxy, adhesive, or solder. Alternatively, the laser diode **110** in FIG. 26 may be located apart from the PIC **170**, and the laser diode may send a beam of light **120** to the PIC **170** via optical fiber. An output end of the optical fiber may be attached or connected to the PIC **170** so that the light is coupled into the PIC waveguide **172**. Light from the laser diode **110** may be amplified by an optical amplifier (not illustrated in FIG. 26) prior to being coupled into the PIC waveguide **172**. For example, the laser diode **110** in FIG. 26 may be a MOPA laser similar to that illustrated in FIG. 21, and the output beam **120** produced by the MOPA laser may be directly coupled into the waveguide **172**.

[0145] FIG. 27 illustrates an example pump laser **110_{pu}** and Stokes laser **110_S** with a photonic integrated circuit (PIC) **170** that produces a combined pump-Stokes beam **140** coupled into an optical waveguide **172** of the PIC. The PIC **170** includes a waveguide combiner **130** and three optical waveguides **172** (two input waveguides for the pump and Stokes beams and one output waveguide for the combined pump-Stokes beam **140**). The waveguide combiner **130** is a waveguide-based optical combiner that combines the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140**. The combined pump-Stokes beam **140** is coupled to an output optical waveguide **172** of the PIC **170**, and the output waveguide may direct the beam to a sample **150**. The pump laser **110_{pu}** or the Stokes laser **110_S** may be a laser diode that is mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached. Alternatively, the pump laser **110_{pu}** or the Stokes laser **110_S** may be a fiber-coupled laser diode that sends a beam of light to the PIC via optical fiber (e.g., an output end of the optical fiber may be attached to the PIC so that the light is coupled into an input optical waveguide **172**).

[0146] In some embodiments, instead of using a single pump laser or a single Stokes laser (as illustrated in FIG. 27), a Raman spectroscopy system may use a pump or Stokes light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. 24). For example, the light source **110** in FIG. 24 may be a pump or Stokes light source that produces a fiber-coupled output beam **120** that is coupled to an input optical waveguide **172** of the PIC **170** in FIG. 27. Alternatively, the multiplexer **118** in FIG. 24 may be a waveguide-based device and the output beam **120** may propagate in a PIC waveguide that directs the light to an input optical waveguide **172** of the PIC **170** in FIG. 27.

[0147] FIG. 28 illustrates an example fiber-optic combiner **130** that combines a Raman signal **160** with a probe beam **120_{pr}**. An optical receiver **200** may include an optical combiner **130** that combines a Raman signal **160** and a probe beam of light **120_{pr}** to produce one or more combined probe-Raman signals **210** that are each directed to a detector **220**. In FIG. 1, the combiner **130_b** may be a free-space optical combiner, and the Raman signal **160**, probe beam **120_{pr}**, and the

combined probe-Raman signal **210** may each be free-space beams. The combiner **130** in FIG. **28** is a fiber-optic combiner that receives the Raman signal **116** and the probe beam **120_{pr}** via two input optical fibers **116** and combines the two beams into a combined probe-Raman signal **210** that is directed to a detector **220** via an output optical fiber **116**. Each of the input or output optical fiber **116** in FIG. **28** may be a single-mode optical fiber or a multi-mode optical fiber.

[0148] The Raman signal **160** in FIG. **28** may be a free-space beam that is coupled into an input optical fiber **116** using one or more lenses. The probe laser **110_{pr}** may be a fiber-coupled laser diode that directs the probe beam **120_{pr}** to the fiber-optic combiner **130** via optical fiber **116**. The probe beam **120_{pr}** may be amplified by an optical amplifier (not illustrated in FIG. **28**) prior to being directed to the combiner **130**. In some embodiments, instead of using a single probe laser **110_{pr}** (as illustrated in FIG. **28**), an optical receiver **200** may use a light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. **24**). For example, the light source **110** in FIG. **24** may be a probe light source that produces a fiber-coupled output beam **120** that is coupled to an input optical fiber **116** of the fiber-optic combiner **130** in FIG. **28**.

[0149] FIG. **29** illustrates an example photonic integrated circuit (PIC) **170** with a waveguide combiner **130** that combines a Raman signal **160** with a probe beam **120_{pr}**. An optical combiner **130** may be a waveguide combiner **130** that is part of a PIC **170** and may combine a Raman signal **160** and a probe beam of light **120_{pr}** to produce one or more combined probe-Raman signals **210** that are each directed to a detector **220**. The waveguide combiner **130** in FIG. **29** receives the Raman signal **160** and the probe beam **120_{pr}** via two input waveguides **172**. A waveguide combiner **130** may produce 1, 2, or 4 combined output beams **210**. The waveguide combiner **130** in FIG. **29** combines the Raman signal **160** and the probe beam **120_{pr}** to produce two combined probe-Raman signals **210_a** and **210_b** which are each directed to a respective detector **220_a** and **220_b** via two output waveguides **172**. The PIC **170** in FIG. **29** may be part of an optical receiver **200**.

[0150] The Raman signal **160** in FIG. **29** may be a free-space beam that is coupled into an input waveguide **172** using one or more lenses. The probe laser **110_{pr}** may be a laser diode that is mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached. Alternatively, the probe laser **110_{pr}** may be a fiber-coupled laser diode that sends the probe beam **120_{pr}** to the PIC **170** via optical fiber (e.g., an output end of the optical fiber may be attached to the PIC so that the light is coupled into an input optical waveguide **172**). The probe beam **120_{pr}** may be amplified by an optical amplifier (not illustrated in FIG. **29**) prior to being coupled into an input optical waveguide **172**. In some embodiments, instead of using a single probe laser **110_{pr}** (as illustrated in FIG. **29**), an optical receiver **200** may use a light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. **24**), and the probe beam **120_{pr}** may be delivered from the multiplexer to the PIC **170** via optical fiber or via a waveguide **172** of the PIC.

[0151] A Raman spectroscopy system **100** may include one or more optical elements that (i) direct a pump beam **120_{pu}** and a Stokes beam **120_S** to a sample **150** and (ii) direct a Raman signal **160** and a probe beam **120_{pr}** to one or more detectors **220**. The optical elements may include one or more PICs **170** that each include one or more optical waveguides **172**. One or more of the optical waveguides **172** may direct the pump beam **120** and the Stokes beam **120_s** to the sample **150**. For example, a PIC **170** may include an optical combiner **130** that produces a combined pump-Stokes beam **140** that is directed to the sample by an optical waveguide **172** of the PIC **170**. One or more other optical waveguides **172** may direct the Raman signal **160** and the probe beam **120_{pr}** to one or more detectors **220**. For example, a PIC **170** may include an optical combiner **130** that combines the Raman signal **160** and the probe beam **120_{pr}** to produce one or more combined probe-Raman signals **210** that are each directed to a detector **220** by an optical waveguide **172** of the PIC **170**.

[0152] FIGS. **30-35** each illustrates example frequency ranges of a pump beam **120_{pu}** and a Stokes beam **120_S**. The pump beam **120_{pu}** may be produced by a pump light source **110_{pu}**, and the Stokes

beam **120S** may be produced by a Stokes light source **110S**. The pump and Stokes light sources may each include one or more fixed wavelength laser diodes or one or more wavelength-tunable laser diodes. In each of FIGS. **30-35**, the pump beam **120pu** and the Stokes beam **120S** each have one or more fixed frequencies or one or more frequencies that are adjustable over a particular frequency range. The corresponding frequency offset Ω between the pump and Stokes beams is indicated as a range of frequencies or a set of discrete frequencies that the frequency offset can be set to, based on the frequencies that are available to the pump and Stokes beams. The frequency offset Ω is determined from $v_{\text{sub.1}} - v_{\text{sub.2}}$, where $v_{\text{sub.1}}$ is the range or set of fixed frequencies for the pump beam **120pu**, and $v_{\text{sub.2}}$ is the range or set of fixed frequencies for the Stokes beam **120S**. The frequency offsets $\Omega_{\text{sub.1}}$, $\Omega_{\text{sub.2}}$, $\Omega_{\text{sub.3}}$, and $\Omega_{\text{sub.4}}$ in FIGS. **30-35** may have any suitable value between approximately 5 THz and approximately 100 THz. The frequency range $\Delta\Omega$ over which a frequency offset may be varied may have any suitable value between approximately 5 THz and approximately 80 THz.

[0153] In FIG. **30**, the pump beam **120pu** has a single fixed frequency $v_{\text{sub.1}}$, and the Stokes beam **120S** has a frequency $v_{\text{sub.2}}$ that is adjustable over a frequency range of width $\Delta v_{\text{sub.2}}$. The adjustable frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S** extends from a low frequency $v_{\text{sub.2L}}$ to a high frequency $v_{\text{sub.2H}}$, where $\Delta v_{\text{sub.2}} = v_{\text{sub.2H}} - v_{\text{sub.2L}}$. The pump laser **110pu** that produces the pump beam **120pu** in FIG. **30** may be a fixed-wavelength laser diode, and the Stokes laser **110S** that produces the Stokes beam **120S** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. In FIG. **30**, the frequency range $\Delta\Omega$ is also equal to the frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S**. When the Stokes beam **120S** is set to the lower frequency $v_{\text{sub.2L}}$, the frequency offset between the pump and Stokes beams is $v_{\text{sub.1}} - v_{\text{sub.2L}}$, which is equal to $\Omega_{\text{sub.2}}$. Similarly, when the Stokes beam **120S** is set to the upper frequency $v_{\text{sub.2H}}$, the frequency offset between the pump and Stokes beams is $v_{\text{sub.1}} - v_{\text{sub.2H}}$, which is equal to $\Omega_{\text{sub.1}}$.

[0154] For example, the pump beam **120pu** in FIG. **30** may have a frequency $v_{\text{sub.1}}$ of 250 THz (corresponding to a wavelength of approximately 1200 nm), and the Stokes beam **120S** may be adjustable from a low frequency $v_{\text{sub.2L}}$ of 195 THz to a high frequency $v_{\text{sub.2H}}$ of 200 THz. This corresponds to a frequency-tuning range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S** of 5 THz (167 cm^{-1} in wavenumbers) and a 38-nm wavelength-tuning range from approximately 1499 nm to approximately 1537 nm. The resulting frequency offset Ω between the pump and Stokes beams may be set to a value between the lower value $\Omega_{\text{sub.1}}$ of 50 THz (1668 cm^{-1} in wavenumbers) and the upper value $\Omega_{\text{sub.2}}$ of 55 THz (1835 cm^{-1} in wavenumbers), corresponding to a frequency range $\Delta\Omega$ of 5 THz (167 cm^{-1} in wavenumbers).

[0155] In FIG. **31**, the pump beam **120pu** has a single fixed frequency $v_{\text{sub.1}}$, and the Stokes beam **120S** has a frequency $v_{\text{sub.2}}$ that is adjustable over a frequency range of width $\Delta v_{\text{sub.2}}$. The adjustable frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S** extends from a low frequency $v_{\text{sub.2L}}$ to a high frequency $v_{\text{sub.2H}}$, where $\Delta v_{\text{sub.2}} = v_{\text{sub.2H}} - v_{\text{sub.2L}}$. The pump laser **110pu** that produces the pump beam **120pu** in FIG. **31** may be a fixed-wavelength laser diode. The Stokes light source **110S** that produces the Stokes beam **120S** may include two wavelength-tunable laser diodes. For example, the Stokes light source **110S** may be similar to the light source in FIG. **24** where two wavelength-tunable laser diodes are combined by a multiplexer **118**. A first wavelength-tunable laser diode may operate from frequency $v_{\text{sub.2L}}$ to frequency $v_{\text{sub.2M}}$, and a second wavelength-tunable laser diode may operate from frequency $v_{\text{sub.2M}}$ to frequency $v_{\text{sub.2H}}$. The total tuning range $\Delta v_{\text{sub.2}}$ of the Stokes laser **110S** equals the sum of the tuning ranges $\Delta v_{\text{sub.2a}}$ and $\Delta v_{\text{sub.2b}}$ of the two wavelength-tunable laser diodes. In other embodiments, if the tuning ranges of the two wavelength-tunable laser diodes overlap, the total tuning range $\Delta v_{\text{sub.2}}$ of the Stokes light source **110S** will be reduced by the amount of frequency overlap between the two lasers.

[0156] In FIG. 31, the frequency offset Ω between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. When the first wavelength-tunable laser diode (with a frequency range from $\nu_{\text{sub.2L}}$ to $\nu_{\text{sub.2M}}$) is selected to operate, the frequency offset Ω may be set to any value between $\Omega_{\text{sub.M}}$ and $\Omega_{\text{sub.2}}$. For example, when the Stokes beam **120S** is set to the lower frequency $\nu_{\text{sub.2L}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1}} - \nu_{\text{sub.2L}}$, which is equal to $\Omega_{\text{sub.2}}$. When the second wavelength-tunable laser diode (with a frequency range from $\nu_{\text{sub.2M}}$ to $\nu_{\text{sub.2H}}$) is selected to operate, the frequency offset Ω may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.M}}$. For example, when the Stokes beam **120S** is set to the upper frequency $\nu_{\text{sub.2H}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1}} - \nu_{\text{sub.2H}}$, which is equal to $\Omega_{\text{sub.1}}$. The total frequency range $\Delta\Omega$ of the frequency offset Ω is equal to the sum of the two frequency ranges $\Delta\Omega_{\text{sub.2a}}$ and $\Delta\Omega_{\text{sub.2b}}$. The frequency range $\Delta\Omega$ is also equal to the overall frequency range $\Delta\nu_{\text{sub.2}}$ of the Stokes beam **120S**.

[0157] In FIG. 32, the pump beam **120pu** can be set to two fixed frequencies $\nu_{\text{sub.1a}}$ and $\nu_{\text{sub.1b}}$, and the Stokes beam **120S** has a frequency $\nu_{\text{sub.2}}$ that is adjustable over a frequency range of width $\Delta\nu_{\text{sub.2}}$. The adjustable frequency range $\Delta\nu_{\text{sub.2}}$ of the Stokes beam **120S** extends from a low frequency $\nu_{\text{sub.2L}}$ to a high frequency $\nu_{\text{sub.2H}}$, where $\Delta\nu_{\text{sub.2}} = \nu_{\text{sub.2H}} - \nu_{\text{sub.2L}}$. The pump light source **110pu** that produces the two pump beams **120pu-a** and **120pu-b** may include two fixed-wavelength laser diodes. For example, the pump light source **110pu** may be similar to the light source in FIG. 24 where two fixed-wavelength laser diodes are combined by a multiplexer **118**. The Stokes laser **110S** that produces the Stokes beam **120S** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. When the pump laser **110pu** produces the pump beam **120pu-a** at frequency $\nu_{\text{sub.1a}}$, the Stokes beam **120S** may be tuned to a frequency between $\nu_{\text{sub.2L}}$ and $\nu_{\text{sub.2H}}$ to produce a frequency offset Ω between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.M}}$. For example, with the Stokes beam **120S** set to the upper frequency $\nu_{\text{sub.2H}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1a}} - \nu_{\text{sub.2H}}$, which is equal to $\Omega_{\text{sub.1}}$. When the pump laser **110pu** produces the pump beam **120pu-b** at frequency $\nu_{\text{sub.1b}}$, the Stokes beam **120S** may be tuned to a frequency between $\nu_{\text{sub.2L}}$ and $\nu_{\text{sub.2H}}$ to produce a frequency offset Ω between $\Omega_{\text{sub.M}}$ and $\Omega_{\text{sub.2}}$. For example, with the Stokes beam **120S** set to the lower frequency $\nu_{\text{sub.2L}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1b}} - \nu_{\text{sub.2L}}$, which is equal to $\Omega_{\text{sub.2}}$.

[0158] FIG. 33 is similar to FIG. 30, except in FIG. 33, the Stokes beam **120S** has a fixed frequency $\nu_{\text{sub.2}}$ and the frequency of the pump beam **120pu** is adjustable. The adjustable frequency range $\Delta\nu_{\text{sub.1}}$ of the pump beam **120pu** extends from a low frequency $\nu_{\text{sub.1L}}$ to a high frequency $\nu_{\text{sub.1H}}$, where $\Delta\nu_{\text{sub.1}} = \nu_{\text{sub.1H}} - \nu_{\text{sub.1L}}$. The Stokes laser **110S** that produces the Stokes beam **120pu** in FIG. 33 may be a fixed-wavelength laser diode, and the pump laser **110pu** that produces the pump beam **120pu** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. In FIG. 33, the frequency range $\Delta\Omega$ is also equal to the frequency range $\Delta\nu_{\text{sub.1}}$ of the pump beam **120pu**. When the pump beam **120pu** is set to the lower frequency $\nu_{\text{sub.1L}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1L}} - \nu_{\text{sub.2}}$, which is equal to $\Omega_{\text{sub.1}}$. Similarly, when the pump beam **120pu** is set to the upper frequency $\nu_{\text{sub.1H}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1H}} - \nu_{\text{sub.2}}$, which is equal to $\Omega_{\text{sub.2}}$.

[0159] In FIG. 34, both the pump beam **120pu** and the Stokes beam **120S** have adjustable frequencies. The pump laser **110pu** that produces the pump beam **120pu** and the Stokes laser **110S** that produces the Stokes beam **120S** may each include a wavelength-tunable laser diode. The adjustable frequency range $\Delta\nu_{\text{sub.1}}$ of the pump beam **120pu** extends from a low frequency $\nu_{\text{sub.1L}}$ to a high frequency $\nu_{\text{sub.1H}}$, where $\Delta\nu_{\text{sub.1}} = \nu_{\text{sub.1H}} - \nu_{\text{sub.1L}}$. The adjustable

frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S** extends from a low frequency $v_{\text{sub.2L}}$ to a high frequency $v_{\text{sub.2H}}$, where $\Delta v_{\text{sub.2}} = v_{\text{sub.2H}} - v_{\text{sub.2L}}$. The frequency offset Ω between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. When the Stokes beam **120S** is set to the lower Stokes frequency $v_{\text{sub.2L}}$ and the pump beam **120pu** is set to the upper pump frequency $v_{\text{sub.1H}}$, the frequency offset between the pump and Stokes beams is $v_{\text{sub.1H}} - v_{\text{sub.2L}}$, which is equal to $\Omega_{\text{sub.2}}$. When the Stokes beam **120S** is set to the upper Stokes frequency $v_{\text{sub.2H}}$ and the pump beam **120pu** is set to the lower pump frequency $v_{\text{sub.1L}}$, the frequency offset between the pump and Stokes beams is $v_{\text{sub.1L}} - v_{\text{sub.2H}}$, which is equal to $\Omega_{\text{sub.1}}$.

[0160] In FIG. 35, both the pump beam **120pu** and the Stokes beam **120S** can be set to two different fixed frequencies. A pump light source **110pu** and a Stokes light source **110S** may each include two or more fixed wavelength laser diodes, and each light source may be similar to the light source in FIG. 24 where multiple fixed-wavelength laser diodes are combined by a multiplexer **118**. In FIG. 35, the pump light source **110pu** that produces the two pump beams **120pu-a** and **120pu-b** may include two fixed-wavelength laser diodes, and the Stokes light source **110S** that produces the two Stokes beams **120S-a** and **120S-b** may include two fixed-wavelength laser diodes. The frequencies of the pump and Stokes beams may be selected to produce four different frequency offsets $\Omega_{\text{sub.1}}$, $\Omega_{\text{sub.2}}$, $\Omega_{\text{sub.3}}$, and $\Omega_{\text{sub.4}}$. For example, selecting Stokes beam **120S-b** at frequency $v_{\text{sub.2b}}$ and pump beam **120pu-a** at frequency $v_{\text{sub.1a}}$ produces a frequency offset $\Omega_{\text{sub.1}}$, which is equal to $v_{\text{sub.1a}} - v_{\text{sub.2b}}$. As another example, selecting Stokes beam **120S-a** at frequency $v_{\text{sub.2a}}$ and pump beam **120pu-b** at frequency $v_{\text{sub.1b}}$ produces a frequency offset $\Omega_{\text{sub.4}}$, which is equal to $v_{\text{sub.1b}} - v_{\text{sub.2a}}$. The frequency offset $\Omega_{\text{sub.2}}$ may be produced by selecting Stokes beam **120S-a** at frequency $v_{\text{sub.2a}}$ and pump beam **120pu-a** at frequency $v_{\text{sub.1a}}$, and the frequency offset $\Omega_{\text{sub.3}}$ may be produced by selecting Stokes beam **120S-b** at frequency $v_{\text{sub.2b}}$ and pump beam **120pu-b** at frequency $v_{\text{sub.1b}}$.

[0161] FIG. 36 illustrates an example optical receiver **200** with two detectors **220a** and **220b**. The optical receivers **200** in FIGS. 29 and 36 are similar, except one difference is that the optical receiver in FIG. 29 is a waveguide-based optical receiver, while the optical receiver in FIG. 36 is a free-space optical receiver **200**. The optical combiner **130** in FIG. 36 may be a 50/50 free-space beamsplitter that reflects approximately 50% of an incident beam of light and transmits approximately 50% of the beam. The optical combiner **130** splits the Raman signal **160** and the probe beam **120pr** into two beams to produce two combined probe-Raman signals **210a** and **210b**. The combined probe-Raman signal **210a** is directed to detector **220a** and includes a transmitted portion of the probe beam **120pr** and a reflected portion of the Raman signal **160** (e.g., approximately 50% of the probe beam and approximately 50% of the Raman signal). Similarly, the combined probe-Raman signal **210b** is directed to detector **220b** and includes a reflected portion of the probe beam **120pr** and a transmitted portion of the Raman signal **160**. The portions of the probe beam **120pr** and the Raman signal **160** that make up the combined probe-Raman signal **210a** may be coherently mixed at detector **220a** to produce the photocurrent signal $i_{\text{sub.a}}$. Similarly, the portions of the probe beam **120pr** and the Raman signal **160** that make up the combined probe-Raman signal **210b** may be coherently mixed at detector **220b** to produce the photocurrent signal $i_{\text{sub.b}}$.

[0162] The two detectors **220a** and **220b** are arranged so that their respective photocurrents $i_{\text{sub.a}}$ and $i_{\text{sub.b}}$ are subtracted. The anode of detector **220a** is electrically connected to the cathode of detector **220b**, and the subtracted photocurrent signal $i_{\text{sub.a}} - i_{\text{sub.b}}$ from the anode-cathode connection is sent to the detection electronics **230**, which produces a digital output signal **240** that corresponds to the subtracted photocurrent signal. The subtracted photocurrent signal may be expressed as $i_{\text{sub.a}} - i_{\text{sub.b}} = 2E_{\text{sub.R}}E_{\text{sub.pr}} \cos[2\pi(v_{\text{sub.R}} - v_{\text{sub.3}})t + \Delta\phi]$, which corresponds to the coherent-mixing term discussed herein. The subtracted photocurrent signal does not include the terms $E_{\text{sub.R}}^2$ and $E_{\text{sub.pr}}^2$ corresponding to the respective optical powers of the

Raman signal **160** and the probe beam **120pr**. By subtracting the two photocurrents $i_{\text{sub.a}}$ and $i_{\text{sub.b}}$, the common-mode terms $E_{\text{sub.R.sup.2}}$ and $E_{\text{sub.pr.sup.2}}$ (as well as common-mode noise) that appear in each of the photocurrent signals $i_{\text{sub.a}}$ and $i_{\text{sub.b}}$ are substantially removed, leaving the coherent-mixing term, which is the quantity of interest. Since subtraction may remove common-mode noise, the subtracted photocurrent signal $i_{\text{sub.a}} - i_{\text{sub.b}}$ may have a reduced noise compared to each of the photocurrent signals $i_{\text{sub.a}}$ and $i_{\text{sub.b}}$ alone. The dual-detector arrangement in FIG. **36** in which the photocurrents are subtracted may be referred to as a balanced optical detector. A balanced detector may be implemented as a free-space device, a fiber-optic-based device, or a waveguide-based device.

[0163] FIG. **37** illustrates an example optical receiver **200** configured for polarization-sensitive detection of a Raman signal **160**. A polarization-sensitive optical receiver **200** may be used to determine the polarization of a Raman signal **160**. The polarization of a Raman signal **160** may be determined by a processor based on one or more digital output signals **240** produced by the polarization-sensitive optical receiver **200**. Determining the polarization of a Raman signal **160** may include determining a relative size or ratio of two orthogonal polarization components of the Raman signal (e.g., horizontal and vertical polarization components of the Raman signal). For example, if the ratio of the horizontal and vertical polarization components of a Raman signal **160** is 100:1, then the Raman signal may be determined to be substantially horizontally polarized. As another example, if the ratio of the horizontal and vertical polarization components of a Raman signal **160** is 1:1, then the horizontal and vertical polarization components of the Raman signal may be determined to be approximately equal (e.g., the Raman signal may be circularly polarized or linearly polarized at a 45-degree angle to the horizontal and vertical directions).

[0164] A polarization-sensitive optical receiver **200** may include a polarization beamsplitter (PBS) **135** that splits an input beam into two output beams, where one output beam is horizontally polarized, and the other output beam is vertically polarized. The horizontally polarized output beam includes the horizontal polarization component of the input beam, and the vertically polarized output beam includes the vertical polarization component of the input beam. The Raman signal **160** in FIG. **37** is directed to a Raman-signal PBS **135R** that splits the Raman signal into a horizontal-polarization Raman signal **160-h** and a vertical-polarization Raman signal **160-v**. Similarly, the probe beam **120pr** is directed to a probe-beam PBS **135pr** that splits the probe beam into a horizontal-polarization probe beam **120pr-h** and a vertical-polarization probe beam **120pr-v**.

[0165] A polarization-sensitive optical receiver **200** may include a waveplate **132** that changes the polarization of the probe beam **120pr** so that the probe beam is split into two polarization components. The two polarization components may each have approximately one-half the power of the probe beam **120pr**. The waveplate **132c** in FIG. **37** may be (i) a half-wave plate that rotates the polarization of the probe beam **120pr** or (ii) a quarter-wave plate that converts the probe beam **120pr** to a circular or elliptical polarization. For example, the probe laser **110pr** may produce a probe beam **120pr** that is vertically polarized, and the waveplate **132c** may be a half-wave plate that rotates the probe-beam polarization by 45 degrees so that the horizontal-polarization and vertical-polarization probe beams **120pr-h** and **120pr-v** each have approximately equal optical powers.

[0166] The optical combiner **130h** in FIG. **37** combines the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** to produce a horizontal probe-Raman signal **210h** that is directed to a horizontal-polarization optical receiver **200h**. Similarly, the optical combiner **130v** combines the vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** to produce a vertical probe-Raman signal **210v** that is directed to a vertical-polarization optical receiver **200v**. The horizontal-polarization optical receiver **200h** may include one or more optical detectors **200**, where each detector is configured to coherently mix at least a portion of the horizontal-polarization Raman signal **160-h** and at least a portion of the horizontal-polarization probe beam **120pr-h** to produce a horizontal-polarization electronic signal. Similarly,

the vertical-polarization optical receiver **200v** may include one or more optical detectors **200**, where each detector is configured to coherently mix at least a portion of the vertical-polarization Raman signal **160-v** and at least a portion of the vertical-polarization probe beam **120pr-v** to produce a vertical-polarization electronic signal. The horizontal-polarization optical receiver **200h** and the vertical-polarization optical receiver **200v** may each include: (i) a single detector **220** (e.g., similar to the optical receiver **200** in FIG. 2), (ii) two detectors **220** (e.g., similar to the balanced optical detector arrangement in FIG. 36), or (iii) four detectors **220** (e.g., similar to the arrangement in FIG. 39). The electronic signals may include a photocurrent signal *i*, and each optical receiver may include an electronic amplifier **232** that produces a corresponding voltage signal **234**. The h-polarization optical receiver **200h** may produce a digital output signal **240-h** corresponding to the horizontal-polarization electronic signal, and the v-polarization optical receiver **200v** may produce a digital output signal **240-v** corresponding to the vertical-polarization electronic signal.

[0167] A processor may determine one or more characteristics of the horizontal-polarization and vertical-polarization electronic signals based on the digital output signals **240-h** and **240-v**. Additionally, a processor may determine a polarization of the Raman signal **160** based on the characteristics of the horizontal-polarization and vertical-polarization electronic signals. For example, the characteristics of the electronic signals may include an amplitude or an area associated with the electronic signals, and the polarization of the Raman signal **160** may be expressed as a relative size or ratio of the amplitudes or areas associated with the horizontal and vertical polarization components of the Raman signal. If the horizontal digital output signal **240-h** includes an amplitude characteristic with value 100 and the vertical digital output signal **240-v** includes a corresponding amplitude characteristic with value 1, then the Raman signal **160** may be determined to be substantially horizontally polarized. If the horizontal and vertical digital output signals each include amplitude characteristics having approximately equal values, then the Raman signal **160** may be determined to have approximately equal horizontal and vertical polarization components.

[0168] A polarization-sensitive optical receiver **200** as illustrated in FIG. 37 may be implemented with free-space optical elements, fiber-optic components, waveguide-based optical elements, a metamaterial-based device, or any suitable combination thereof. For example, the two PBSs **135** in FIG. 37 may be free-space polarization beamsplitter cubes, and the Raman signal **160** and the probe beam **120pr** may be free-space optical beams. Alternatively, the two PBSs **135** may be fiber-optic components, and the Raman signal **160** and the probe beam **120pr** may be conveyed to the PBSs **135** via optical fiber (e.g., single-mode optical fiber or polarization-maintaining optical fiber). Additionally, the horizontally and vertically polarized probe-Raman signals **210h** and **210v** may be conveyed to the respective h-polarization and v-polarization optical receivers via polarization-maintaining optical fiber. The h-polarization and v-polarization optical receivers may each preserve the polarization of the respective horizontally and vertically polarized probe-Raman signals. For example, the h-polarization and v-polarization optical receivers may each include polarization-maintaining optical fiber that maintains the polarization of the beams. Alternatively, the h-polarization and v-polarization receivers may each include a PIC with optical waveguides configured to maintain the polarization of the beams.

[0169] FIG. 38 illustrates an example optical receiver **200** configured to detect in-phase and quadrature components of a Raman signal **160**. The optical receiver **200** includes a 90-degree optical hybrid **250** and four detectors **220I+**, **220I-**, **220Q+**, and **220Q-**. A 90-degree optical hybrid **250** is an optical-combiner component with two input ports and four output ports. Input light received at each of the two input ports is split, combined, and directed to each of the four output ports, and a 90-degree phase shift is imparted to one of the split beams before the Raman signal **160** and probe beam **120pr** are combined. The 90-degree optical hybrid **250** in FIG. 38 combines a Raman signal **160** and a probe beam **120pr** to produce four combined output beams: two in-phase combined beams **210I+** and **210I-**, and two quadrature combined beams **210Q+** and **210Q-**. Each

of the four combined beams **210** may include a portion of the Raman signal **160** and a portion of the probe beam **120_{pr}**, and each of the combined beams is directed to one of the four detectors of the optical receiver **200**. In FIG. **38**, each of the four detectors produces a photocurrent signal that corresponds to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120_{pr}**.

[0170] A 90-degree optical hybrid **250** may be configured so that the combined beams directed to each of the output ports have approximately the same optical power or energy. For example, the 90-degree optical hybrid **250** in FIG. **38** may split the Raman signal **160** into four approximately equal portions and direct each of the Raman-signal portions to one of the detectors. Similarly, the probe beam **120_{pr}** may be split into four approximately equal portions directed to each of the four detectors. In the example of FIG. **38**, the combined beam **210I+**, which is directed to detector **220I+**, may include approximately one-quarter of the power of the Raman signal **160** and approximately one-quarter of the power of the probe beam **120_{pr}**. Similarly, each of the three other combined beams (**210I-**, **210Q+**, **210Q-**) in FIG. **38** may also include approximately one-quarter of the Raman signal **160** and approximately one-quarter of the probe beam **120_{pr}**.

[0171] A 90-degree optical hybrid **250** may be implemented as a waveguide-based device in a PIC. The 90-degree optical hybrid **250** in FIG. **38** is a waveguide-based optical device that includes two waveguide-based optical splitters (**252a**, **252b**) and two waveguide-based optical combiners (**130I**, **130Q**). Splitter **252a** may split the Raman signal **160** into two portions having substantially equal optical power, a first portion directed to combiner **130I** and a second portion directed to combiner **130Q**. Similarly, splitter **252b** may split the probe beam **120_{pr}** into two portions having substantially equal power, a first portion directed to combiner **130I** and a second portion directed to combiner **130Q**. Each optical combiner **130** combines a portion of the Raman signal **160** with a portion of the probe beam **120_{pr}**, and the combined portions are split into a first combined beam (e.g., combined beam **210I+**) and a second combined beam (e.g., combined beam **210I-**). The combined beam **210I+** is directed to detector **220I+** and includes portions of the Raman signal **160** and the probe beam **120_{pr}** (e.g., approximately 25% of the Raman signal **160** and approximately 25% of the probe beam **120_{pr}**). The combined beam **210I-** is directed to detector **220I-** and may include approximately 25% of the Raman signal **160** and approximately 25% of the probe beam **120_{pr}**.

[0172] In other embodiments, all or part of a 90-degree optical hybrid **250** may be implemented as a free-space optical device. For example, a free-space 90-degree optical hybrid **250** may include one or more free-space beamsplitters or combiners that receive the Raman signal **160** and probe beam **120_{pr}** as free-space beams and produce four free-space combined beams (**210I+**, **210I-**, **210Q+**, **210Q-**). Alternatively, all or part of a 90-degree optical hybrid **250** may be implemented as a fiber-optic device. For example, a 90-degree optical hybrid **250** may be contained in a package with two input optical fibers that direct the Raman signal **160** and probe beam **120_{pr}** into the package and four output optical fibers that direct the four combined beams to four respective detectors.

[0173] A 90-degree optical hybrid **250** may include an optical phase shifter **254** that imparts a 90-degree phase change ($\Delta\phi$) to a portion of the probe beam **120_{pr}** or to a portion of the Raman signal **160**. The phase shifter **254** may apply the 90-degree phase change after a beam of light is split by an optical splitter **252** and prior to combining the Raman signal with the probe beam at an optical combiner **130**. For example, a splitter **252a** may split the Raman signal **160** into two portions, and a phase shifter **254** may impart a 90-degree phase change to one portion of the Raman signal with respect to the other portion, after which the two portions are sent to two different optical combiners. As another example, a splitter **252b** may split the probe beam **120_{pr}** into two portions, and a phase shifter **254** may impart a 90-degree phase change to one portion of the probe beam with respect to the other portion. In FIG. **38**, the phase shifter is located after the splitter **252b** and before the combiner **130Q**. The splitter **252b** splits the probe beam **120_{pr}** into two portions, and the

phase shifter **254** imparts a 90-degree phase change to the probe-beam portion directed to combiner **130Q**. The other portion of the probe beam **120pr** directed to combiner **130I** does not pass through the phase shifter **254** and does not receive a phase shift from the phase shifter **254**.

[0174] An optical phase shifter **254** may be implemented as a part of a waveguide-based 90-degree optical hybrid **250**. For example, a phase shifter **254** may be implemented as part of an optical waveguide that only one portion of the probe beam **120pr** propagates through. That part of the optical waveguide may be temperature controlled to adjust the refractive index of the waveguide and produce a relative phase delay of approximately 90 degrees between two portions of the probe beam **120pr**. Additionally or alternatively, the 90-degree optical hybrid **250** as a whole may be temperature controlled to set and maintain a 90-degree phase delay. As another example, a phase shifter **254** may be implemented by applying an external electric field to part of an optical waveguide to change the refractive index of the waveguide and produce a 90-degree phase delay. In other embodiments, a phase shifter **254** may be implemented as a part of a free-space or fiber-coupled 90-degree optical hybrid **250**. For example, the input and output beams in a free-space 90-degree optical hybrid **250** may be reflected by or transmitted through the optical surfaces of a free-space optical hybrid **250** so that a relative phase shift of 90 degrees is imparted to one portion of the probe beam **120pr** with respect to another portion of the probe beam.

[0175] In FIG. **38**, each of the four detectors produces a photocurrent signal that corresponds to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**. The photocurrents are subtracted in a manner similar to that illustrated in FIG. **36**. The photocurrents $i_{\text{sub.I}+}$ and $i_{\text{sub.I}-}$ from detectors **220I+** and **220I-** are subtracted to produce the subtracted in-phase photocurrent signal $i_{\text{sub.I}}$ which is equal to $i_{\text{sub.I}+} - i_{\text{sub.I}-}$. Similarly, the photocurrents $i_{\text{sub.Q}+}$ and $i_{\text{sub.Q}-}$ from detectors **220Q+** and **220Q-** are subtracted to produce the subtracted quadrature photocurrent signal to which is equal to $i_{\text{sub.Q}+} - i_{\text{sub.Q}-}$. Each of the subtracted photocurrent signals represents a coherent-mixing term corresponding to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**. The two subtracted photocurrent signals $i_{\text{sub.I}}$ and $i_{\text{sub.Q}}$ are similar, except the in-phase photocurrent signal $i_{\text{sub.I}}$ includes a cosine function, while the quadrature photocurrent signal $i_{\text{sub.Q}}$ includes a sine function. This difference between the two subtracted photocurrent signals arises from the 90-degree phase shift provided by the phase shifter **254**. Because a 90-degree phase shift is imparted to the probe beam **120pr** directed to the combiner **130Q**, the subtracted quadrature photocurrent signal $i_{\text{sub.Q}}$ includes a sine function (which has a 90-degree phase offset with respect to a cosine function).

[0176] Each of the subtracted photocurrent signals $i_{\text{sub.I}}$ and $i_{\text{sub.Q}}$ may be sent to detection electronics **230** that produce voltage signals and digital output signals corresponding to the subtracted photocurrent signals. Based on the digital output signals (which result from the four photocurrent signals $i_{\text{sub.I}+}$, $i_{\text{sub.I}-}$, $i_{\text{sub.Q}+}$, and $i_{\text{sub.Q}-}$), a processor may determine an in-phase portion IP associated with the Raman signal **160** and a quadrature portion Q associated with the Raman signal. Additionally or alternatively, the processor may determine a phase associated with the Raman signal **160**. For example, the processor may determine a phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr**. A phase difference may be referred to as a phase offset or a relative phase between the Raman signal **160** and the probe beam **120pr**.

[0177] The in-phase portion IP associated with the Raman signal **160** may be determined from a characteristic (e.g., an amplitude or an area) of an electronic signal associated with the in-phase photocurrent signal i , and the quadrature portion Q may be determined from a characteristic associated with the quadrature photocurrent signal $i_{\text{sub.Q}}$. The in-phase portion IP may correspond to an amount of the Raman signal **160** that is in-phase with the probe beam **120pr**, and the quadrature portion Q may represent an amount of the Raman signal that is out of phase (i.e., 90-degrees phase-shifted) with the probe beam. For example, the in-phase portion IP and the quadrature portion Q , may each have values from -1 to 1 . If the Raman signal **160** is in-phase with

the probe beam **120pr**, then the in-phase portion IP may have a value of approximately 1, and the quadrature portion Q may have a value of approximately 0. Similarly, if the Raman signal **160** is out of phase by +90 degrees with respect to the probe beam **120pr**, then the in-phase portion IP may have a value of 0, and the quadrature portion Q may have a value of +1. The phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr** may be determined from the expression $\Delta\phi = \arctan(Q/IP)$. For example, if Q is 0 and IP is 1, then the Raman signal and the probe beam are substantially in phase, with a phase difference $\Delta\phi$ of 0 degrees. As another example, if Q is 1 and IP is 0, then the Raman signal and the probe beam are substantially out of phase, with a phase difference $\Delta\phi$ of 90 degrees.

[0178] FIG. **39** illustrates an example optical receiver **200** configured to detect polarization as well as in-phase and quadrature components of a Raman signal **160**. The optical receiver **200** in FIG. **39** is similar the optical receiver **200** in FIG. **37**, where the horizontal-polarization optical receiver **200h** and the vertical-polarization optical receiver **200v** each includes a 90-degree optical hybrid **250**. Each of the 90-degree optical hybrids in FIG. **39** may be similar to the 90-degree optical hybrid **250** in FIG. **38**. In FIG. **39**, the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** are directed to a horizontal-polarization optical receiver **200h** that includes a 90-degree optical hybrid **250h** and four detectors **220h-I+**, **220h-I-**, **220h-Q+**, and **220h-Q-**. The vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-h** are directed to a vertical-polarization optical receiver **200v** that includes a 90-degree optical hybrid **250v** and four detectors **220v-I+**, **220v-I-**, **220v-Q+**, and **220v-Q-**. The h-polarization optical receiver **200h** may be used to determine the relative size of the horizontal-polarization Raman signal **160-h** as well as the in-phase and quadrature components of the horizontal-polarization Raman signal **160-h**. Similarly, the v-polarization optical receiver **200h** may be used to determine the relative size of the vertical-polarization Raman signal **160-v** as well as the in-phase and quadrature components of the vertical-polarization Raman signal **160-v**.

[0179] The 90-degree optical hybrid **250h** in FIG. **39** combines the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** to produce four horizontally polarized combined output beams: two in-phase combined beams **210h-I+** and **210h-I-**, and two quadrature combined beams **210h-Q+** and **210h-Q-**. Each of the four horizontally polarized combined beams includes a portion of the horizontal-polarization Raman signal **160-h** and a portion of the horizontal-polarization probe beam **120pr-h**. The four combined beams are directed to four respective detectors (**220h-I+**, **220h-I-**, **220h-Q+**, **220h-Q-**), and each detector produces a respective photocurrent signal (i.sub.h-I+, i.sub.h-I-, i.sub.h-Q+, i.sub.h-Q-) that corresponds to the coherent mixing of a portion of the horizontal Raman signal **160-h** and a portion of the horizontal probe beam **120pr-h**. Each of the four photocurrents i.sub.h-I+, i.sub.h-I-, i.sub.h-Q+, and i.sub.h-Q- may be referred to as a horizontal-polarization electronic signal. The photocurrents from the detectors are subtracted to produce a subtracted horizontal in-phase photocurrent signal in-which is equal to i.sub.h-I+ - i.sub.h-I- and a subtracted horizontal quadrature photocurrent signal i.sub.h-Q which is equal to i.sub.h-Q+ - i.sub.h-Q-.

[0180] The 90-degree optical hybrid **250v** in FIG. **39** combines the vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** to produce four vertically polarized combined output beams: two in-phase combined beams **210v-I+** and **210v-I-**, and two quadrature combined beams **210v-Q+** and **210v-Q-**. Each of the four vertically polarized combined beams includes a portion of the vertical-polarization Raman signal **160-v** and a portion of the vertical-polarization probe beam **120pr-v**. The four combined beams are directed to four respective detectors (**220v-I+**, **220v-I-**, **220v-Q+**, **220v-Q-**), and each detector produces a respective photocurrent signal (i.sub.v-I+, i.sub.v-I-, i.sub.v-Q+, i.sub.v-Q-) that corresponds to the coherent mixing of a portion of the vertical Raman signal **160-v** and a portion of the vertical probe beam **120pr-v**. Each of the four photocurrents i.sub.v-I+, i.sub.v-I-, i.sub.v-Q+, and i.sub.v-Q- may be referred to as a vertical-polarization electronic signal. The photocurrents from the detectors are

subtracted to produce a subtracted vertical in-phase photocurrent signal $i_{\text{sub.v-I}}$ which is equal to $i_{\text{sub.v-I}} - i_{\text{sub.v-I}}$ and a subtracted vertical quadrature photocurrent signal $i_{\text{sub.v-Q}}$ which is equal to $i_{\text{sub.v-I}} - i_{\text{sub.v-I}}$.

[0181] Each of the subtracted photocurrent signals $i_{\text{sub.h-I}}$, $i_{\text{sub.h-Q}}$, $i_{\text{sub.v-I}}$, and $i_{\text{sub.v-Q}}$ may be sent to detection electronics **230** that produces voltage signals and digital output signals corresponding to the subtracted photocurrent signals. Based on the digital output signals (which are determined from the four horizontal-polarization electronic signals and the four vertical-polarization electronic signals), a processor may determine (i) the polarization of the Raman signal **160** and (ii) a phase associated with the Raman signal (e.g., a phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120_{pr}**). Determining the polarization of a Raman signal **160** may include determining a relative size or ratio of the horizontal and vertical polarization components of the Raman signal. For example, the relative size of the horizontal polarization component of the Raman signal **160** may be determined by adding characteristics (e.g., areas or amplitudes) associated with the two horizontal photocurrent signals $i_{\text{sub.h-I}}$ and $i_{\text{sub.h-Q}}$. Similarly, the relative size of the vertical polarization component of the Raman signal **160** may be determined by adding characteristics associated with the two vertical photocurrent signals $i_{\text{sub.v-I}}$ and $i_{\text{sub.v-Q}}$. As an example, if the relative size of the horizontal polarization component of the Raman signal **160** is 1 and the relative size of the vertical polarization component of the Raman signal **160** is 100, then the Raman signal **160** may be determined to be substantially vertically polarized.

[0182] Based on the digital output signals, a processor may determine a phase associated with the Raman signal. For example, the processor may determine (i) a phase difference $\Delta\phi_{\text{sub.h}}$ between the horizontal Raman signal **160-h** and the horizontal probe beam **120_{pr-h}** and (ii) a phase difference $\Delta\phi_{\text{sub.v}}$ between the vertical Raman signal **160-v** and the vertical probe beam **120_{pr-v}**. Based on the digital output signals, a processor may determine in-phase and quadrature portions associated with each of the horizontal Raman signal **160-h** and vertical Raman signal **160-v**. The phase difference $\Delta\phi_{\text{sub.h}}$ between the horizontal Raman signal **160-h** and the horizontal probe beam **120_{pr-h}** may be determined from the expression $\Delta\phi_{\text{sub.h}} = \arctan(Q_{\text{sub.h}}/I_{\text{sub.h}})$, where $Q_{\text{sub.h}}$ and $I_{\text{sub.h}}$ are the quadrature and in-phase portions associated with the horizontal Raman signal. The phase difference $\Delta\phi_{\text{sub.v}}$ between the vertical Raman signal **160-v** and the vertical probe beam **120_{pr-v}** may be determined from the expression $\Delta\phi_{\text{sub.h}} = \arctan(Q_{\text{sub.v}}/I_{\text{sub.v}})$, where $Q_{\text{sub.v}}$ and $I_{\text{sub.v}}$ are the quadrature and in-phase portions associated with the vertical Raman signal.

[0183] An optical receiver **200** may include one or more detectors **220**. An optical receiver **200** may include one detector **220** (e.g., as illustrated in FIGS. **1**, **2**, and **18**), or an optical receiver **200** may include multiple detectors **220** (e.g., as illustrated in FIGS. **17**, **36**, **38**, and **39**). An optical receiver **200** with multiple detectors **220** may include 2, 3, 4, 8, 16, or any other suitable number of detectors. For example, an optical receiver **200** may include two detectors **220** arranged so that their respective photocurrents are subtracted (e.g., as illustrated in FIG. **36**). As another example, an optical receiver **200** may include four detectors **220** (e.g., as illustrated in FIG. **38**) or eight detectors **220** (e.g., as illustrated in FIG. **39**). In an optical receiver **200** with multiple detectors **220**, portions of a probe beam **120_{pr}** and a Raman signal **160** may be coherently mixed together at one or more of the multiple detectors **220**, and each of these one or more detectors may produce a photocurrent signal i corresponding to the coherent mixing of the probe beam and the Raman signal. Any of the optical receivers **200** described herein as having a single detector **220** may also be configured to have two or more detectors. For example, the optical receiver in FIG. **1** (which includes one detector **220**) may include a second detector (not illustrated in FIG. **1**), and the detection electronics **230** may be configured to receive and process photocurrent signals from each of the two detectors.

[0184] FIG. **40** illustrates an example Raman spectroscopy system **100** with standoff detection. A

Raman spectroscopy system **100** with standoff detection refers to a Raman spectroscopy system that measures a Raman signal **160** produced by an object **150** located external to the system. For example, a Raman spectroscopy system **100** may be contained within an enclosure, and the object **150** may be located outside of the enclosure. The object **150** in FIG. **40** is located external to or outside of the Raman spectroscopy system **100** and may be referred to as an external object or a remote object. A Raman spectroscopy system **100** with standoff detection may be referred to as a standoff Raman spectroscopy system or a Raman spectroscopy system with remote detection, and standoff detection may be referred to as remote detection.

[0185] The object **150** in FIG. **40** is located a distance D from the Raman spectroscopy system **100**. The distance D from the Raman spectroscopy system **100** to the object **150** may be any suitable distance between approximately 1 centimeter (cm) and approximately 1 meter (m) or any suitable distance between approximately 1 m and approximately 10 kilometers (km). For example, the distance D may be approximately 1 cm, 10 cm, 1 m, 5 m, 10 m, 20 m, 50 m, 100 m, 200 m, 500 m, 1 km, 5 km, or 10 km. Herein, an object **150** may be referred to as a sample **150**. For example, the sample **150** in each FIGS. **1**, **2**, **17**, and **18** may correspond to or may be referred to as an object **150**.

[0186] The Raman spectroscopy system **100** in FIG. **40** is similar to the Raman spectroscopy system illustrated in FIG. **1**, except the standoff Raman spectroscopy system in FIG. **40** operates in a backward-scattered or reflection mode, while the system in FIG. **1** operates in a forward-scattered or transmission mode. In FIG. **1**, the combined pump-Stokes beam **140** is directed to one side of the sample **150**, and the Raman signal **160** that is measured by the Raman spectroscopy system **100** is emitted from the opposite side of the sample. In FIG. **40**, the combined pump-Stokes beam **140** is directed to one side of the object **150**, and the Raman signal **160** that is measured by the Raman spectroscopy system is emitted from the same side of the object. In FIG. **40**, the combined pump-Stokes beam **140** may penetrate at least partially into the object **150**, and the Raman signal **160** may be produced from within the object. Additionally or alternatively, the Raman signal **160** may be produced by material located on the surface of the object or by material (e.g., a gas or aerosol) located around the object.

[0187] The object **150** in FIG. **40** may be any suitable type of object and may include any suitable material. For example, the object **150** may be or may include a package (e.g., luggage, a purse, an envelope, a box, a suspicious package, or an explosive device), a person, a vehicle (e.g., a car, truck, or bus), or an aircraft (e.g., an airplane, helicopter, or unmanned aerial vehicle). As another example, the object **150** may include a hazardous material, a chemical weapon, an explosive material or explosive precursor components, water, wastewater, food, air, or exhaled breath. The Raman spectroscopy system **100** in FIG. **40** may be used: (i) to investigate packages, people, vehicles, or aircraft; (ii) for real-time monitoring of a manufacturing process (e.g., a chemical or biological manufacturing process); (iii) for luggage or package screening (e.g., for presence of explosives or contraband); (iv) to monitor food or water for the presence of poison, drugs, or contaminants (e.g., bacteria); (v) to measure wastewater for indication of disease (e.g., virus); (v.sub.1) to measure a person's skin for indication of disease (e.g., skin cancer); (vii) or to measure a person's exhaled breath for indication of disease (e.g., digestive issues, cancer, or diabetes).

[0188] The Raman spectroscopy system **100** in FIG. **40** is configured to measure a Raman signal **160** produced by an object **150** in response to the pump beam of light **120_{pu}** and the Stokes beam of light **120_S**. The pump light source **110_{pu}** produces a pump beam of light **120_{pu}** at a pump frequency (which may be referred to as a first frequency and may be represented by $\nu_{\text{sub.pu}}$, $\nu_{\text{sub.1}}$, $\omega_{\text{sub.pu}}$, or $\omega_{\text{sub.1}}$), and the Stokes light source **110_S** produces a Stokes beam of light **120_S** at a Stokes frequency (which may be referred to as a second frequency and may be represented by $\nu_{\text{sub.S}}$, $\nu_{\text{sub.2}}$, $\omega_{\text{sub.S}}$, or $\omega_{\text{sub.2}}$). The pump and Stokes frequencies may be offset by a frequency offset Ω , where Ω equals $\nu_{\text{sub.pu}} - \nu_{\text{sub.S}}$ (or equivalently, $\Omega = \nu_{\text{sub.1}} - \nu_{\text{sub.2}}$). The pump light source **110_{pu}** may be referred to as a first light source, and the

pump beam of light **120_{pu}** may be referred to as a first beam of light. The Stokes light source **110S** may be referred to as a second light source, and the Stokes beam of light **120S** may be referred to as a second beam of light.

[0189] In FIG. **40**, the pump and Stokes beams are directed to the object **150**, which is located a distance *D* from the Raman spectroscopy system **100**, and the object produces a Raman signal **160** in response to the pump and Stokes beams. The Raman signal **160** may be produced by coherent Raman scattering of the pump and Stokes beams at the object **150**. For example, the Raman signal **160** may be produced by coherent Raman scattering that occurs within the object **150**, at the surface of the object, or near the object (e.g., the Raman signal may be produced by a gas or aerosol originating from the object and located near the object).

[0190] A Raman spectroscopy system **100** may include one or more optical elements that direct the pump beam **120_{pu}** and Stokes beam **120S** to an object **150**. For example, the pump and Stokes beams may be directed to the object **150** by an optical scanner (e.g., as illustrated in FIG. **49**). Alternatively, the pump and Stokes beams may be directed to the object **150** by an operator of the Raman spectroscopy system **100** (e.g., an operator may use one or more optical elements to direct the pump and Stokes beams to the object). For example, an operator may move or position the Raman spectroscopy system **100** so that the combined pump-Stokes beam **140** is directed to the object. As another example, the Raman spectroscopy system **100** may include a moveable mirror that is adjusted by an operator to direct the combined pump-Stokes beam **140** to the object. As another example, the combined pump-Stokes beam **140** may be coupled into an optical fiber with a terminal end that produces a free-space optical beam (e.g., the terminal end may include a lens that produces a collimated free-space optical beam), and an operator may move or adjust the angle or location of the terminal end so that the combined pump-Stokes beam **140** is directed to the object.

[0191] A Raman spectroscopy system **100** may include one or more optical elements that collect a Raman signal **160** produced by an object **150** in response to the pump and Stokes beams.

Additionally, the optical elements may direct the Raman signal **160** to an optical receiver **200**. The Raman spectroscopy system **100** in FIG. **40** includes an optical receiver **200** that detects the Raman signal **160** produced by the object **150**. The optical receiver **200** detects the Raman signal **160** using an optical heterodyne technique in which the Raman signal **160** is coherently mixed with a probe beam of light **120_{pr}**. An optical receiver **200** may include one or more optical detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam **120_{pr}** to produce an electronic signal. The optical receiver **200** in FIG. **40** includes one optical detector **220** that coherently mixes the Raman signal **160** with the probe beam **120_{pr}** to produce the photocurrent signal *i*. In other embodiments, an optical receiver **200** may include two or more optical detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with a portion of a probe beam **120_{pr}** to produce a corresponding photocurrent signal.

[0192] In FIG. **40**, the probe light source **110_{pr}** produces a probe beam of light **120_{pr}** at a probe frequency (which may be referred to as a third frequency and may be represented by $\nu_{\text{sub.pr}}$, $\omega_{\text{sub.pr}}$, or $\omega_{\text{sub.3}}$). The probe beam **120_{pr}** is combined with the Raman signal **160**, and the combined probe-Raman signal **210** is directed to an optical detector **220**. The Raman signal **160** and the probe beam **120_{pr}** are coherently mixed at the detector **220** to produce a corresponding photocurrent signal *i*. Coherent mixing of the Raman signal **160** and the probe beam **120_{pr}** may refer to coherently mixing at least a portion of the Raman signal **160** with at least a portion of the probe beam **120_{pr}**. For example, an optical receiver **200** may include multiple detectors **220**, and each detector may coherently mix a portion of a Raman signal with a portion of a probe beam **120_{pr}**. As another example, a detector **220** may coherently mix at least a portion of a probe beam **120_{pr}** with a spectral portion of a Raman signal **160**, where the spectral portion of the Raman signal includes optical frequency components of the Raman signal that are within a particular frequency range of the frequency $\nu_{\text{sub.pr}}$ of the probe beam.

[0193] In FIG. 40, the detection electronics 230 (which may include or may be referred to as an electronic circuit) produces a digital output signal 240 corresponding to the photocurrent signal i . The electronic circuit 230 in FIG. 40 may include an electronic amplifier 232 and a digitizer 236 (e.g., like that illustrated in FIG. 2). The electronic amplifier 232 may amplify the photocurrent signal i to produce a voltage signal 234 that corresponds to the photocurrent signal, and the digitizer 236 may produce a digital representation of the voltage signal. The digital representation of the voltage signal 234 may include digital values that approximate the shape of the voltage signal. The digital representation of the voltage signal 234 may be sent to a processor as part of the digital output signal 240. The digital output signal 240 may be referred to as corresponding to the photocurrent signal i , since the digital output signal 240 includes a digital representation of the voltage signal 234, and the voltage signal corresponds to the photocurrent signal i .

[0194] The digital output signal 240 in FIG. 40 may be sent to a processor, and the processor may determine a characteristic 162 of the photocurrent signal i based on the digital output signal. For example, the digital output signal may include a digital representation of a voltage signal 234. The processor may determine a characteristic of the digital representation of the voltage signal 234, and that characteristic may be referred to as being a characteristic of the corresponding photocurrent signal i . The characteristic 162 of a photocurrent signal may include one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization. Additionally, a processor may associate a Raman frequency shift with a determined characteristic 162 of a photocurrent signal i . The Raman frequency shift may equal $v_{\text{sub.pr}} - v_{\text{sub.pu}}$, where $v_{\text{sub.pu}}$ is the pump frequency, and $v_{\text{sub.pr}}$ is the probe frequency.

[0195] The probe light source 110_{pr} in FIG. 40 may include a wavelength-tunable laser configured to sequentially change the probe-beam frequency $v_{\text{sub.pr}}$ to multiple different frequencies. At each of the different probe-beam frequencies, the optical detector 220 may coherently mix the probe beam 120_{pr} and the Raman signal 160 to produce a corresponding photocurrent signal, and a processor may determine a characteristic of each of the photocurrent signals. In some embodiments, the frequency offset Ω between the pump and Stokes frequencies may be approximately equal to a vibrational frequency of a particular material, and a processor may determine, based on a determined characteristic 162 of a photocurrent signal i , (i) whether the particular material is present in an object or (ii) an amount or a concentration of the particular material in the object. A processor of a Raman spectroscopy system 100 may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. A processor may be similar to the computer system 6000 illustrated in FIG. 60 and described herein.

[0196] A technical advantage of a Raman spectroscopy system 100 with standoff detection as described herein is the ability to measure objects at a distance. A Raman spectroscopy system 100 with standoff detection may allow an operator to make measurements in a “point-and-shoot” manner rather than first having to collect a sample. Instead of collecting a sample that is put into a system for measurement, a Raman spectroscopy system 100 with standoff detection allows a Raman signal 160 produced by an object located some distance from the system to be measured. For example, if the object 150 is potentially hazardous or may include an explosive material, an operator of the Raman spectroscopy system may measure the object while remaining at a safe distance from the object. As another example, an object 150 that may be difficult to collect a sample from (e.g., a vehicle, a package, a person's skin, a manufacturing process, or wastewater) may be measured in place without having to first collect a sample.

[0197] FIG. 41 illustrates an example Raman spectroscopy system 100 with a balanced-detection optical receiver 200. The optical receiver 200 includes two detectors (signal detector 220-sig and reference detector 220-ref) arranged in a balanced-detection configuration. The Raman

spectroscopy system **100** in FIG. **41** may detect a Raman signal **160** by coherently mixing the Raman signal **160** with the probe beam of light **120_{pr}** at the signal detector **220-sig**. Additionally, the reference detector **220-ref** may be used to reduce or remove common-mode noise that is present in both the signal beam **210-sig** and the reference beam **210-ref**. The Raman spectroscopy system **100** in FIG. **41** is similar to the Raman spectroscopy system in FIG. **40**, except the optical receiver **200** in FIG. **41** includes two detectors arranged for balanced detection. Additionally, the system in FIG. **41** is configured to produce a Stokes reference beam **120S-ref** and a probe reference beam **120_{pr}-ref** that are detected by the reference detector **220-ref**. The Raman spectroscopy system **100** in FIG. **41** may be referred to as a Raman spectroscopy system with balanced detection and standoff detection or as a standoff Raman spectroscopy system with balanced detection.

[0198] In FIG. **41**, the combined pump-Stokes beam **140** is directed to an object **150**, and the return light that is collected by the Raman spectroscopy system **100** includes a Raman signal **160** as well as residual light from the pump beam **120_{pu}** and Stokes beam **120S**. The residual light from the pump beam and Stokes beam is light that is “leftover” after the pump and Stokes beams have interacted with the object **150** to produce the Raman signal **160**. The residual Stokes beam of light **120S'** may be produced as light from the Stokes beam **120S** that is reflected from, transmitted through, or scattered by the object **150**. The optical filter **134** blocks the residual light from the pump beam **120_{pu}** and transmits the Raman signal **160** and the residual Stokes beam of light **120S'**. The Raman signal **160** and the residual Stokes beam of light **120S'** are combined with the probe beam **120_{pr}** at the optical combiner **130_b** to produce the combined signal beam **210-sig**, which is directed to the signal detector **220-sig**.

[0199] In addition to producing the Stokes beam of light **120S**, the Stokes light source **110S** in FIG. **41** produces a Stokes reference beam of light **120S-ref**. Similarly, in addition to producing the probe beam of light **120_{pr}**, the probe light source **110_{pr}** produces a probe reference beam of light **120_{pr}-ref**. The probe and Stokes reference beams are combined at the optical combiner **130-ref** to produce the reference beam **210-ref**, which is directed to the reference detector **220-ref**. The probe and Stokes reference beams may each be produced by splitting off a portion of a primary beam. For example, the Stokes light source **110S** may include a laser that produces a primary output beam of light, and a small portion (e.g., between 1% and 10%) of light from the primary output beam may be split off to produce the Stokes reference beam of light **120S-ref**. The Stokes beam of light **120S** may be produced from the remaining portion (e.g., between 90% and 99%) of light from the primary output beam that is not split off.

[0200] The signal detector **220-sig** in FIG. **41** receives the signal beam **210-sig** (which includes the Raman signal **160**, probe beam **120_{pr}**, and residual Stokes beam **120S'**) and produces a signal photocurrent $i_{\text{sub.sig}}$ corresponding to the Raman signal, probe beam, and residual Stokes beam, where a portion of the signal photocurrent corresponds to coherent mixing between the Raman signal and the probe beam. The reference detector **220-ref** receives the reference beam **210-ref** (which includes the probe reference beam **120_{pr}-ref** and the Stokes reference beam **120S-ref**) and produces a reference photocurrent $i_{\text{sub.ref}}$ corresponding to the probe and Stokes reference beams.

[0201] The detection electronics **230** in FIG. **41** receives the signal photocurrent $i_{\text{sub.sig}}$ and the reference photocurrent $i_{\text{sub.ref}}$ and produces a digital output signal **240** that corresponds to the two photocurrents. For example, the detection electronics **230** may include a subtraction module that determines a subtraction signal that corresponds to or that equals a difference between (i) a signal corresponding to the signal photocurrent $i_{\text{sub.sig}}$ and (ii) a signal corresponding to the reference photocurrent $i_{\text{sub.ref}}$. The detection electronics **230** may include two electronic amplifiers **232**, where each electronic amplifier is configured to produce a voltage signal corresponding to one of the photocurrents. The subtraction signal may be determined as the difference between a first voltage signal corresponding to the signal photocurrent $i_{\text{sub.sig}}$ and a second voltage signal corresponding to the reference photocurrent $i_{\text{sub.ref}}$. The detection electronics **230** may include a digitizer that produces a digital representation of the subtraction signal, and the digital output signal

240 may include the digital representation of the subtraction signal. The digital output signal **240** may be sent to a processor, and the processor may determine a characteristic **162** of the subtraction signal based on the digital output signal **240**. In other embodiments, the detection electronics **230** may include: two electronic amplifiers **232**; a first digitizer that produces a first digital signal corresponding to the signal photocurrent $i_{\text{sub.sig}}$; and a second digitizer that produces a second digital signal corresponding to the reference photocurrent $i_{\text{sub.ref}}$. For example, the first digital signal may include a digital representation of a first voltage signal that corresponds to the signal photocurrent $i_{\text{sub.sig}}$, and the second digital signal may include a digital representation of a second voltage signal that corresponds to the reference photocurrent $i_{\text{sub.ref}}$. The digital output signal **240** may include the two digital signals corresponding to the two photocurrents. The processor may determine a digital subtraction signal from the two digital signals, and a characteristic **162** of a subtraction signal may be determined from the digital subtraction signal.

[0202] The balanced-detection configuration in FIG. **41** uses two detectors to substantially reduce or remove intensity noise that may be present in the probe beam **120_{pr}** or the Stokes beam **120_S**. The signal beam **210-sig** detected by the signal detector **220-sig** includes the Raman signal **160** along with the probe beam **120_{pr}** and the residual Stokes beam **120_{S'}**. The reference beam **210-ref** detected by the reference detector **220-ref** includes the probe reference beam **120_{pr-ref}** and the Stokes reference beam **120_{S-ref}**. Since the probe beam **120_{pr}** and probe reference beam **120_{pr-ref}** may be derived from the same light source, the two beams may each include correlated (or, common-mode) noise signals. Similarly, the residual Stokes beam **120_{S'}** and the Stokes reference beam **120_{S-ref}** may each include correlated noise signals. In a balanced-detection optical receiver **200**, a signal corresponding to the reference photocurrent $i_{\text{sub.ref}}$ may be subtracted from a signal corresponding to the signal photocurrent $i_{\text{sub.sig}}$ to produce a subtraction signal. The subtraction signal may include a signal associated with the Raman signal **160**, while the common-mode noise present in the probe beam and the Stokes beam may be substantially removed by the subtraction operation that produces the subtraction signal. As a result, the subtraction signal may have a reduced noise compared to each of the photocurrent signals $i_{\text{sub.sig}}$ and $i_{\text{sub.ref}}$ alone.

[0203] FIG. **42** illustrates an example light detection and ranging (lidar) system **101**. The lidar system **101** includes a lidar light source **310** that produces a lidar output beam **320** that includes pulses of light **321**. The pulses of light **321** may be referred to as lidar pulses of light or output pulses of light. The lidar output beam **320** is directed to an optical scanner **330** that scans the output beam across a field of regard (FOR) of the lidar system **101**, where the FOR encompasses a region around the lidar system. The optical scanner **330** may scan the output beam **320** across the FOR along a horizontal direction, a vertical direction, or any other suitable direction or combination of directions. The optical scanner **330** in FIG. **42** includes two scan mirrors: scan mirror **331_x** pivots to scan the output beam **320** along a O_x (or horizontal) direction, and scan mirror **331_y** pivots to scan the output beam **320** along a O_y (or vertical) direction. An optical scanner **330** may scan an output beam **320** across a region around the lidar system **101** within a FOR having any suitable angular size. For example, the optical scanner **330** in FIG. **42** may scan the output beam **320** along a FOR with an angular size of 100° along the O_x direction and 40° along the O_y direction. An optical scanner **330** may include a galvanometer-driven scan mirror, a microelectromechanical systems (MEMS) device with a mirror, a rotating polygon mirror driven by a motor, or a solid-state scanning device (e.g., an optical phased array or a liquid-crystal scanner).

[0204] As the lidar output beam **320** (which includes output pulses of light **321**) is scanned across the region around the lidar system **101**, a corresponding lidar input beam **322** is directed back to the lidar system **101**. The lidar input beam **322** includes input pulses of light **323**, where each input pulse of light includes light from a corresponding output pulse of light **321** that is scattered from an object located in the region around the lidar system **101**. The lidar input beam **322** is directed to the lidar optical receiver **300**, which detects the input pulses of light **323** of the lidar input beam.

[0205] The output pulse of light **321** in FIG. **42** is directed to the object **150**, and the input pulse of

light **323** includes light from the output pulse of light that is scattered or reflected from the object. The dashed-line inset in FIG. **42** includes a time-domain plot of the output pulse of light **321** (which has a time of emission $t_{\text{sub.1}}$) and the input pulse of light **323** (which has a time of arrival $t_{\text{sub.2}}$). The lidar optical receiver **300** detects the input pulse of light **323** and produces a digital output signal **340**, where the digital output signal may be referred to as corresponding to the input pulse of light (or, corresponding to an associated pulse of photocurrent or an associated voltage pulse). The lidar optical receiver **300** may include an optical detector that produces a pulse of photocurrent corresponding to the input pulse of light **323**, and the time of arrival $t_{\text{sub.2}}$ may be determined from the pulse of photocurrent. For example, the lidar receiver **300** may include an electronic amplifier that amplifies the pulse of photocurrent to produce a corresponding voltage pulse, and a digitizer may produce a digital representation of the voltage pulse, from which the time of arrival $t_{\text{sub.2}}$ may be determined. The digital representation of the voltage pulse may include (i) digital values that approximate the shape of the voltage pulse or (ii) a digital value representing a time associated with a rising edge, falling edge, peak, or center of the voltage pulse. For example, the time of arrival $t_{\text{sub.2}}$ may be determined based on the time associated with a rising edge of the voltage pulse (e.g., as illustrated in the dashed-line inset in FIG. **42**). The digital output signal **340** (which may be referred to as corresponding to the input pulse of light **323**, the pulse of photocurrent, or the voltage pulse) may include one or more of the following: a digital representation of the input pulse of light **323** (e.g., a digital representation of a voltage pulse corresponding to the pulse of photocurrent, which in turn corresponds to the input pulse of light); a time of emission $t_{\text{sub.1}}$ of the output pulse of light **321**; a time of arrival $t_{\text{sub.2}}$ of the input pulse of light **323**; a difference Δt between the time of arrival and the time of emission. The digital output signal **340** may be sent to a processor which determines the distance D from the lidar system **101** to the object **150**. The distance D may be determined based on the difference Δt between the time of arrival $t_{\text{sub.2}}$ of the input pulse of light **323** and the time of emission $t_{\text{sub.1}}$ of the corresponding output pulse of light **321**. For example, the distance D may be determined from the expression $D=c \times \Delta t / 2$, where $\Delta t = t_{\text{sub.2}} - t_{\text{sub.1}}$. The time interval Δt may be referred to as the round-trip time of flight for the output pulse of light **321** to travel to the object **150** and back.

[0206] The lidar system **101** may determine the location of the object **150** by scanning the region around the lidar system. For example, the location of the object **150** may be determined based on the input pulse of light **323** that was scattered from the object. The location of the object **150** may include coordinates of the object with respect to the lidar system **101** (e.g., (x, y, z) coordinates of the object with respect to the lidar system; or the distance, azimuth angle, and elevation angle of the object with respect to the lidar system). For example, the distance D to the object **150** may be determined from the time of arrival of the input pulse of light **323**. Additionally, the azimuth and elevation angles of the object **150** may be determined from the position of the optical scanner **330** at the time when the output pulse of light **321** is emitted or when the input pulse of light **323** is received. For example, the azimuth angle may be determined based on the angular position of the O_x scan mirror **331x**, and the elevation angle may be determined based on the angular position of the O_y scan mirror **331y**. In addition to determining the location of an object **150**, a lidar system **101** may also determine a shape of an object or what type of object it is. By sending multiple pulses of light **321** to an object, a three-dimensional shape of the object may be obtained, and based on the shape of the object, the lidar system **101** may determine that the object is of a particular type (e.g., a package, luggage, a box, a person, a vehicle, an aircraft).

[0207] FIG. **43** illustrates example pulses of light **321a** and **321b** produced by the lidar system **101** of FIG. **42**. Each output pulse of light **321** has a pulse duration of τ , which is represented in FIG. **43** as the full width at half maximum (FWHM) of the optical power or intensity of the pulse. The pulse duration τ may have any suitable value between approximately 0.1 nanoseconds (ns) and 10 ns, and the optical energy of each output pulse of light **321** may be between approximately 1 microjoule and 1 millijoule. The time T_L represents the time between pulse **321a** and pulse **321b**

and may be referred to as the pulse period of the lidar system. The pulse period TL may be adjustable or fixed and have any suitable value between approximately 0.1 microseconds (μs) and approximately 1 millisecond (ms), which corresponds to a pulse repetition frequency between approximately 1 kilohertz (kHz) and approximately 10 megahertz (MHz). For example, the pulse period TL in FIG. 43 may be 1 μs , which corresponds to a pulse repetition frequency of 1 MHz. [0208] The lidar system 101 in FIG. 42, which may be referred to as a pulsed-lidar system, uses pulses of light to determine the distance to an object. Another type of lidar system is a frequency-modulated continuous-wave (FMCW) lidar system that uses a frequency-modulated output beam to determine the distance to an object. For example, the output beam may have a substantially constant optical power and a frequency that varies approximately linearly over a particular time interval. An input beam that includes a portion of the output beam scattered from an object may be received by the FMCW lidar system. The frequency of the input beam may be compared to the frequency of a portion of the output beam to determine the distance to the object. For example, the distance to the object may be proportional to the frequency difference between the input beam and the portion of the emitted output beam, and the distance may be determined by measuring this frequency difference.

[0209] FIGS. 44-45 each illustrates an example Raman spectroscopy system 100 and lidar system 101. In some embodiments, a system may include a Raman spectroscopy system 100 and a lidar system 101 that are configured to operate together or in a synchronized manner. In FIG. 44, the lidar system 101 scans a region around the lidar and Raman spectroscopy systems and determines a location of an object 150, and in FIG. 45, the Raman spectroscopy system directs a combined pump-Stokes beam 140 to the object and measures a Raman signal 160 produced by the object. [0210] In FIG. 44, the lidar system 101 scans the output beam 320 across a field of regard (FOR) and determines a location of the object 150 based on the scan of the FOR. The lidar system 101 detects an input beam 322 that includes scattered light from the output beam. When the output beam 320 is directed to the object 150, the input beam 322 may include light from the output beam that is scattered from the object. The lidar system 101 may detect the scattered light from the object 150 and determine the location of the object based on the detected scattered light. The lidar system 101 in FIG. 44 may be a pulsed lidar system (e.g., as illustrated in FIG. 42) or a FMCW lidar system.

[0211] In FIG. 45, the Raman spectroscopy system 100 produces a combined pump-Stokes beam 140 that is directed to the object 150 and then detects a Raman signal 160 produced by the object in response to the pump and Stokes beams. The operation of the lidar system 101 and the Raman spectroscopy system 100 may be synchronized so that prior to the Raman spectroscopy system producing the pump and Stokes beams of light, the lidar system 101 first scans the FOR. Based on one or more scans of the FOR, the lidar system 101 may determine that an object 150 is present in the FOR and may determine the location of the object. After the location of the object has been determined, the Raman spectroscopy system 100 may then produce the pump and Stokes beams of light, and one or more optical elements of the Raman spectroscopy system may direct the pump and Stokes beams of light to the object 150 based on the determined location of the object. For example, the Raman spectroscopy system 100 may include an optical scanner 330 that directs the pump and Stokes beams to the object 150. The Raman spectroscopy system 100 then receives the Raman signal 160 produced by the object 150, and an optical receiver 200 detects the Raman signal. The Raman spectroscopy system 100 may select a pulse energy or an optical power of the pump beam or the Stokes beam based on the distance to the object 150, as determined by the lidar system 101. For example, the pump energy or the optical power of the pump or Stokes beam may be selected to be higher for larger values of distance.

[0212] A lidar system 101 may scan a region around a Raman spectroscopy system 100 and determine the locations or shapes of one or more objects 150 located within the scan region. Additionally, the lidar system 101 may scan the region to identify or classify the one or more

objects. For example, each object may be classified as an object of interest or as a benign object. An object of interest may refer to an object that may be hazardous or may require additional study or measurement (e.g., a package). A benign object may refer to an object that is nonhazardous or that does not require additional study or measurement (e.g., a rock or a piece of wood). After the locations of the one or more objects have been determined, the Raman spectroscopy system **100** may produce pump and Stokes beams of light and direct the beams to one of the objects for measurement of a Raman signal **160** produced by the object. For example, in FIG. **45**, the pump and Stokes beams may be directed to the object **150** in response to the object being classified as an object of interest.

[0213] FIGS. **46-49** each illustrates an example combined Raman spectroscopy and lidar system **100A**. The system **100A** in FIGS. **46-49** is configured to operate as a Raman spectroscopy system and a lidar system. In each of FIGS. **46-48**, the combined Raman spectroscopy and lidar system **100A** is in a lidar operation mode, and in FIG. **49**, the system is in a Raman-spectroscopy operation mode. Instead of having a separate Raman spectroscopy system **100** and lidar system **101** (e.g., as illustrated in FIGS. **44-45**), a single system **100A** may be configured to operate as both a Raman spectroscopy system and a lidar system. In each of FIGS. **46-48**, the system **100A** operates as a lidar system (e.g., similar to that illustrated in FIG. **42**). The system **100A** scans a lidar output beam **320** across a FOR and determines the location of an object **150**. In FIG. **49**, the system **100A** operates as a Raman spectroscopy system (e.g., similar to that illustrated in FIG. **40**, with the addition of an optical scanner **330**). The optical scanner **330** directs a combined pump-Stokes beam **140** to the object **150**, and the optical receiver **200** measures a Raman signal **160** produced by the object.

[0214] The system **100A** in FIGS. **46-49** includes a pump light source **110pu**, a Stokes light source **110S**, an optical scanner **330**, and an optical receiver **200** (which includes a probe light source **110pr**, an optical detector **220**, and detection electronics **230**). Each of FIGS. **46-48** provides an example of the system **100A** operating in a lidar operation mode. When operating as a lidar system, one of the light sources of the system **100A** may act as a lidar light source (e.g., similar to lidar light source **310** in FIG. **42**). In FIG. **46**, the pump light source **110pu** acts as a lidar light source and produces a lidar output beam **320** that includes pulses of light **321**. Additionally, the Stokes and probe light sources are turned off or their respective beams are blocked. In FIGS. **47-48**, the Stokes light source **110S** acts as a lidar light source and produces a lidar output beam **320** that includes pulses of light **321**. In FIG. **47**, the pump and probe light sources are turned off or their respective beams are blocked. In FIG. **48**, the pump light source **110pu** is turned off, and the probe light source **110pr** produces a local-oscillator (LO) beam **324**. The pump light source **110pu** or the Stokes light source **110S** may include one or more laser diodes, and the lidar output beam **320** in FIGS. **46-48** may be produced by one of those laser diodes. Similarly, the probe light source **110pr** may include one or more laser diodes, and the LO beam **324** in FIG. **48** may be produced by one of those laser diodes.

[0215] In each of FIGS. **46-48**, the pump light source **110pu** or the Stokes light source **110S** produces a lidar output beam **320** that includes pulses of light **321**. Each output pulse of light **321** may have a duration between approximately 0.1 nanoseconds (ns) and 10 ns. The optical scanner **330** scans the lidar output beam **320** across a field of regard (FOR) that encompasses a region around the system **100A**. As the lidar output beam is scanned across the region around the system **100A**, a corresponding lidar input beam **322** is directed back to the system. The lidar input beam **322** includes input pulses of light **323**, where each input pulse of light includes light from a corresponding output pulse of light **321** that is scattered from an object located in the region around the system **100A**. The lidar input beam **322** is directed to the optical receiver **200**, which detects the input pulses of light **323** of the lidar input beam.

[0216] The output pulse of light **321** in each of FIGS. **46-48** is directed to the object **150**, and the input pulse of light **323** includes light from the output pulse of light that is scattered or reflected

from the object. The lidar input beam **322** is directed to the optical detector **220**, and the detector produces a photocurrent signal i that may include a pulse of photocurrent that corresponds to the input pulse of light **323**. The detection electronics **230** produces a digital output signal **240** that corresponds to the input pulse of light **323**. For example, the detection electronics **230** may include (i) an electronic amplifier **232** that amplifies the pulse of photocurrent to produce a corresponding voltage pulse and (ii) a digitizer **236** that produces a digital representation of the voltage pulse (from which a time of arrival of the input pulse of light **323** may be determined).

[0217] The digital output signal **240** may include one or more of the following: a digital representation of the input pulse of light **323**; a time of emission of the output pulse of light **321**; a time of arrival of the input pulse of light **323**; a difference Δt between the time of arrival and the time of emission. The digital output signal **240** may be sent to a processor which determines a location of the object based on the digital output signal. Determining the location of the object **150** may include determining the distance D from the combined Raman spectroscopy and lidar system **100A** to the object based on the time of arrival of the input pulse of light **323**. For example, the distance D may be determined based on the difference Δt between the time of arrival $t_{\text{sub.2}}$ of the input pulse of light **323** and the time of emission $t_{\text{sub.1}}$ of the corresponding output pulse of light **321**. The distance D may be determined from the expression $D = C \times \Delta t / 2$, where $\Delta t = t_{\text{sub.2}} - t_{\text{sub.1}}$. Additionally, determining the location of the object **150** may include determining an azimuth angle and an elevation angle of the object with respect to the system **100A** (e.g., based on a position of the optical scanner **330**). In addition to determining the location of an object **150**, the system **100A** (when operating in a lidar operation mode) may also determine a shape of the object or what type of object it is. For example, the system **100A** may send multiple pulses of light **321** to the object **150** and determine a three-dimensional shape of the object based on the determined locations where each of the pulses of light are scattered from the object. Based on the shape of the object **150**, the system **100A** may determine what type of object it is or whether it is an object of interest.

[0218] In FIGS. **46-47**, the photocurrent signal i is produced by the optical detector **220** in direct response to the input pulse of light **323**. The detection process in FIGS. **46-47**, which does not involve coherent mixing of the lidar input beam **322**, may be referred to as direct detection of the input pulse of light **323**. In FIG. **48**, detection of the input pulse of light **323** includes a coherent-mixing process that occurs at the optical detector **220**. The probe light source **110pr** produces a local-oscillator (LO) beam **324** (which may have a substantially constant optical power), and the LO beam is combined with the lidar input beam **322** to produce a combined LO-lidar beam **211** that is directed to the optical detector **220**. The input pulse of light **323** is coherently mixed with the LO beam **324** at the detector **220** to produce a photocurrent signal i that may include a pulse of photocurrent that corresponds to the input pulse of light. The detection electronics **230** may include (i) an electronic amplifier **232** that amplifies the pulse of photocurrent to produce a corresponding voltage pulse and (ii) a digitizer **236** that produces a digital representation of the voltage pulse. The coherent mixing of the input pulse of light **323** and the LO beam **324** may provide an increased signal as compared to the direct-detection approach illustrated in FIGS. **46-47**. For example, the photocurrent signal that results from coherent mixing of the input pulse of light **323** and the LO beam **324** may include a coherent-mixing term that is proportional to the product of the amplitudes of the electric fields of the input pulse of light and the LO beam. Providing a LO beam **324** with a particular amount of power may result in a photocurrent signal associated with the coherent mixing that is effectively increased or boosted as compared to a direct detection of the input pulse of light **323**.

[0219] In FIG. **48**, a processor may determine, based on the digital output signal **240** (which corresponds to a pulse of photocurrent, which in turn corresponds to the input pulse of light **323**), that the input pulse of light **323** includes light from the output pulse of light **321**. For example, coherent mixing of the input pulse of light **323** and the LO beam **324** may produce a photocurrent signal i that includes a pulse of photocurrent having a characteristic that is uniquely associated with

the output pulse of light **321**. Each output pulse of light **321** may include a particular characteristic that results in a particular corresponding characteristic in the photocurrent signal that is produced through coherent mixing. For example, each output pulse of light **321** may have a particular frequency offset with respect to the frequency of the LO beam **324**, and the frequency offset may produce a photocurrent signal with an amplitude modulation at a frequency corresponding to the frequency offset. Measuring the frequency of the amplitude modulation of the photocurrent signal may allow an input pulse of light **323** to be uniquely associated with a corresponding output pulse of light **321**. By providing one of a number of different frequency offsets to each output pulse of light **321**, an input pulse of light **323** may be determined to be associated with a particular output pulse of light. For example, each emitted pulse of light **321** produced by the Stokes light source **110S** in FIG. **48** may have a frequency offset with respect to the LO beam **324** of 1 GHz, 2 GHz, or 3 GHz. If the input pulse of light **323** produces a photocurrent signal having a 2-GHz modulation frequency, then a processor may determine that the input pulse of light **323** includes light from a corresponding output pulse of light **321** with a 2-GHz frequency offset. Determining that an input pulse of light **323** includes light from a particular output pulse of light **321** may prevent ambiguity in identifying which output pulse of light an input pulse of light is associated with by allowing the system **100A** to unambiguously associate input and output pulses of light.

[0220] In FIG. **49**, the combined Raman spectroscopy and lidar system **100A** operates as a Raman spectroscopy system. The operation of the system **100A** in FIG. **49** is similar to that of the Raman spectroscopy system **100** in FIG. **40**, except the system **100A** includes an optical scanner **330**. In FIGS. **46-48**, when the system **100A** operates as a lidar system, the optical scanner **330** is used to scan the lidar output beam **320** across the FOR. In FIG. **49**, when the system **100A** operates as a Raman spectroscopy system, the pump light source **110pu** produces the pump beam **120pu**, the Stokes light source **110S** produces the Stokes beam **120S**, and the optical scanner **330** directs the pump and Stokes beams to the object **150**. The resulting Raman signal **160** produced by the object **150** is directed to the receiver **200**, and the Raman signal is coherently mixed with the probe beam **120pr** at the detector **220**.

[0221] The combined Raman spectroscopy and lidar system **100A** may first operate as a lidar system (e.g., as illustrated in each of FIGS. **46-48**). The system **100A** may scan a lidar output beam **320** across a FOR that encompasses a region around the system and may determine the locations or shapes of one or more objects **150** located within the FOR. Additionally, the system **100A** may scan the region to identify or classify the objects (e.g., each object may be classified as an object of interest or a benign object). In each of FIGS. **46-48**, the system **100A** may determine the location of the object **150**. After the location of the object **150** has been determined, the system **100A** may then operate as a Raman spectroscopy system and produce the pump and Stokes beams of light, as illustrated in FIG. **49**. Additionally, the optical scanner **330** may direct the combined pump-Stokes beam **140** to the object **150** based on the determined location of the object. The pump and Stokes beams may be directed to the object **150** in response to the object being classified as an object of interest. In FIG. **49**, the Raman signal **160** produced by coherent Raman scattering of the pump and Stokes beams at the object **150** is directed to the optical receiver **200**, which detects the Raman signal in the same manner as illustrated in FIG. **40**. In some embodiments, the optical receiver **200** in FIG. **49** may be an optical receiver with balanced detection (e.g., as illustrated in FIG. **41**).

[0222] In FIG. **49**, the optical scanner **330** may remain in a substantially fixed configuration while directing the pump and Stokes beams to the object **150** and while measuring the resulting Raman signal **160**. For example, the optical scanner **330** may direct the combined pump-Stokes beam **140** to the object **150** without scanning the beam. Alternatively, the optical scanner **330** may scan the combined pump-Stokes beam **140**, and the Raman signal **160** may be measured at a time when the beam is directed to the object **150**.

[0223] In FIG. **49**, the pulse energy or the optical power of the pump beam or the Stokes beam may be selected based on the distance D to the object **150**, as determined when the combined Raman

spectroscopy and lidar system **100A** operates as a lidar system. For example, the system **100A** may select the pulse energy or optical power of the pump or Stokes beam to be higher for larger values of distance **D**. As another example, the pump energy or the optical power of both the pump and Stokes beams may be selected to increase in proportion to the distance **D**. The higher value of pulse energy or optical power for larger values of distance may allow the system **100A** to compensate for: optical losses that occur in propagating to and from the object; expansion of the pump and Stokes beams while propagating to the object; or a reduction in the amount of Raman signal **160** collected by the system for objects located a greater distance away.

[0224] In some embodiments, a combined Raman spectroscopy and lidar system **100A** may operate without an optical scanner. For example, the combined Raman spectroscopy and lidar system **100A** illustrated in FIGS. **46-49** may not include an optical scanner **330**, and the lidar output beam **320** and the combined pump-Stokes beam **140** may not be scanned. A combined Raman spectroscopy and lidar system **100A** that does not include an optical scanner may be referred to as a combined Raman spectroscopy and rangefinder system. A rangefinder, which may be referred to as a scannerless lidar system, may determine the distance to a remote object without scanning a lidar output beam **320** across a field of regard. A Raman spectroscopy and rangefinder system may include a pump light source **110_{pu}**, a Stokes light source **110_S**, and an optical receiver **200** (which includes a probe light source **110_{pr}**, an optical detector **220**, and detection electronics **230**), and the system may not include an optical scanner. For example, the Raman spectroscopy system **100** in FIG. **40** or FIG. **41** may be configured to operate as a combined Raman spectroscopy and rangefinder system. As another example, the combined Raman spectroscopy and lidar system **100A** in FIGS. **46-49**, without the optical scanner **330**, may be configured to operate as a combined Raman spectroscopy and rangefinder system.

[0225] The lidar output beam **320** produced by a combined Raman spectroscopy and rangefinder system may be directed to an object **150**, and the system may determine the distance **D** to the object. The lidar output beam **320** may be directed to an object **150** by an operator of the system. For example, an operator may move or position the combined Raman spectroscopy and rangefinder system or may use one or more optical elements to direct the lidar output beam **320** to the object **150**. Similarly, an operator may move or position the combined Raman spectroscopy and rangefinder system or may use one or more optical elements to direct the pump and Stokes beams to the object **150**.

[0226] The system **100A** in FIGS. **46-49**, with the optical scanner **330** removed, may correspond to a combined Raman spectroscopy and rangefinder system. The pump light source **110_{pu}** or the Stokes light source **110_S** may produce an output pulse of light **321** (e.g., as illustrated in FIGS. **46-48**) that is directed to an object **150** (e.g., by an operator of the system **100A**). The optical receiver **200** of a combined Raman spectroscopy and rangefinder system may detect an input pulse of light **323** in a manner similar to that illustrated in FIGS. **46-48**. For example, the optical detector **220** may produce a photocurrent signal **i** that includes a pulse of photocurrent that corresponds to the input pulse of light **323**, and the detection electronics **230** may produce a digital output signal **240** that corresponds to the pulse of photocurrent (which in turn corresponds to the input pulse of light). The digital output signal **240** may be sent to a processor which determines a distance **D** to the object. For example, the distance to the object **150** may be determined based on the difference between the time of arrival of the input pulse of light **323** and the time of emission of the corresponding output pulse of light **321**. In some embodiments, a combined Raman spectroscopy and rangefinder system may use a coherent-detection process to detect an input pulse of light **323**. For example, the input pulse of light **323** may be coherently mixed at the optical detector **220** with a LO beam **324** produced by the probe light source **110_{pr}** (e.g., as illustrated in FIG. **49**).

[0227] In addition to determining a distance **D** to an object **150**, a combined Raman spectroscopy and rangefinder system may measure a Raman signal **160** produced by the object. For example, the system **100A** in FIG. **49**, without the optical scanner **330**, may produce a combined pump-Stokes

beam **140** that is directed to the object **150** (e.g., by an operator of the system), and an optical receiver **200** of the combined Raman spectroscopy and rangefinder system may measure the Raman signal **160** produced by the object **150** in a manner similar to that illustrated in FIG. **49**. [0228] A combined Raman spectroscopy and rangefinder system may first measure a Raman signal **160** produced by an object **150** and then determine the distance to the object. Alternatively, a combined Raman spectroscopy and rangefinder system may first determine the distance to an object **150** and then measure a Raman signal **160** produced by the object. For example, the system **100A** in FIGS. **46-49** (without the optical scanner **330**) may first determine the distance D to the object **150** (e.g., as illustrated in FIGS. **46-48**). After measuring the distance, the system **100A** may select a pulse energy or an optical power of the pump beam **120_{pu}** or the Stokes beam **120_S** based on the distance to the object. For example, the selected pulse energy or optical power of the pump or Stokes beam may be higher for larger values of distance D (e.g., the selected pulse energy or optical power may increase in proportion to the distance D , in proportion to $D^{\sup.2}$, or in any other suitable monotonically increasing manner). The higher value of pulse energy or optical power for larger values of distance may allow a system **100A** to compensate for: optical losses that occur in propagating to and from the object; expansion of the optical beams while propagating between the system and object; or a reduction in the amount of Raman signal **160** collected by the system for objects located a greater distance away. The pump and Stokes light sources may produce the pump and Stokes beams based on the selected pulse energy or optical power, and the pump and Stokes beams may be directed to the object **150**. The optical receiver **200** may measure the resulting Raman signal **160** produced by coherent Raman scattering of the pump and Stokes beams at the object (e.g., as illustrated in FIG. **49**).

[0229] FIG. **50** illustrates an example combined Raman spectroscopy and rangefinder system **100A** configured to perform a distance measurement with a Stokes pulse of light **331_S**. In some embodiments, a Raman spectroscopy and rangefinder system **100A**, as discussed above, may (i) produce an output pulse of light **321** to measure the distance to an object **150** and (ii) produce, at a different time, a combined pump-Stokes beam **140** to measure a Raman signal **160** produced by the object. In other embodiments (e.g., as illustrated in FIG. **50**), instead of producing two separate optical signals (e.g., an output pulse of light **321** and a pump-Stokes beam **140**), the combined Raman spectroscopy and rangefinder system **100A** produces a Stokes pulse of light **331_S** that is used to (i) measure the distance to an object **150** and (ii) cause (along with the pump beam **120_{pu}**) the object to produce a Raman signal **160**. In the Raman spectroscopy and rangefinder system **100A** of FIG. **50**, the measurement of the distance to the object and the measurement of the Raman signal may be performed at approximately the same time.

[0230] In FIG. **50**, the pump light source **110_{pu}** produces a pump beam **120_{pu}** (which may include pulses of light or light having a substantially constant optical power), and the Stokes light source **110_S** produces a Stokes beam **120_S** that includes a Stokes pulse of light **331_S**. The Stokes beam of light **120_S** may include multiple Stokes pulses of light **331_S**, and the optical frequencies of the pump beam **120_{pu}** and each Stokes pulse of light **331_S** may be offset by a particular frequency offset Ω . The pump and Stokes beams are directed to an object **150**, and the object produces a Raman signal **160** by coherent Raman scattering of the pump beam **120_{pu}** and the Stokes pulse of light **331_S**. The light returning from the object **150** includes the Raman signal **160** as well as a residual pump beam of light **120_{pu}'** (which is blocked by the optical filter **134**) and a residual Stokes beam of light **120_S'**. The residual Stokes beam of light **120_S'** includes a Stokes input pulse of light **333_S** (which may be referred to as an input pulse of light), where the Stokes input pulse of light **333_S** includes light from the Stokes pulse of light **331_S** that is scattered from the object **150**. The input pulse of light **333_S** may also include light from the Raman signal **160**.

[0231] The Raman signal **160** and the residual Stokes beam of light **120_S'** (which includes the input pulse of light **333_S**) are combined with the probe beam **120_{pr}** at the optical combiner **130_b** to produce a combined beam **210**, which is directed to the optical detector **220**. The optical detector

220 receives the combined beam **210** (which includes the Raman signal **160**, probe beam **120pr**, and input pulse of light **333S**) and produces a photocurrent signal *i* corresponding to the Raman signal, probe beam, and input pulse of light. A portion of the photocurrent signal *i* may correspond to coherent mixing between the Raman signal **160** and the probe beam **120pr**, and another portion of the photocurrent signal *i* may include a pulse of photocurrent that corresponds to the input pulse of light **333S**. The coherent-mixing portion and the pulse portion of the photocurrent signal may be substantially overlapped in time. The detection electronics **230** receives the photocurrent signal *i* and produces a digital output signal **240** corresponding to the photocurrent signal. For example, the detection electronics **230** may include (i) an electronic amplifier **232** that amplifies the photocurrent signal to produce a corresponding voltage signal **234** and (ii) a digitizer **236** that produces a digital representation of the voltage signal. The voltage signal **234** may include a voltage pulse that corresponds to the pulse of photocurrent, and the digital representation of the voltage signal may include a digital representation of the voltage pulse.

[0232] The digital output signal **240** produced by the detection electronics **230** may include a portion that corresponds to coherent mixing between the Raman signal **160** and the probe beam **120pr** and another portion that corresponds to the pulse of photocurrent. The two portions may be substantially overlapped in time. A processor may determine the distance *D* to the object **150** based on the portion of the digital output signal that corresponds to the pulse of photocurrent. For example, the digital output signal **240** produced by the detection electronics **230** may include one or more of the following: a digital representation of the voltage signal **234** or the voltage pulse; a time of emission $t_{\text{sub.1}}$ of the Stokes pulse of light **331S**; a time of arrival $t_{\text{sub.2}}$ of the input pulse of light **333S**; a difference Δt between the time of arrival and the time of emission. The times $t_{\text{sub.1}}$ and $t_{\text{sub.2}}$ may be determined based on times associated with a rising edge of signals associated with the pulses of light **331S** and **333S** (e.g., as illustrated in the dashed-line inset in FIG. **50**). For example, the time of arrival $t_{\text{sub.2}}$ of the input pulse of light **333S** may be determined based on a time associated with a rising edge of the voltage signal **234** or the voltage pulse. The distance *D* to the object **150** may be determined from the expression $D = C \times \Delta t / 2$, where $\Delta t = t_{\text{sub.2}} - t_{\text{sub.1}}$.

[0233] FIG. **51** illustrates an example combined Raman spectroscopy and rangefinder system **100A** configured to perform a distance measurement with a pump pulse of light **331pu**. In FIG. **50**, the Stokes light source **110S** produces a Stokes pulse of light **331S** that is used to determine the distance to an object, while in FIG. **51**, the pump light source **110pu** produces a pump pulse of light **331pu** that is used to determine the distance to an object. In other embodiments, both the Stokes light source **110S** and the pump light source **110pu** may produce pulses of light that are used to determine the distance to an object. In FIG. **51**, the combined Raman spectroscopy and rangefinder system **100A** produces a pump pulse of light **331pu** that is used to (i) measure the distance to an object **150** and (ii) cause (along with the Stokes beam **120S**) the object to produce a Raman signal **160**. The measurement of the distance to the object and the measurement of the Raman signal may be performed at approximately the same time.

[0234] In FIG. **51**, the pump light source **110pu** produces a pump beam **120pu** that includes a pump pulse of light **331pu**, and the Stokes light source **110S** produces a Stokes beam **120S** (which may include pulses of light or light having a substantially constant optical power). The pump beam of light **120pu** may include multiple pump pulses of light **331pu**, and the optical frequencies of each pump pulse of light **331pu** and the Stokes beam **120S** may be offset by a particular frequency offset Ω . The pump and Stokes beams are directed to an object **150**, and the object produces a Raman signal **160** by coherent Raman scattering of the pump pulse of light **331pu** and the Stokes beam **120S**. The light returning from the object **150** includes the Raman signal **160** as well as a residual pump beam of light **120pu'** and a residual Stokes beam of light **120S'**.

[0235] The Raman signal **160** and the residual Stokes beam of light **120S'** (which includes light from the Stokes beam **120S** that is scattered from the object) are directed to the optical receiver

200. The Raman signal **160** and the residual Stokes beam of light **120S'** are combined with the probe beam **120pr** at the optical combiner **130b** to produce a combined beam **210**, which is directed to the optical detector **220**. The Raman signal **160** and the probe beam **120pr** are coherently mixed at the optical detector **220** to produce a corresponding photocurrent signal **i**. The detection electronics **230** produces a digital output signal **240** corresponding to the photocurrent signal, and a processor may determine a characteristic of the photocurrent signal based on the digital output signal. In some embodiments, an optical receiver **200** of a combined Raman spectroscopy and rangefinder system **100A** may be an optical receiver with balanced detection (e.g., as illustrated in FIG. **41**).

[0236] In FIG. **51**, the residual pump beam of light **120pu'** includes a pump input pulse of light **333pu** (which may be referred to as an input pulse of light), where the pump input pulse of light **333pu** includes light from the pump pulse of light **331pu** that is scattered from the object **150**. The combined Raman spectroscopy and rangefinder system **100A** in FIG. **51** includes an optical beamsplitter **130e** that directs the input pulse of light **333pu** to a pump detector **220-P**. The optical beamsplitter **130e** may be (i) a dichroic beamsplitter that reflects light at the pump-beam wavelength and transmits light at the Stokes and Raman wavelengths or (ii) a non-dichroic beamsplitter that splits off at least a portion of the pump input pulse of light **333pu**. Alternatively, the optical beamsplitter **130e** may be a polarization beamsplitter that splits off light with a polarization of the pump beam **120pu** and transmits light with a polarization of the Raman signal **160**. The pump detector **220-P** is an optical detector and may include a PN photodiode, PIN photodiode, APD, or SPAD. The pump detector **220-P** may produce a pulse of photocurrent corresponding to the input pulse of light **333pu**, and a processor may determine a distance **D** to the object based on a time of arrival $t_{sub.2}$ of the input pulse of light. For example, the pump detector **220-P** may be coupled to an electronic amplifier that amplifies the pulse of photocurrent to produce a corresponding voltage pulse, and the time of arrival $t_{sub.2}$ of the input pulse of light **333pu** may be determined from a rising edge (as illustrated in the dashed-line inset in FIG. **51**), falling edge, peak, or center of the voltage pulse. The distance **D** may be determined based on the difference Δt between the time of arrival $t_{sub.2}$ of the input pulse of light **333pu** and the time of emission $t_{sub.1}$ of the corresponding pump pulse of light **331pu**. For example, the distance **D** may be determined from the expression $D=c \times \Delta t / 2$, where $\Delta t = t_{sub.2} - t_{sub.1}$.

[0237] FIGS. **52-54** each illustrates example time-domain plots of pump and Stokes beams of light. The combined pump-Stokes beam **140** in each of FIGS. **52-54** includes a pump beam of light **120S** and a Stokes beam of light **120S**. Each of the combined pump-Stokes beam **140** may be used in any suitable Raman spectroscopy system **100**, combined Raman spectroscopy and lidar system **100A**, or combined Raman spectroscopy and rangefinder system **100A**, as described or illustrated herein. In some embodiments, the pump and Stokes beams of light may each include light having a substantially constant optical power. In other embodiments, the pump beam of light **120pu** or the Stokes beam of light **120S** may include pulses of light, as illustrated in FIGS. **52-54**. In FIG. **52**, the pump beam **120pu** and the Stokes beam **120S** each includes pulses of light. In FIG. **53**, the pump beam **120pu** has a substantially constant optical power, and the Stokes beam **120S** includes pulses of light **331S**. In FIG. **54**, the pump beam **120pu** includes pulses of light **331pu**, and the Stokes beam **120S** has a substantially constant optical power.

[0238] The pump beam **120pu** in each of FIGS. **52** and **54** includes two pump pulses of light **331pu-1** and **331pu-2**. The duration $\Delta T_{sub.pu}$ of each pump pulse of light **331pu** may be between approximately 50 ns and approximately 5 μ s, and each pump pulse of light **331pu** may have a peak power between approximately 1 watt (W) and approximately 1 KW. The Stokes beam **120S** in each of FIGS. **52** and **53** includes two Stokes pulses of light **331S-1** and **331S-2**. The duration $\Delta T_{sub.S}$ of each Stokes pulse of light **331S** may be between approximately 50 ns and approximately 5 μ s, and each Stokes pulse of light **331S** may have a peak power between approximately 1 W and approximately 1 KW. The pulse period $T_{sub.pu}$ of the pump pulses of light **331pu** and the pulse

period $T_{\text{sub.S}}$ of the Stokes pulses of light **331S** may be adjustable or fixed and may be between approximately 100 ns and approximately 10 μs . In FIG. 52, the pump and Stokes pulses may be synchronized to overlap temporally. Additionally, in FIG. 52, the pulse durations $\Delta T_{\text{sub.pu}}$ and $\Delta T_{\text{sub.S}}$ may have approximately the same values, and the pulse periods $T_{\text{sub.pu}}$ and $T_{\text{sub.S}}$ may have approximately the same values. For example, the pulse durations $\Delta T_{\text{sub.pu}}$ and $\Delta T_{\text{sub.S}}$ may be 200 ns, and the pulse periods $T_{\text{sub.pu}}$ and $T_{\text{sub.S}}$ may be 400 ns, which corresponds to a 50% duty cycle.

[0239] The pulse energy, peak power, or average power of a pump beam **120pu** or a Stokes beam **120S** may be selected to be higher for larger values of distance D to a remote object **150**. For example, a lidar or rangefinder measurement may determine the distance D to an object **150**, and the power or energy of the pump and Stokes beams used for a corresponding Raman-signal measurement may be selected based on the distance to the object. In FIG. 52, the pulse energy or peak power of the pump and Stokes pulses of light may be selected to be approximately two times higher for an object located a distance $2D$ than for another object located a distance D (e.g., the energy or power may increase in proportion to the distance). In FIG. 53, the average power of the pump beam **120pu** and the pulse energy or peak power of the Stokes pulses of light **331S** may be selected to be approximately four times higher for an object located a distance $2D$ than for another object located a distance D (e.g., the energy or power may increase in proportion to $D^{\text{sup.2}}$).

[0240] FIGS. 55-56 each illustrates example time-domain plots of pump and Stokes beams of light along with a Raman signal **160** and a probe beam of light **120pr**. The signals in each of FIGS. 55-56 may be associated with a Raman spectroscopy system **100**, combined Raman spectroscopy and lidar system **100A**, or combined Raman spectroscopy and rangefinder system **100A**, as described or illustrated herein. The pump beam **120pu** and Stokes beam **120S** in each of FIGS. 55 and 56 includes pulses of light (similar to FIG. 52). In other embodiments, the pump beam **120** may have a substantially constant optical power (e.g., as illustrated in FIG. 53), or the Stokes beam **120S** may have a substantially constant optical power (e.g., as illustrated in FIG. 54).

[0241] The combined pump-Stokes beam **140** in each of FIGS. 55 and 56 is directed to an object **150**, and a Raman signal **160** is produced by coherent Raman scattering of the pump and Stokes pulses of light at the object. The Raman signal **160** includes pulses of light, where each Raman-signal pulse of light **335R** is produced by a corresponding pair of pump and Stokes pulses of light. For example, Raman-signal pulse **335R** is produced by coherent Raman scattering of pump pulse **331pu** and Stokes pulse **331S**. The duration $\Delta T_{\text{sub.R}}$ of the Raman-signal pulse **335R** may be approximately equal to the duration $\Delta T_{\text{sub.pu}}$ of the corresponding pump pulse **331pu** or the duration $\Delta T_{\text{sub.S}}$ of the corresponding Stokes pulse **331S**. Each Raman-signal pulse **335R** is received and detected a particular time delay Δt after the corresponding pump and Stokes pulses of light, where the time delay depends on the distance D from the system **100** to the object **150**. The time delay Δt may be approximately equal to the round-trip time of flight for light to travel to the object **150** and back to the system **100**. The time delay Δt may be related to the distance D by the expression $\Delta t = 2D/c$.

[0242] The Raman signal **160** is coherently mixed with a probe beam of light **120pr** at an optical detector **220**. In some embodiments, the probe beam **120pr** may have a substantially constant optical power, and each pulse of light **335R** from the Raman signal **160** may be coherently mixed with the probe beam. Alternatively, as illustrated in FIGS. 55-56, the probe beam **120pr** may include pulses of light **335pr**. The probe pulses of light **335pr** may be synchronized with the Raman-signal pulses **335R** so they overlap temporally. Additionally, the pulse periods $T_{\text{sub.pu}}$, $T_{\text{sub.S}}$, and $T_{\text{sub.pr}}$ may be approximately equal.

[0243] In FIG. 55, the probe pulse **335pr** begins at approximately the same time as the pump and Stokes pulses of light **331pu** and **331S**. The duration $\Delta T_{\text{sub.pr}}$ of the probe pulse **335pr** in FIG. 55 is greater than the durations of the other pulses of light, which may allow for the probe pulse to temporally overlap the Raman-signal pulse **335R**. For example, the duration $\Delta T_{\text{sub.pr}}$ of the probe

pulse **335pr** may be (i) greater than or equal to $\Delta t + \Delta T_{\text{sub.pu}}$, $\Delta t + \Delta T_{\text{sub.S}}$, or $\Delta t + \Delta T_{\text{sub.R}}$ and (ii) less than the pulse period $T_{\text{sub.pu}}$, $T_{\text{sub.S}}$, or $T_{\text{sub.pr}}$.

[0244] In some embodiments, the probe pulse **335pr** may be emitted a particular time delay after the corresponding pump pulse **331pu** or Stokes pulse **331S**, where the time delay is based on the distance from the system **100** to the object **150**. In FIG. **56**, the probe pulse **335pr** is emitted by a time delay Δt after the pump and Stokes pulses, and the time delay Δt may be approximately equal to the round-trip time of flight for light to travel to the object **150** and back to the system **100**. For example, the time delay Δt may equal the time for the pump and Stokes pulses **331pu** and **331S** to travel to the object **150** plus the time for the corresponding Raman-signal pulse **335R** to travel from the object to the system **100**. The time delay Δt may be related to the distance D by the expression $\Delta t = 2D/c$. A lidar or rangefinder measurement may determine the distance D to an object **150**, and the time delay Δt of the probe pulse **335pr** may be selected based on the determined distance. The duration $\Delta T_{\text{sub.pr}}$ of the probe pulse **335pr** may be greater than or equal to the pulse duration $\Delta T_{\text{sub.pu}}$, $\Delta T_{\text{sub.S}}$, or $\Delta T_{\text{sub.R}}$. In FIG. **56**, the duration $\Delta T_{\text{sub.pr}}$ of the probe pulse **335pr** is approximately equal to the duration $\Delta T_{\text{sub.R}}$ of the Raman pulse **335R** (which may be approximately equal to the pulse durations $\Delta T_{\text{sub.pu}}$ and $\Delta T_{\text{sub.S}}$).

[0245] FIG. **57** illustrates an example Raman spectroscopy system **100** that includes a visible light source **110v**. The Raman spectroscopy system **100** in FIG. **57** is similar to that illustrated in FIG. **40**, with the addition of the visible light source **110v**. The system **100** in FIG. **57** may be configured to operate as a Raman spectroscopy system, a combined Raman spectroscopy and lidar system, or a combined Raman spectroscopy and rangefinder system.

[0246] The visible light source **110v** produces a visible beam of light **120v** that is directed to the object **150** along with the pump and Stokes beams. In FIG. **57**, the visible beam of light **120v** is combined at the optical combiner **130v** with the combined pump-Stokes beam **140** to produce a combined pump-Stokes-visible beam **142**, which is directed to the object **150**. The combined pump-Stokes-visible beam **142** includes the pump beam **120pu**, the Stokes beam **120S**, and the visible beam **120v**. The visible beam **120v** may have a wavelength that is visible to the human eye. For example, the visible light source **110v** may include a laser that produces a visible beam of light **120v** having a wavelength between approximately 380 nm (blue) and approximately 780 nm (red). As another example, the visible light source **110v** may include (i) a laser diode with an operating wavelength of approximately 450-490 nm or 635 nm or (ii) a solid-state laser with an operating wavelength of approximately 532 nm.

[0247] The pump and Stokes beams may each have wavelengths that are not visible to the human eye (e.g., the wavelengths may be greater than approximately 900 nm), and the pump and Stokes beams may not produce a visible spot of light at the object **150**. In FIG. **57**, the visible beam **120v** is directed to the object **150** and produces a visible alignment spot **144** at the object that substantially overlaps the location of the pump and Stokes beams at the object. The visible alignment spot **144** includes light from the visible beam **120v** that is scattered or reflected from the object **150**. The visible alignment spot **144** may be used by an operator of the system **100** to aim the combined pump-Stokes beam **140** to the object **150**. For example, an operator may use one or more optical elements (e.g., a moveable mirror or the terminal end of an optical fiber) to direct the combined pump-Stokes-visible beam **142** to the object **150** using the location of the visible alignment spot **144** as an alignment aid.

[0248] The visible alignment spot **144** may substantially overlap the location of the pump and Stokes beams at the object **150**. For example, the pump beam **120pu**, Stokes beam **120S**, and visible beam **120v** may be combined so the three beams are substantially overlapped with one another and propagate in the same direction along approximately the same optical axis. In other embodiments, the visible beam **120v** may be combined with the pump and Stokes beams so that the visible beam is located directly adjacent to the pump and Stokes beams and propagates along an optical axis that is approximately parallel to the optical axis along which the pump and Stokes

beams propagate. The dashed-line inset in FIG. 57 illustrates example beam profiles of the combined pump-Stokes beam **140** and the visible alignment spot **144** produced by the visible beam **120v** at the object. The combined pump-Stokes beam **140** is centered at location $x_{sub.1}$ with a beam diameter of $\Delta x_{sub.1}$, and the visible alignment spot **144** is centered at location $x_{sub.2}$ with a beam diameter of $\Delta x_{sub.2}$. The visible alignment spot **144** may be substantially overlapped with the location of the pump and Stokes beams at the object **150** if the separation distance ΔX between the beams at the object is less than $\Delta x_{sub.1} + \Delta x_{sub.2}$.

[0249] FIG. 58 illustrates an example method **5800** for measuring a Raman signal **160**. The method may be performed by a Raman spectroscopy system as described herein. For example, the method may be performed by the Raman spectroscopy system **100** illustrated in FIG. 1 or 2 and may be used to measure a Raman signal **160**. The method may begin at step **5810** by producing a first beam of light at a first frequency ($\nu_{sub.1}$). At step **5820**, a second beam of light at a second frequency ($\nu_{sub.2}$) is produced. The first beam of light may be referred to as a pump beam of light **120pu**, and the second beam of light may be referred to as a Stokes beam of light **120S**. The first and second frequencies may be offset by a frequency offset Ω , where $\Omega = \nu_{sub.1} - \nu_{sub.2}$. For example, the first and second frequencies may each correspond to a wavelength between approximately 300 nm and approximately 5,000 nm, and the frequency offset Ω may be between approximately 5 THz and approximately 100 THz. The first and second beams of light may each be produced by a light source that includes one or more laser diodes, where each laser diode is a fixed-wavelength laser diode or a wavelength-tunable laser diode.

[0250] At step **5830**, the first and second beams of light are directed to a sample **150**, and at step **5840**, a Raman signal **160** is collected, where the Raman signal is produced by the sample in response to the first and second beams of light. For example, the Raman signal may be produced by coherent Raman scattering of the first and second beams of light within the sample. The first and second beams of light may be directed to the sample by one or more optical elements that include a free-space optical element, an optical fiber, or an optical waveguide. Similarly, one or more optical elements (which may include a free-space optic, an optical fiber, or an optical waveguide) may collect the Raman signal and direct the Raman signal to an optical receiver.

[0251] At step **5850**, the Raman signal is detected. Detection of the Raman signal **160**, which may be performed by an optical receiver **200**, includes steps **5852** and **5854**. At step **5852**, a third beam of light at a third frequency $\nu_{sub.3}$ is produced. The third beam of light (which may be referred to as a probe beam of light **120pr**) may be produced by a wavelength-tunable laser, where the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser. At step **5854**, a portion of the Raman signal is coherently mixed with at least a portion of the third beam of light to produce an electronic signal (e.g., a photocurrent signal *i*). The Raman signal and the third beam of light may be coherently mixed at a detector of an optical receiver. The portion of the Raman signal that is coherently mixed with the third beam of light may include optical frequency components of the Raman signal that are within a particular frequency range of the third frequency, and the particular frequency range may depend on the electronic bandwidth of the detector. For example, the particular frequency range may extend from approximately $\nu_{sub.3} - \Delta f$ to approximately $\nu_{sub.3} + \Delta f$, where $\nu_{sub.3}$ is the third frequency, and Δf is the electronic bandwidth of the detector.

[0252] At step **5860**, a characteristic of the electronic signal is determined, at which point the method may end. For example, a processor may receive a digital signal corresponding to the electronic signal, and the processor may determine the characteristic of the electronic signal based on the digital signal. The characteristic of the electronic signal may be associated with the Raman signal **160** and may include a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, an area, a frequency, a phase, or a polarization. Additionally, the characteristic of the electronic signal may be associated with a Raman shift at a frequency $\nu_{sub.1} - \nu_{sub.3}$, where

v.sub.1 is the first frequency, and v.sub.3 is the third frequency. Based on one or more determined signal characteristics **162**, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0253] FIG. **59** illustrates another example method **5900** for measuring a Raman signal **160**. The method may be performed by a Raman spectroscopy system as described herein. For example, the method may be performed by the Raman spectroscopy system with standoff detection **100** illustrated in FIG. **40**, **41**, **44**, **45**, **50**, or **51** or by the combined Raman spectroscopy and lidar system **100A** illustrated in FIGS. **46-49**.

[0254] At step **5910**, a pump light source **110_{pu}** produces a pump beam of light **120_{pu}** at a pump frequency (v.sub.pu). At step **5920**, a Stokes light source **110_S** produces a Stokes beam of light **120_S** at a Stokes frequency (v.sub.S), where (i) the pump and Stokes frequencies are offset by a frequency offset Ω and (ii) the pump and Stokes beams are directed to an object **150**. The object **150** (which may be referred to as a sample **150**) may be located external to the Raman spectroscopy system **100**, and the pump and Stokes beams may be directed to the object by an optical scanner **330** or by an operator of the Raman spectroscopy system. The pump beam of light may be referred to as a first beam of light at a first frequency (v.sub.1), and the Stokes beam of light may be referred to as a second beam of light at a second frequency (v.sub.2).

[0255] At step **5930**, a Raman signal **160** produced by coherent Raman scattering of the pump and Stokes beams of light at the object **150** is collected by one or more optical elements. The Raman signal **160** being produced at the object **150** may refer to the Raman signal being produced within the object, at the surface of the object, or near the object. For example, the Raman signal **160** may be produced by a particular material located within the object, on the surface of the object, or emitted by the object.

[0256] At step **5940**, an optical receiver **200** detects the Raman signal **160**, where detection of the Raman signal includes steps **5942**, **5944**, and **5946**. At step **5942**, a probe light source **110_{pr}** produces a probe beam of light **120_{pr}** at a probe frequency v.sub.pr. The probe light source **110_{pr}** may include a wavelength-tunable laser, where the probe frequency v.sub.pr is adjustable by changing the wavelength of light produced by the wavelength-tunable laser. The probe beam of light may be referred to as a third beam of light at a third frequency (v.sub.3). At step **5944**, the Raman signal **160** and the probe beam of light **120_{pr}** are coherently mixed at an optical detector **220** which produces a corresponding photocurrent signal i. At step **5946**, an electronic circuit **230** produces a digital output signal **240** that corresponds to the photocurrent signal i. For example, the electronic circuit **230** may include: (i) an electronic amplifier **232** that amplifies the photocurrent signal to produce a corresponding voltage signal **234** and (ii) a digitizer that produces a digital representation of the voltage signal.

[0257] At step **5950**, a processor determines a characteristic of the photocurrent signal i based on the digital output signal **240**, at which point the method may end. The digital output signal may include a digital representation of a voltage signal **234** that corresponds to the photocurrent signal i, and the characteristic of the photocurrent signal may be determined based on the digital representation of the voltage signal. The characteristic of the photocurrent signal may be associated with the Raman signal **160** and may include a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, or a polarization.

Additionally, the characteristic of the photocurrent signal may be associated with a Raman shift at a frequency v.sub.pu-v.sub.pr. Based on one or more determined signal characteristics **162**, a processor may determine (i) whether a particular material is present in or around an object or (ii) an amount or a concentration of the particular material located in or around the object. A processor of a Raman spectroscopy system **100** may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. A processor may be similar to the computer system **6000** illustrated in FIG. **60** and described herein.

[0258] Various example aspects directed to a Raman spectroscopy system are described below.

[0259] Aspect 1. A system comprising: a first light source configured to produce a first beam of light at a first frequency; a second light source configured to produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; one or more optical elements configured to: direct the first and second beams of light to a sample; and collect a Raman signal produced by the sample in response to the first and second beams of light; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a third light source configured to produce a third beam of light at a third frequency, wherein the third light source comprises a wavelength-tunable laser, wherein the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser; and an optical detector configured to coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and a processor configured to determine a characteristic of the electronic signal.

[0260] Aspect 2. The system of aspect 1, wherein: the third frequency is $\nu_{\text{sub.3}}$; the third light source is further configured to change the frequency of the third beam of light by a frequency change ΔF to a frequency $\nu_{\text{sub.3}} + \Delta F$; the detector is further configured to coherently mix another portion of the Raman signal with at least a portion of the third beam of light at the frequency $\nu_{\text{sub.3}} + \Delta F$ to produce another electronic signal; and the processor is further configured to determine a characteristic of the another electronic signal.

[0261] Aspect 3. The system of aspect 2, wherein: the characteristic of the electronic signal is associated with a Raman shift at a frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3}}$, wherein $\nu_{\text{sub.1}}$ is the first frequency; and the characteristic of the another electronic signal is associated with another Raman shift at a frequency $\nu_{\text{sub.1}} - (\nu_{\text{sub.3}} + \Delta F)$.

[0262] Aspect 4. The system of aspect 2, wherein the frequency change ΔF is between approximately 10 megahertz (MHz) and approximately 10 gigahertz (GHz).

[0263] Aspect 5. The system of aspect 1, wherein: the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and the processor is further configured to determine an amount or a concentration of the particular material in the sample based on the characteristic of the electronic signal.

[0264] Aspect 6. The system of aspect 1, wherein the frequency offset Ω is between approximately 5 terahertz (THz) and approximately 100 THz.

[0265] Aspect 7. The system of aspect 1, wherein the first light source or the second light source comprises a wavelength-tunable laser, wherein the frequency offset Ω is adjustable by changing a wavelength of the wavelength-tunable laser.

[0266] Aspect 8. The system of aspect 1, wherein the first light source or the second light source comprises two or more fixed-wavelength lasers, each of the fixed-wavelength lasers having a different operating wavelength, wherein the frequency offset Ω is adjustable by selecting one of the fixed-wavelength lasers for operation.

[0267] Aspect 9. The system of aspect 1, wherein: the frequency offset Ω is a first frequency offset $\Omega_{\text{sub.1}}$, and the Raman signal is a first Raman signal; subsequent to the optical receiver detecting the first Raman signal: the first light source is further configured to change the first frequency to produce a second frequency offset $\Omega_{\text{sub.2}}$ different from the first frequency offset $\Omega_{\text{sub.1}}$; and the optical receiver is further configured to detect a second Raman signal produced by the sample in response to the first and second beams of light with the second frequency offset $\Omega_{\text{sub.2}}$, wherein the detector is configured to coherently mix a portion of the second Raman signal with at least a portion of the third beam of light to produce a second electronic signal; and the processor is further configured to determine a characteristic of the second electronic signal.

[0268] Aspect 10. The system of aspect 1, wherein: the frequency offset Ω is a first frequency offset $\Omega_{\text{sub.1}}$, and the Raman signal is a first Raman signal; subsequent to the optical receiver detecting the first Raman signal: the second light source is further configured to change the second

frequency to a new second frequency to produce a second frequency offset $\Omega_{\text{sub.2}}$ different from the first frequency offset $\Omega_{\text{sub.1}}$; the third light source is further configured to change the third frequency to a new third frequency, wherein the new third frequency is within 200 GHz of the new second frequency; and the optical receiver is further configured to detect a second Raman signal produced by the sample in response to the first and second beams of light with the second frequency offset $\Omega_{\text{sub.2}}$, wherein the detector is configured to coherently mix a portion of the second Raman signal with at least a portion of the third beam of light to produce a second electronic signal; and the processor is further configured to determine a characteristic of the second electronic signal.

[0269] Aspect 11. The system of aspect 1, wherein the wavelength of light produced by the wavelength-tunable laser is adjustable over a wavelength range having a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0270] Aspect 12. The system of aspect 1, wherein the wavelength-tunable laser comprises a wavelength-tunable laser diode having a wavelength-tuning range with a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0271] Aspect 13. The system of aspect 1, wherein the wavelength-tunable laser comprises a sampled-grating distributed Bragg reflector (SG-DBR) laser.

[0272] Aspect 14. The system of aspect 1, wherein the wavelength of light produced by the wavelength-tunable laser is between approximately 1490 nanometers (nm) and approximately 1570 nm.

[0273] Aspect 15. The system of aspect 1, wherein the wavelength of light produced by the wavelength-tunable laser is between approximately 1000 nanometers (nm) and approximately 1100 nm.

[0274] Aspect 16. The system of aspect 1, wherein the third light source comprises: two or more laser diodes, wherein each of the laser diodes is a fixed-wavelength laser diode or a wavelength-tunable laser diode; and an optical multiplexer configured to combine light produced by each of the laser diodes into a single output beam of light.

[0275] Aspect 17. The system of aspect 16, wherein the third light source is configured to operate only one laser diode at a time, wherein: the third light source comprises N laser diodes, wherein N is an integer greater than or equal to 2; during a first measurement period, a first one of the N laser diodes is configured to produce light, and $N-1$ of the laser diodes, excluding the first one, are configured to not produce light; and during a second measurement period, a second one of the N laser diodes is configured to produce light, and $N-1$ of the laser diodes, excluding the second one, are configured to not produce light.

[0276] Aspect 18. The system of aspect 1, wherein the third light source comprises N wavelength-tunable laser diodes, wherein N is an integer greater than or equal to 2, and the third light source is configured to tune over one or more wavelength ranges having a total width between $(0.7)N \cdot \Delta\lambda_{\text{sub.av}}$ and $N \cdot \Delta\lambda_{\text{sub.av}}$, wherein $\Delta\lambda_{\text{sub.av}}$ is an average wavelength-tuning range of the N laser diodes.

[0277] Aspect 19. The system of aspect 1, wherein the second frequency and the third frequency differ by greater than 100 megahertz (MHz) and less than 200 gigahertz (GHz).

[0278] Aspect 20. The system of aspect 1, wherein the first or second light source is a wavelength-tunable light source, wherein the respective first or second frequency is adjustable over a frequency range corresponding to a wavelength range having a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0279] Aspect 21. The system of aspect 1, wherein the first frequency corresponds to a wavelength between approximately 1220 nanometers (nm) and approximately 1450 nm, and the second frequency corresponds to a wavelength between approximately 1490 nm and approximately 1570 nm.

[0280] Aspect 22. The system of aspect 1, wherein each of the first, second, and third beams of

light has a spectral linewidth of less than 200 megahertz (MHz).

[0281] Aspect 23. The system of aspect 1, wherein one or more of the first, second, and third beams of light have a spectral linewidth of less than 1 megahertz (MHz).

[0282] Aspect 24. The system of aspect 1, wherein each of the first light source, the second light source, and the third light source comprises one or more laser diodes, wherein each laser diode is a fixed-wavelength laser diode or a wavelength-tunable laser diode.

[0283] Aspect 25. The system of aspect 1, wherein the first light source, the second light source, or the third light source comprises a seed laser configured to produce seed light and an optical amplifier configured to amplify the seed light to produce an output beam of light, wherein the optical amplifier comprises a semiconductor optical amplifier (SOA) or a fiber-optic amplifier.

[0284] Aspect 26. The system of aspect 1, wherein the portion of the Raman signal that is coherently mixed with the third beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the third frequency, wherein the particular frequency range is based on an electronic bandwidth of the optical detector.

[0285] Aspect 27. The system of aspect 1, wherein the detector has an electronic bandwidth of approximately Δf , and the electronic signal produced by the detector comprises one or more electronic frequency components, each electronic frequency component having a frequency less than or equal to approximately Δf .

[0286] Aspect 28. The system of aspect 1, wherein the characteristic of the electronic signal comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, an area, a frequency, a phase, and a polarization.

[0287] Aspect 29. The system of aspect 1, wherein the frequency offset Ω equals $v_{\text{sub.1}} - v_{\text{sub.2}}$, wherein $v_{\text{sub.1}}$ is the first frequency, and $v_{\text{sub.2}}$ is the second frequency.

[0288] Aspect 30. The system of aspect 1, wherein each of the first, second, and third frequencies corresponds to a wavelength between approximately 300 nanometers (nm) and approximately 5,000 nm.

[0289] Aspect 31. The system of aspect 1, wherein the first frequency corresponds to a wavelength between approximately 1300 nanometers (nm) and approximately 1400 nm.

[0290] Aspect 32. The system of aspect 1, wherein the first frequency corresponds to a wavelength between approximately 890 nanometers (nm) and approximately 920 nm.

[0291] Aspect 33. The system of aspect 1, wherein the first frequency corresponds to a wavelength between approximately 700 nanometers (nm) and approximately 850 nm.

[0292] Aspect 34. The system of aspect 1, wherein the third frequency corresponds to a wavelength between approximately 1500 nanometers (nm) and approximately 1600 nm.

[0293] Aspect 35. The system of aspect 1, wherein the detector has an electronic bandwidth between approximately 100 megahertz (MHz) and approximately 50 gigahertz (GHz).

[0294] Aspect 36. The system of aspect 1, wherein the first light source, the second light source, or the third light source comprises a distributed feedback (DFB) laser diode.

[0295] Aspect 37. The system of aspect 1, wherein the first light source, the second light source, or the third light source comprises a wavelength-tunable laser diode comprising an external-cavity laser diode, a thermally tuned laser diode, or a sampled-grating distributed Bragg reflector (SG-DBR) laser.

[0296] Aspect 38. The system of aspect 1, wherein the first light source, the second light source, or the third light source comprises a light-emitting diode (LED), super-luminescent light source, short-pulse laser, broadband light source, fiber laser, or solid-state laser.

[0297] Aspect 39. The system of aspect 1, wherein the Raman signal is produced by coherent Raman scattering of the first and second beams of light within the sample.

[0298] Aspect 40. The system of aspect 1, wherein the sample comprises a biological material.

[0299] Aspect 41. The system of aspect 1, wherein the sample comprises an inorganic material.

[0300] Aspect 42. The system of aspect 1, wherein the sample comprises a crystalline material.

[0301] Aspect 43. The system of aspect 1, further comprising a half-wave plate configured to rotate a polarization of the first or second beam of light prior to being directed to the sample.

[0302] Aspect 44. The system of aspect 1, further comprising a quarter-wave plate configured to convert a polarization of the first or second beam of light to a circular or elliptical polarization prior to being directed to the sample.

[0303] Aspect 45. The system of aspect 1, further comprising a half-wave plate configured to rotate a polarization of the third beam of light.

[0304] Aspect 46. The system of aspect 1, further comprising an optical filter located between the sample and the optical receiver, the optical filter configured to transmit one or more wavelengths associated with the Raman signal and block one or more wavelengths associated with the first or second beam of light.

[0305] Aspect 47. The system of aspect 1, further comprising an optical polarizer located between the sample and the optical receiver, wherein the optical polarizer is oriented to transmit light with a polarization associated with the Raman signal.

[0306] Aspect 48. A method for measuring a Raman signal, the method comprising: producing a first beam of light at a first frequency; producing a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; directing the first and second beams of light to a sample; collecting a Raman signal produced by the sample in response to the first and second beams of light; detecting the Raman signal, comprising: producing a third beam of light at a third frequency, wherein the third beam of light is produced by a wavelength-tunable laser, wherein the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser; and coherently mixing a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and determining a characteristic of the electronic signal.

[0307] Aspect 50. One or more computer-readable non-transitory storage media embodying software that is operable when executed to: produce a first beam of light at a first frequency; produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; direct the first and second beams of light to a sample; collect a Raman signal produced by the sample in response to the first and second beams of light; detect the Raman signal, comprising: produce a third beam of light at a third frequency, wherein the third beam of light is produced by a wavelength-tunable laser, wherein the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser; and coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and determine a characteristic of the electronic signal.

[0308] Various example aspects directed to another Raman spectroscopy system are described below.

[0309] Aspect 1. A system comprising: a first light source configured to produce a first beam of light at a first frequency; a second light source configured to produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; one or more optical elements configured to: direct the first and second beams of light to a sample; and collect a Raman signal produced by coherent Raman scattering of the first and second beams of light within the sample; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a third light source configured to produce a third beam of light at a third frequency; and an optical detector configured to coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal, wherein the portion of the Raman signal that is coherently mixed with the third beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the third frequency, wherein the particular frequency range is based on an electronic bandwidth of the detector; and a processor configured to determine a characteristic of the electronic signal.

[0310] Aspect 2. The system of aspect 1, wherein the particular frequency range extends from approximately $v_{\text{sub.3}} - \Delta f$ to approximately $v_{\text{sub.3}} + \Delta f$, wherein $v_{\text{sub.3}}$ is the third frequency, and Δf is the electronic bandwidth of the detector.

[0311] Aspect 3. The system of aspect 1, wherein the electronic signal produced by the detector comprises one or more electronic frequency components, each electronic frequency component having a frequency less than or equal to Δf , wherein Δf is the electronic bandwidth of the detector.

[0312] Aspect 4. The system of aspect 1, wherein the detector has an electronic bandwidth between approximately 100 megahertz (MHz) and approximately 10 gigahertz (GHz).

[0313] Aspect 5. The system of aspect 1, wherein the electronic signal comprises one or more electronic frequency components less than or equal to approximately 10 gigahertz (GHz).

[0314] Aspect 6. The system of aspect 1, wherein the detector comprises a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT).

[0315] Aspect 7. The system of aspect 1, wherein: the electronic signal comprises a photocurrent signal produced by the detector; and the optical receiver further comprises: an electronic amplifier configured to amplify the photocurrent signal to produce a voltage signal corresponding to the photocurrent signal; and a digitizer configured to produce a digital representation of the voltage signal.

[0316] Aspect 8. The system of aspect 7, wherein the processor is configured to determine the characteristic of the electronic signal based on the digital representation of the voltage signal, wherein the characteristic of the electronic signal comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, an area, a frequency, a phase, and a polarization.

[0317] Aspect 9. The system of aspect 7, wherein the voltage signal is a time-domain signal, and the processor is further configured to determine a Fourier transform of the digital representation of the voltage signal to determine a frequency-domain representation of the voltage signal.

[0318] Aspect 10. The system of aspect 1, wherein the processor is further configured to associate a Raman frequency shift with the determined characteristic of the electronic signal, wherein the Raman frequency shift equals $v_{\text{sub.1}} - v_{\text{sub.3}}$, wherein $v_{\text{sub.1}}$ is the first frequency, and $v_{\text{sub.3}}$ is the third frequency.

[0319] Aspect 11. The system of aspect 1, wherein the electronic signal comprises a photocurrent signal corresponding to the coherent mixing of the portion of the Raman signal and the third beam of light.

[0320] Aspect 12. The system of aspect 1, wherein the electronic signal comprises a coherent-mixing term that is proportional to a product of (i) an amplitude of an electric field of the Raman signal and (ii) an amplitude of an electric field of the third beam of light.

[0321] Aspect 13. The system of aspect 12, wherein the coherent-mixing term is proportional to $E_{\text{sub.R}} \cdot E_{\text{sub.3}} \cdot \cos[2\pi(v_{\text{sub.R}} - v_{\text{sub.3}})t + \Delta\phi]$, wherein: $E_{\text{sub.R}}$ is the amplitude of the electric field of the Raman signal; $E_{\text{sub.3}}$ is the amplitude of the electric field of the third beam of light; $v_{\text{sub.R}}$ is a frequency of the electric field of the Raman signal; $v_{\text{sub.3}}$ is a frequency of the electric field of the third beam of light; and $\Delta\phi$ is a phase difference between the electric field of the Raman signal and the electric field of the third beam of light.

[0322] Aspect 14. The system of aspect 1, wherein the first, second, or third light source is a wavelength-tunable light source, wherein the first, second, or third frequency is adjustable over a frequency range corresponding to a wavelength range having a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0323] Aspect 15. The system of aspect 1, wherein the second frequency and the third frequency differ by greater than 100 megahertz (MHz) and less than 200 gigahertz (GHz).

[0324] Aspect 16. The system of aspect 1, wherein: the third frequency is $v_{\text{sub.3}}$; the third light

source is further configured to change the frequency of the third beam of light by a frequency change ΔF to a frequency $\nu_{.3} + \Delta F$; the detector is further configured to coherently mix another portion of the Raman signal with at least a portion of the third beam of light at the frequency $\nu_{.3} + \Delta F$ to produce another electronic signal; and the processor is further configured to determine a characteristic of the another electronic signal.

[0325] Aspect 17. The system of aspect 1, wherein: the third light source comprises a wavelength-tunable laser, wherein the third light source is further configured to sequentially change the frequency of the third beam of light to a plurality of different frequencies; the detector is further configured to coherently mix another portion of the Raman signal with at least a portion of one of the different frequencies of the third beam of light to produce a corresponding one of a plurality of electronic signals; and the processor is further configured to determine a characteristic of each of the plurality of electronic signals.

[0326] Aspect 18. The system of aspect 17, wherein the processor is further configured to determine, based on the determined characteristics of the electronic signals, (i) whether a particular material is present in the sample or (ii) an amount or a concentration of the particular material in the sample.

[0327] Aspect 19. The system of aspect 17, wherein each of the different frequencies to which the third light source is changed is offset from an adjacent one of the different frequencies by between approximately 10 megahertz (MHz) and approximately 10 gigahertz (GHz).

[0328] Aspect 20. The system of aspect 17, wherein the processor is further configured to determine a Raman spectrum based on the determined characteristics of the electronic signals.

[0329] Aspect 21. The system of aspect 20, wherein: the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and the processor is further configured to determine an amount or a concentration of the particular material in the sample based on the determined Raman spectrum.

[0330] Aspect 22. The system of aspect 1, wherein the first light source or the second light source comprises a wavelength-tunable laser, wherein the frequency offset Ω is adjustable by changing a wavelength of the wavelength-tunable laser.

[0331] Aspect 23. The system of aspect 1, wherein the frequency offset Ω is between approximately 5 terahertz (THz) and approximately 100 THz.

[0332] Aspect 24. The system of aspect 1, wherein the frequency offset Ω is approximately equal to a vibrational frequency of a particular material.

[0333] Aspect 25. The system of aspect 24, wherein the processor is further configured to determine whether the particular material is present in the sample based on the characteristic of the electronic signal.

[0334] Aspect 26. The system of aspect 24, wherein the processor is further configured to determine an amount or a concentration of the particular material in the sample based on the characteristic of the electronic signal.

[0335] Aspect 27. The system of aspect 1, wherein the Raman signal is an optical signal having a spectral linewidth between approximately 30 gigahertz (GHz) and approximately 300 GHz.

[0336] Aspect 28. The system of aspect 1, wherein the Raman signal is an optical signal having a center frequency approximately equal to $2\nu_{.1} - \nu_{.2}$, wherein $\nu_{.1}$ is the first frequency, and $\nu_{.2}$ is the second frequency.

[0337] Aspect 29. The system of aspect 1, wherein the Raman signal is an optical signal having a center frequency within 200 gigahertz (GHz) of the first frequency or the second frequency.

[0338] Aspect 30. The system of aspect 1, wherein: the optical receiver is further configured to detect residual light from the first beam of light after the first beam of light has interacted with the sample, wherein the optical receiver further comprises: a fourth light source configured to produce a fourth beam of light at a fourth frequency, wherein the fourth frequency is within 50 GHz of the first frequency; and an additional optical detector configured to coherently mix at least a portion of

the residual light with at least a portion of the fourth beam of light to produce an additional electronic signal; and the processor is further configured to determine a characteristic of the additional electronic signal.

[0339] Aspect 31. A method for measuring a Raman signal, the method comprising: producing a first beam of light at a first frequency; producing a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; directing the first and second beams of light to a sample; collecting a Raman signal produced by coherent Raman scattering of the first and second beams of light within the sample; detecting the Raman signal, comprising: producing a third beam of light at a third frequency; and coherently mixing, by an optical detector, a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal, wherein the portion of the Raman signal that is coherently mixed with the third beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the third frequency, wherein the particular frequency range is based on an electronic bandwidth of the detector; and determining a characteristic of the electronic signal.

[0340] Aspect 32. One or more computer-readable non-transitory storage media embodying software that is operable when executed to: produce a first beam of light at a first frequency; produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; direct the first and second beams of light to a sample; collect a Raman signal produced by coherent Raman scattering of the first and second beams of light within the sample; detect the Raman signal, comprising: produce a third beam of light at a third frequency; and coherently mix, by an optical detector, a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal, wherein the portion of the Raman signal that is coherently mixed with the third beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the third frequency, wherein the particular frequency range is based on an electronic bandwidth of the detector; and determine a characteristic of the electronic signal.

[0341] Various example aspects directed to another Raman spectroscopy system are described below.

[0342] Aspect 1. A system comprising: a first light source configured to produce a first beam of light at a first frequency; a second light source configured to produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; one or more optical elements configured to: direct the first and second beams of light to a sample; and collect a Raman signal produced by the sample in response to the first and second beams of light; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a third light source configured to produce a third beam of light at a third frequency; and one or more optical detectors, wherein each detector is configured to coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and a processor configured to determine a characteristic of the electronic signal.

[0343] Aspect 2. The system of aspect 1, wherein the processor is further configured to determine a phase difference between the Raman signal and the third beam of light.

[0344] Aspect 3. The system of aspect 1, wherein the processor is further configured to determine an in-phase portion and a quadrature portion associated with the Raman signal.

[0345] Aspect 4. The system of aspect 1, wherein the optical receiver further comprises a 90-degree optical hybrid configured to: combine the Raman signal with the third beam of light to produce four combined beams, the four combined beams comprising two in-phase combined beams and two quadrature combined beams, wherein each combined beam comprises a portion of the Raman signal and a portion of the third beam of light; and direct each of the combined beams to one of four detectors of the optical receiver.

[0346] Aspect 5. The system of aspect 4, wherein: prior to combining the Raman signal with the third beam of light, the 90-degree optical hybrid is configured to split the Raman signal or the third

beam of light into a first part and a second part; and the 90-degree optical hybrid comprises a phase shifter configured to impart a 90-degree phase change to the first part with respect to the second part. Aspect 6. The system of aspect 4, wherein: the four detectors are each configured to coherently mix the portion of the Raman signal and the portion of the third beam of light to produce one of four electronic signals; and the processor is further configured to determine a phase difference between the Raman signal and the third beam of light.

[0347] Aspect 7. The system of aspect 1, wherein the processor is further configured to determine a polarization of the Raman signal.

[0348] Aspect 8. The system of aspect 1, wherein the optical receiver further comprises: a Raman-signal polarization beamsplitter configured to split the Raman signal into a horizontal-polarization Raman signal and a vertical-polarization Raman signal; a third-beam polarization beamsplitter configured to split the third beam into a horizontal-polarization third beam and a vertical-polarization third beam; a horizontal-polarization optical receiver comprising one or more of the optical detectors, wherein each detector is configured to coherently mix at least a portion of the horizontal-polarization Raman signal and horizontal-polarization third beam to produce a horizontal-polarization electronic signal; and a vertical-polarization optical receiver comprising another one or more of the optical detectors, wherein each detector is configured to coherently mix at least a portion of the vertical-polarization Raman signal and vertical-polarization third beam to produce a vertical-polarization electronic signal.

[0349] Aspect 9. The system of aspect 8, wherein determining the characteristic of the electronic signal comprises determining one or more characteristics of the horizontal-polarization and vertical-polarization electronic signals.

[0350] Aspect 10. The system of aspect 9, wherein the processor is further configured to determine a polarization of the Raman signal based on the characteristics of the horizontal-polarization and vertical-polarization electronic signals.

[0351] Aspect 11. The system of aspect 8, wherein: the horizontal-polarization optical receiver comprises: four optical detectors; and a first 90-degree optical hybrid configured to: combine the horizontal-polarization Raman signal with the horizontal-polarization third beam to produce four combined horizontal-polarization beams, each combined horizontal-polarization beam comprising a portion of the horizontal-polarization Raman signal and a portion of the horizontal-polarization third beam; and direct each of the combined horizontal-polarization beams to one of the four detectors; and the vertical-polarization optical receiver comprises: another four optical detectors; and a second 90-degree optical hybrid configured to: combine the vertical-polarization Raman signal with the vertical-polarization third beam to produce four combined vertical-polarization beams, each combined vertical-polarization beam comprising a portion of the vertical-polarization Raman signal and a portion of the vertical-polarization third beam; and direct each of the combined vertical-polarization beams to one of the another four optical detectors.

[0352] Aspect 12. The system of aspect 11, wherein: the four detectors of the horizontal-polarization optical receiver are each configured to coherently mix the portion of the horizontal-polarization Raman signal and the portion of the horizontal-polarization third beam to produce one of four horizontal-polarization electronic signals; the four detectors of the vertical-polarization optical receiver are each configured to coherently mix the portion of the vertical-polarization Raman signal and the portion of the vertical-polarization third beam to produce one of four vertical-polarization electronic signals; and based on the four horizontal-polarization electronic signals and the four vertical-polarization electronic signals, the processor is further configured to determine (i) a polarization of the Raman signal and (ii) a phase difference between the Raman signal and the third beam of light.

[0353] Aspect 13. The system of aspect 1, wherein the optical elements comprise an optical combiner configured to combine the first and second beams of light to produce a combined beam that is directed to the sample.

[0354] Aspect 14. The system of aspect 13, wherein the optical combiner is part of a photonic integrated circuit (PIC), wherein the first and second beams of light are combined into an optical waveguide of the PIC.

[0355] Aspect 15. The system of aspect 13, wherein the optical combiner is a fiber-optic combiner, wherein the first and second beams of light are combined into an optical fiber.

[0356] Aspect 16. The system of aspect 1, wherein the optical receiver further comprises an optical combiner configured to combine the Raman signal and the third beam of light to produce one or more combined beams that are each directed to one of the optical detectors.

[0357] Aspect 17. The system of aspect 16, wherein the optical combiner is part of a photonic integrated circuit (PIC), wherein the portion of the Raman signal and the portion of the third beam of light are combined into an optical waveguide of the PIC.

[0358] Aspect 18. The system of aspect 16, wherein the optical combiner is a fiber-optic combiner, wherein the portion of the Raman signal and the portion of the third beam of light are combined into an optical fiber.

[0359] Aspect 19. The system of aspect 1, wherein the optical elements comprise a photonic integrated circuit (PIC) comprising one or more optical waveguides, wherein: one or more optical waveguides are configured to direct the first and second beams of light to the sample; and one or more other optical waveguides are configured to direct the Raman signal and the third beam of light to the one or more of the detectors.

[0360] Aspect 20. The system of aspect 1, wherein the first, second, or third light source is a wavelength-tunable light source, wherein the first, second, or third frequency is adjustable over a frequency range corresponding to a wavelength range having a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0361] Aspect 21. The system of aspect 1, wherein the frequency offset Ω is between approximately 5 terahertz (THz) and approximately 100 THz.

[0362] Aspect 22. The system of aspect 1, wherein: the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and the processor is further configured to determine, based on the characteristic of the electronic signal, (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material in the sample.

[0363] Aspect 23. The system of aspect 1, wherein: the electronic signal comprises a photocurrent signal produced by the detector; and the optical receiver further comprises: an electronic amplifier configured to amplify the photocurrent signal to produce a voltage signal corresponding to the photocurrent signal; and a digitizer configured to produce a digital representation of the voltage signal.

[0364] Aspect 24. The system of aspect 23, wherein the processor is configured to determine the characteristic of the electronic signal based on the digital representation of the voltage signal, wherein the characteristic of the electronic signal comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, an area, a frequency, a phase, and a polarization.

[0365] Aspect 25. The system of aspect 1, wherein each detector has an electronic bandwidth of approximately Δf , and the electronic signal produced by each detector comprises one or more electronic frequency components, each electronic frequency component having a frequency less than or equal to Δf .

[0366] Aspect 26. The system of aspect 1, wherein the one or more detectors comprise a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT).

[0367] Aspect 27. A method for measuring a Raman signal, the method comprising: producing a first beam of light at a first frequency; producing a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; directing the first and

second beams of light to a sample; collecting a Raman signal produced by the sample in response to the first and second beams of light; detecting the Raman signal, comprising: producing a third beam of light at a third frequency; and coherently mixing a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and determining a characteristic of the electronic signal.

[0368] Aspect 28. One or more computer-readable non-transitory storage media embodying software that is operable when executed to: produce a first beam of light at a first frequency; produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; direct the first and second beams of light to a sample; collect a Raman signal produced by the sample in response to the first and second beams of light; detect the Raman signal, comprising: produce a third beam of light at a third frequency; and coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and determine a characteristic of the electronic signal.

[0369] Various example aspects directed to another Raman spectroscopy system are described below.

[0370] Aspect 1. A system comprising: a pump light source configured to produce a pump beam of light at a pump frequency; one or more optical elements configured to: direct the pump beam to a sample; and collect a Raman signal produced by the sample in response to the pump beam, wherein the Raman signal is produced by spontaneous Raman scattering of the pump beam within the sample; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a probe light source configured to produce a probe beam of light at a probe frequency; and one or more optical detectors, wherein each detector is configured to coherently mix a portion of the Raman signal with at least a portion of the probe beam of light to produce an electronic signal; and a processor configured to determine a characteristic of the electronic signal.

[0371] Aspect 2. The system of aspect 1, wherein the probe light source comprises a wavelength-tunable laser configured to tune the probe frequency of the probe beam of light to a plurality of frequencies across at least a portion of the Raman signal.

[0372] FIG. 60 illustrates an example computer system 6000. One or more computer systems 6000 may perform one or more steps of one or more methods described or illustrated herein. One or more computer systems 6000 may provide functionality described or illustrated herein. Software running on one or more computer systems 6000 may perform one or more steps of one or more methods described or illustrated herein or may provide functionality described or illustrated herein. A computer system 6000 may include or may be referred to as a processor, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. Herein, reference to a computer system may encompass one or more computer systems, where appropriate. A processor of a Raman spectroscopy system 100 may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus.

[0373] Computer system 6000 may take any suitable physical form. As an example, computer system 6000 may be an embedded computer system, a system-on-chip (SOC), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a single-board computer system (SBC), a desktop computer system, a laptop or notebook computer system, a graphics processing unit (GPU), a mainframe, a mesh of computer systems, a server, a tablet computer system, or any suitable combination of two or more of these. As another example, all or part of computer system 6000 may be combined with, coupled to, or integrated into a variety of devices, including, but not limited to, a medical device, wearable device, camera, camcorder, personal digital assistant (PDA), mobile telephone, smartphone, electronic reading device (e.g., an e-reader), game console, smart watch, clock, calculator, television monitor, flat-panel display, computer monitor, eyewear, or head-mounted display. Where appropriate, computer system 6000 may include one or more computer systems 6000; be unitary or distributed; span multiple

locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **6000** may perform without substantial spatial or temporal limitation one or more steps of one or more methods described or illustrated herein. As an example, one or more computer systems **6000** may perform in real time or in batch mode one or more steps of one or more methods described or illustrated herein. One or more computer systems **6000** may perform at different times or at different locations one or more steps of one or more methods described or illustrated herein, where appropriate.

[0374] As illustrated in the example of FIG. **60**, computer system **6000** may include a processor **6010**, memory **6020**, storage **6030**, an input/output (I/O) interface **6040**, a communication interface **6050**, or a bus **6060**. Computer system **6000** may include any suitable number of any suitable components in any suitable arrangement.

[0375] Processor **6010** may include hardware for executing instructions, such as those making up a computer program. As an example, to execute instructions, processor **6010** may retrieve (or fetch) the instructions from an internal register, an internal cache, memory **6020**, or storage **6030**; decode and execute them; and then write one or more results to an internal register, an internal cache, memory **6020**, or storage **6030**. A processor **6010** may include one or more internal caches for data, instructions, or addresses. A processor **6010** may include one or more internal registers for data, instructions, or addresses. Processor **6010** may include any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor **6010** may include one or more arithmetic logic units (ALUs); may be a multi-core processor; or may include one or more processors **6010**.

[0376] Memory **6020** may include main memory for storing instructions for processor **6010** to execute or data for processor **6010** to operate on. As an example, computer system **6000** may load instructions from storage **6030** or another source (such as, for example, another computer system **6000**) to memory **6020**. Processor **6010** may then load the instructions from memory **6020** to an internal register or internal cache. To execute the instructions, processor **6010** may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor **6010** may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor **6010** may then write one or more of those results to memory **6020**. One or more memory buses (which may each include an address bus and a data bus) may couple processor **6010** to memory **6020**. Bus **6060** may include one or more memory buses. Memory **6020** may include random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Memory **6020** may include one or more memories **6020**, where appropriate.

[0377] Storage **6030** may include mass storage for data or instructions. As an example, storage **6030** may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage **6030** may include removable or non-removable (or fixed) media, where appropriate. Storage **6030** may be internal or external to computer system **6000**, where appropriate. Storage **6030** may be non-volatile, solid-state memory. Storage **6030** may include read-only memory (ROM). Where appropriate, this ROM may be mask ROM (MROM), programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), flash memory, or a combination of two or more of these. Storage **6030** may include one or more storage control units facilitating communication between processor **6010** and storage **6030**, where appropriate. Where appropriate, storage **6030** may include one or more storages **6030**.

[0378] I/O interface **6040** may include hardware, software, or both, providing one or more interfaces for communication between computer system **6000** and one or more I/O devices. Computer system **6000** may include one or more of these I/O devices, where appropriate. One or

more of these I/O devices may enable communication between a person and computer system **6000**. As an example, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, camera, stylus, tablet, touch screen, trackball, another suitable I/O device, or any suitable combination of two or more of these. An I/O device may include one or more sensors. Where appropriate, I/O interface **6040** may include one or more device or software drivers enabling processor **6010** to drive one or more of these I/O devices. I/O interface **6040** may include one or more I/O interfaces **6040**, where appropriate.

[0379] Communication interface **6050** may include hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computer system **6000** and one or more other computer systems **6000** or one or more networks. As an example, communication interface **6050** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network or a wireless NIC (WNIC); a wireless adapter for communicating with a wireless network, such as a WI-FI network; or an optical transmitter (e.g., a laser or a light-emitting diode) or an optical receiver (e.g., a photodetector) for communicating using fiber-optic communication or free-space optical communication. Computer system **6000** may communicate with an ad hoc network, a personal area network (PAN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), or one or more portions of the Internet or a combination of two or more of these. One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **6000** may communicate with a wireless PAN (WPAN) (such as, for example, a BLUETOOTH WPAN), a WI-FI network, a Worldwide Interoperability for Microwave Access (WiMAX) network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network), or other suitable wireless network or a combination of two or more of these. As another example, computer system **6000** may communicate using fiber-optic communication based on 100 Gigabit Ethernet (100 GbE), 10 Gigabit Ethernet (10 GbE), or Synchronous Optical Networking (SONET). Computer system **6000** may include any suitable communication interface **6050** for any of these networks, where appropriate. Communication interface **6050** may include one or more communication interfaces **6050**, where appropriate.

[0380] Bus **6060** may include hardware, software, or both coupling components of computer system **6000** to each other. As an example, bus **6060** may include an Accelerated Graphics Port (AGP) or other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCIe) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local bus (VLB), or another suitable bus or a combination of two or more of these. Bus **6060** may include one or more buses **6060**, where appropriate.

[0381] Various modules, circuits, systems, methods, or algorithm steps described in connection with the implementations disclosed herein may be implemented as electronic hardware, computer software, or any suitable combination of hardware and software. Computer software (which may be referred to as software, computer-executable code, computer code, a computer program, computer instructions, or instructions) may be used to perform various functions described or illustrated herein, and computer software may be configured to be executed by or to control the operation of computer system **6000**. As an example, computer software may include instructions configured to be executed by processor **6010**. Owing to the interchangeability of hardware and software, the various illustrative logical blocks, modules, circuits, or algorithm steps have been described generally in terms of functionality. Whether such functionality is implemented in hardware, software, or a combination of hardware and software may depend upon the particular application or design constraints imposed on the overall system.

[0382] A computing device may be used to implement various modules, circuits, systems, methods,

or algorithm steps disclosed herein. As an example, all or part of a module, circuit, system, method, or algorithm disclosed herein may be implemented or performed by a general-purpose single- or multi-chip processor, a digital signal processor (DSP), an ASIC, a FPGA, a GPU, any other suitable programmable-logic device, discrete gate or transistor logic, discrete hardware components, or any suitable combination thereof. A general-purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, GPU, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0383] One or more implementations of the subject matter described herein may be implemented as one or more computer programs (e.g., one or more modules of computer-program instructions encoded or stored on a computer-readable non-transitory storage medium). As an example, the steps of a method or algorithm disclosed herein may be implemented in a processor-executable software module which may reside on a computer-readable non-transitory storage medium. A computer-readable non-transitory storage medium may include any suitable storage medium that may be used to store or transfer computer software and that may be accessed by a computer system. Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such, as for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives (HHDs), optical discs (e.g., compact discs (CDs), CD-ROM, digital versatile discs (DVDs), Blu-ray discs, or laser discs), optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, flash memories, solid-state drives (SSDs), RAM, RAM-drives, ROM, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

[0384] Certain features described herein in the context of separate implementations may also be combined and implemented in a single implementation. Conversely, various features that are described in the context of a single implementation may also be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

[0385] While operations may be depicted in the drawings as occurring in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all operations be performed. Further, the drawings may schematically depict one more example processes or methods in the form of a flow diagram or a sequence diagram. However, other operations that are not depicted may be incorporated in the example processes or methods that are schematically illustrated. For example, one or more additional operations may be performed before, after, simultaneously with, or between any of the illustrated operations. Moreover, one or more operations depicted in a diagram may be repeated, where appropriate. Additionally, operations depicted in a diagram may be performed in any suitable order. Furthermore, although particular components, devices, or systems are described herein as carrying out particular operations, any suitable combination of any suitable components, devices, or systems may be used to carry out any suitable operation or combination of operations. In certain circumstances, multitasking or parallel processing operations may be performed. Moreover, the separation of various system components in the implementations described herein should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems may be integrated together in a single software

product or packaged into multiple software products.

[0386] Various embodiments have been described in connection with the accompanying drawings. However, it should be understood that the figures may not necessarily be drawn to scale. As an example, distances or angles depicted in the figures are illustrative and may not necessarily bear an exact relationship to actual dimensions or layout of the devices illustrated.

[0387] One or more of the figures described herein may include example data that is prophetic. For example, the graphs illustrated FIGS. 3-16, 19, 42, 43, and 50-56 may include or may be referred to as prophetic examples.

[0388] The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes or illustrates respective embodiments herein as including particular components, elements, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend.

[0389] The term “or” as used herein is to be interpreted as an inclusive or meaning any one or any combination, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, the expression “A or B” means “A, B, or both A and B.” As another example, herein, “A, B, or C” means at least one of the following: A; B; C; A and B; A and C; B and C; A, B, and C. An exception to this definition will occur if a combination of elements, devices, steps, or operations is in some way inherently mutually exclusive.

[0390] As used herein, words of approximation such as, without limitation, “approximately,” “substantially,” or “about” refer to a condition that when so modified is understood to not necessarily be absolute or perfect but would be considered close enough to those of ordinary skill in the art to warrant designating the condition as being present. The extent to which the description may vary will depend on how great a change can be instituted and still have one of ordinary skill in the art recognize the modified feature as having the required characteristics or capabilities of the unmodified feature. In general, but subject to the preceding discussion, a numerical value herein that is modified by a word of approximation such as “approximately” may vary from the stated value by $\pm 0.5\%$, $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$, $\pm 10\%$, $\pm 12\%$, or $\pm 15\%$. The term “substantially constant” refers to a value that varies by less than a particular amount over any suitable time interval. For example, a value that is substantially constant may vary by less than or equal to 20%, 10%, 1%, 0.5%, or 0.1% over a time interval of approximately 104 s, 103 s, 102 s, 10 s, 1 s, 100 ms, 10 ms, 1 ms, 100 μ s, 10 μ s, or 1 μ s. The term “substantially constant” may be applied to any suitable value, such as for example, an optical power, an electrical current, a wavelength, an optical or electrical frequency, or an optical or electrical phase.

[0391] As used herein, the terms “first,” “second,” “third,” etc. may be used as labels for nouns that they precede, and these terms may not necessarily imply a particular ordering (e.g., a particular spatial, temporal, or logical ordering). As an example, a system may be described as determining a “first result” and a “second result,” and the terms “first” and “second” may not necessarily imply that the first result is determined before the second result.

[0392] As used herein, the terms “based on” and “based at least in part on” may be used to describe or present one or more factors that affect a determination, and these terms may not exclude additional factors that may affect a determination. A determination may be based solely on those factors which are presented or may be based at least in part on those factors. The phrase “determine A based on B” indicates that B is a factor that affects the determination of A. In some instances, other factors may also contribute to the determination of A. In other instances, A may be determined based solely on B.

[0393] Various example aspects included in this disclosure may be presented in a range format. It

should be understood that a description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of a disclosed aspect. Accordingly, the description of a range should be considered to have specifically disclosed all suitable sub-ranges as well as individual numerical values within that range, unless expressly indicated otherwise or indicated otherwise by context. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed sub-ranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6, etc., as well as individual values within that range, for example, 1.1, 2, 2.3, 5, and 5.9.

[0394] Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There may be numerous alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

Claims

1. A system comprising: a pump light source configured to produce a pump beam of light at a pump frequency; a Stokes light source configured to produce a Stokes beam of light at a Stokes frequency, wherein the pump and Stokes frequencies are offset by a frequency offset Ω , and wherein the pump and Stokes beams are directed to an object; one or more optical elements configured to collect a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the object; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a probe light source configured to produce a probe beam of light at a probe frequency; an optical detector configured to coherently mix the Raman signal with the probe beam of light to produce a corresponding photocurrent signal; and an electronic circuit configured to produce a digital output signal corresponding to the photocurrent signal; and a processor configured to determine a characteristic of the photocurrent signal based on the digital output signal.
2. The system of claim 1, wherein the object is located external to the system.
3. The system of claim 2, wherein the object is located between 1 meter and 10 kilometers from the system.
4. The system of claim 2, wherein the optical elements comprise an optical scanner configured to direct the pump and Stokes beams of light to the object.
5. The system of claim 2, wherein the pump and Stokes beams of light are directed to the object by an operator of the system.
6. The system of claim 2, wherein the processor is further configured to select a pulse energy or an optical power of the pump beam of light or the Stokes beam of light based on a distance from the system to the object, wherein the selected pulse energy or optical power is higher for larger values of distance.
7. The system of claim 1, further comprising a lidar system, wherein: the object is located external to the system; the lidar system is configured to scan a region around the system and determine a location of the object.
8. The system of claim 7, wherein: prior to the system producing the pump and Stokes beam of light, the lidar system is configured to scan the region around the system and determine the location of the object; and after the location of the object has been determined, the optical elements are further configured to direct the pump and Stokes beams of light to the object based on the determined location of the object.
9. The system of claim 7, wherein: the lidar system is further configured to classify one or more objects in the region around the system, the one or more objects comprising the object, wherein the object is classified as an object of interest; and the optical elements are further configured to direct the pump and Stokes beams of light to the object in response to the object being classified as an

object of interest.

10. The system of claim 1, wherein: the object is located external to the system; prior to producing the pump and Stokes beams of light, the pump light source or the Stokes light source is configured to produce a lidar output beam comprising a plurality of output pulses of light; the optical elements comprise an optical scanner configured to scan the lidar output beam across a region around the system; the optical receiver is further configured to detect a lidar input beam comprising input pulses of light, each input pulse of light including scattered light from one of the output pulses of light, wherein: the optical detector is further configured to produce a pulse of photocurrent corresponding to an input pulse of light, the input pulse of light including light from a corresponding output pulse of light scattered from the object; and the electronic circuit is further configured to produce a digital output signal that corresponds to the pulse of photocurrent; and the processor is further configured to determine a location of the object based on the digital output signal that corresponds to the pulse of photocurrent, wherein determining the location of the object comprises determining a distance to the object based on a time of arrival of the input pulse of light.

11. The system of claim 10, wherein: the processor is further configured to classify one or more objects in the region around the system based on one or more scans of the lidar output beam across the region, wherein the one or more objects comprise the object, and the object is classified as an object of interest; and the pump and Stokes beams of light are directed to the object in response to the object being classified as an object of interest.

12. The system of claim 10, wherein, after the location of the object has been determined, the optical scanner is further configured to direct the pump and Stokes beams of light to the object based on the determined location of the object.

13. The system of claim 10, wherein determining the distance to the object comprises determining a difference between the time of arrival of the input pulse of light and a time of emission of the corresponding output pulse of light.

14. The system of claim 10, wherein each of the output pulses of light has a duration between 0.1 nanosecond and 10 nanoseconds and a pulse energy between 1 microjoule and 1 millijoule.

15. The system of claim 10, wherein: the probe light source is further configured to produce a local-oscillator (LO) beam of light; and the optical detector is further configured to coherently mix the LO beam of light and the input pulse of light to produce the pulse of photocurrent.

16. The system of claim 15, wherein the processor is further configured to determine, based on the digital output signal that corresponds to the pulse of photocurrent, that the input pulse of light includes light from the corresponding output pulse of light.

17. The system of claim 1, wherein: the object is located external to the system; the pump light source or the Stokes light source is further configured to produce an output pulse of light that is directed to the object; the optical receiver is further configured to detect an input pulse of light that includes light from the output pulse of light scattered from the object, wherein: the optical detector is further configured to produce a pulse of photocurrent corresponding to the input pulse of light; and the electronic circuit is further configured to produce a digital output signal that corresponds to the pulse of photocurrent; and the processor is further configured to determine a distance to the object based on the digital output signal that corresponds to the pulse of photocurrent, wherein determining the distance to the object comprises determining a time of arrival of the input pulse of light.

18. The system of claim 17, wherein: the probe light source is further configured to produce a local-oscillator (LO) beam of light; and the optical detector is further configured to coherently mix the LO beam of light and the input pulse of light to produce the pulse of photocurrent.

19. The system of claim 1, wherein: the object is located external to the system; the Stokes beam of light comprises a Stokes pulse of light, wherein the Raman signal is produced by coherent Raman scattering of the pump beam of light and the Stokes pulse of light; the optical receiver is further configured to detect an input pulse of light, the input pulse of light including light from the Stokes

pulse of light scattered from the object, wherein: the photocurrent signal produced by the optical detector comprises a pulse of photocurrent corresponding to the input pulse of light; and at least a portion of the digital output signal produced by the electronic circuit corresponds to the pulse of photocurrent; and the processor is further configured to determine a distance to the object based on the portion of the digital output signal that corresponds to the pulse of photocurrent.

20. The system of claim 1, wherein: the object is located external to the system; the pump beam of light comprises a pump pulse of light, wherein the Raman signal is produced by coherent Raman scattering of the pump pulse of light and the Stokes beam of light; the system further comprises: an optical beamsplitter configured to direct an input pulse of light to a pump detector, the input pulse of light including light from the pump pulse of light scattered from the object; and the pump detector, wherein the pump detector is configured to produce a pulse of photocurrent corresponding to the input pulse of light; and the processor is further configured to determine a distance to the object based on a time of arrival of the input pulse of light.

21. The system of claim 1, wherein the pump beam of light and the Stokes beam of light each comprises pulses of light, each pulse of light having a duration between 50 nanoseconds and 5 microseconds and a peak power between 1 watt and 1 kilowatt.

22. The system of claim 1, wherein the pump beam of light has a substantially constant optical power and the Stokes beam of light comprises pulses of light.

23. The system of claim 1, wherein the pump beam of light comprises pulses of light and the Stokes beam of light has a substantially constant optical power.

24. The system of claim 1, wherein: the pump beam of light or the Stokes beam of light comprises pulses of light; the probe beam of light comprises pulses of light, wherein each probe-beam pulse of light is emitted a particular time delay after a corresponding pump-beam or Stokes-beam pulse of light, the particular time delay based on a distance from the system to the object.

25. The system of claim 1, wherein: the Stokes light source is further configured to produce a Stokes reference beam of light; the one or more optical elements are further configured to collect residual light from the Stokes beam of light after the Stokes beam of light has interacted with the object; the probe light source is further configured to produce a probe reference beam of light; the optical detector is a signal detector, wherein the photocurrent signal produced by the signal detector is a signal photocurrent that corresponds to the Raman signal, the probe beam of light, and the residual Stokes beam of light, wherein a portion of the signal photocurrent corresponds to the coherent mixing between the Raman signal and the probe beam of light; the optical receiver further comprises a reference detector configured to produce a reference photocurrent corresponding to the probe reference beam of light and the Stokes reference beam of light; the electronic circuit is further configured to determine a subtraction signal that equals a difference between a signal corresponding to the signal photocurrent and a signal corresponding to the reference photocurrent; and the processor is further configured to determine a characteristic of the subtraction signal.

26. The system of claim 1, further comprising a visible light source configured to produce a visible beam of light, wherein the visible beam of light is directed to the object and is configured to produce a visible alignment spot at the object that substantially overlaps a location of the pump and Stokes beams at the object.

27. The system of claim 1, wherein the object comprises a package, a person, a vehicle, or an aircraft.

28. The system of claim 1, wherein: the probe light source comprises a wavelength-tunable laser, wherein the probe light source is further configured to sequentially change the probe frequency of the probe beam of light to a plurality of different frequencies; the optical detector is further configured to coherently mix the Raman signal with the probe beam of light at each of the different frequencies to produce a plurality of corresponding photocurrent signals; and the processor is further configured to determine a characteristic of each of the plurality of corresponding photocurrent signals.

- 29.** The system of claim 1, wherein the pump light source or the Stokes light source comprises a wavelength-tunable laser, wherein the frequency offset Ω is adjustable by changing a wavelength of the wavelength-tunable laser.
- 30.** The system of claim 1, wherein the detector comprises a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT).
- 31.** The system of claim 1, wherein the optical detector is one of a plurality of optical detectors, wherein each detector is configured to coherently mix a portion of the Raman signal with a portion of the probe beam of light to produce a corresponding photocurrent signal.
- 32.** The system of claim 1, wherein the electronic circuit comprises: an electronic amplifier configured to amplify the photocurrent signal to produce a voltage signal corresponding to the photocurrent signal; and a digitizer configured to produce a digital representation of the voltage signal, wherein the digital representation of the voltage signal is part of the digital output signal.
- 33.** The system of claim 1, wherein the characteristic of the photocurrent signal comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization.
- 34.** The system of claim 1, wherein the processor is further configured to associate a Raman frequency shift with the determined characteristic of the photocurrent signal, wherein the Raman frequency shift equals $v_{\text{sub.pu}} - v_{\text{sub.pr}}$, wherein $v_{\text{sub.pu}}$ is the pump frequency, and $v_{\text{sub.pr}}$ is the probe frequency.
- 35.** The system of claim 1, wherein: the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and the processor is further configured to determine, based on the characteristic of the photocurrent signal, (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material in the sample.
- 36.** A method for measuring a Raman signal, the method comprising: producing a pump beam of light at a pump frequency; producing a Stokes beam of light at a Stokes frequency, wherein the pump and Stokes frequencies are offset by a frequency offset Ω , and wherein the pump and Stokes beams are directed to an object; collecting a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the object; detecting the Raman signal, comprising: producing a probe beam of light at a probe frequency; coherently mixing the Raman signal with the probe beam of light to produce a corresponding photocurrent signal; and producing a digital output signal corresponding to the photocurrent signal; and determining a characteristic of the photocurrent signal based on the digital output signal.
- 37.** One or more computer-readable non-transitory storage media embodying software that is operable when executed to: produce a pump beam of light at a pump frequency; produce a Stokes beam of light at a Stokes frequency, wherein the pump and Stokes frequencies are offset by a frequency offset Ω , and wherein the pump and Stokes beams are directed to an object; collect a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the object; detect the Raman signal, comprising: produce a probe beam of light at a probe frequency; coherently mix the Raman signal with the probe beam of light to produce a corresponding photocurrent signal; and produce a digital output signal corresponding to the photocurrent signal; and determine a characteristic of the photocurrent signal based on the digital output signal.
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